

Curriculum and Syllabus

For

Bachelor of Technology in
Computer Science & Engineering
B.Tech (CSE)

Under

Choice Based Credit System (CBCS)
[With effect from the Session 2024-25]



USHA MARTIN UNIVERSITY
RANCHI, JHARKHAND
(Recognized by UGC under Sec. 2(f) of UGC Act 1956)

Course Structure of B.Tech (CSE)

SI No.	Course Code	Title of the Course	Credits
Semester-I			
1	GSC101	Physics	04
2	GSC102	Mathematics – I	04
3	GEC101	Basic Electrical Engineering	04
4	GEC102	Engineering Graphics and Design	01
5	GSC111	Physics Lab	1.5
6	GEC111	Basic Electrical Engineering Lab	1
7	GEC112	Engineering Graphics and Design Lab	2
Semester-II			
1	GSC201	Chemistry	4
2	GSC202	Mathematics-II	4
3	GEC201	Programming for Problem Solving	3
4	GEC202	Workshop/ Manufacturing Process	1
5	HSC201	English	2
6	GSC211	Chemistry-I Lab	1.5
7	GEC211	Programming for Problem Solving Lab	2
8	GEC212	Workshop/ Manufacturing Process Lab	2
9	HSC211	English Lab	1
Semester-III			
1	GSC301	Mathematics-III	2
2	GEC301	Analog Electronic Circuit	3
3	GEC302	Digital Electronics	3
4	CSC301	Data Structures and Algorithms	3
5	CSC302	IT Workshop	1
6	HSC301	Development of Life Skills – I	3
7	GEC311	Analog Electronic Circuit Lab	2
8	GEC312	Digital Electronics Lab	2
9	CSC311	Data Structures and Algorithms Lab	2
10	CSC312	IT Workshop Lab	2
Semester-IV			
1	CSC401	Discrete Mathematics	4
2	CSC402	Computer Organization and Architecture	3
3	CSC403	Operating Systems	3
4	CSC404	Design and Analysis of Algorithms	3
5	HSC401	Management I	3
6	CSC411	Computer Organization and Architecture Lab	2
7	CSC412	Operating Systems Lab	2
8	CSC413	Design and Analysis of Algorithms Lab	2
9	MDC401	Environmental Sciences	NC
Semester-V			
1	GEC501	Signals and Systems	3

2	CSC501	Database Management Systems	3
3	CSC502	Formal Language and Automata Theory	3
4	CSC503	Object Oriented Programming	2
5	HSC501	Development of Life Skills – II	3
6	CSC511	Database Management Systems Lab	2
7	CSC512	Object Oriented Programming Lab	2
8	MDC501	Constitution of India	NC
9	ECD/ECS/E CT/ECA	Elective-I	3
Semester-VI			
1	CSC601	Compiler Design	3
2	CSC602	Computer Networks	3
3	CSC611	Compiler Design Lab	2
4	CSC612	Computer Networks Lab	2
5	CSC613	Project – I	3
6	ECD/ECS/E CT/ECA	Elective-II	3
7	ECD/ECS/E CT/ECA	Elective-III	3
8	OEC	Open Elective-I	3
Semester-VII			
1	ECD/ECS/E CT/ECA	Elective-IV	3
2	ECD/ECS/E CT/ECA	Elective-V	3
3	OEC	Open Elective-II	3
4	GSC701	Biology	3
5	CSC711	Project – II	6
Semester-VIII			
1	ECD/ECS/E CT/ECA	Elective-VI	3
2	OEC	Open Elective-II	3
3	OEC	Open Elective-III	3
4	CSC811	Project – III	6
OPEN ELECTIVES			
1	OEC001	Soft Skills and Interpersonal development	3
2	OEC002	Human Resource Development and Organizational Behavior	3
3	OEC003	Cyber Law and Ethics	3
4	OEC004	Indian Music System	3
5	OEC005	History of Science	3
6	OEC006	Introduction to Arts and Aesthetics	3
7	OEC007	Economic Policies in India	3
8	OEC008	Environmental Engineering and Disaster Management	3
9	OEC009	Sustainable Engineering	3

10	OEC010	Water Pollution Control Engineering	3
11	OEC011	Intellectual Property Rights	3
ELECTIVE COURSES			
1	ECT001	Theory of Computation	3
2	ECT002	Graph Theory	3
3	ECT003	Advanced Algorithms	3
4	ECT004	Parallel and Distributed Algorithms	3
5	ECT005	Computational Complexity	3
6	ECS001	Advanced Computer Architecture	3
7	ECS002	Software Engineering	3
8	ECS003	Distributed Systems	3
9	ECS004	Embedded Systems	3
10	ECS005	Advanced Operating Systems	3
11	ECS008	Real Time Systems	3
12	ECS011	Internet of Things	3
13	ECD001	Artificial Intelligence	3
14	ECD002	Machine Learning	3
15	ECD003	Data Mining	3
16	ECD004	Soft Computing	3
17	ECD006	Data Analytics	3
18	ECD008	Neural Networks and Deep Learning	3
19	ECA001	Image Processing	3
20	ECA002	Digital Signal Processing	3
21	ECA003	Cloud Computing	3
22	ECA004	Human Computer Interaction	3
23	ECA006	Computer Graphics	3
24	ECA008	Optimization Technique	3
25	ECA010	Cryptography and Network Security	3



USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Faculty of Computing and Information Technology

Program: Bachelor of Technology (Computer Science and Engineering)

SEMESTER I

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: I

Session: ODD

Subject Code & Name: GSC101 Physics

Credits: 4

Course Outcome: At the end of the course, the students will be able to:

GSC101.CO1	Explain the principles of Newton's laws of motion, coordinate transformations, and rotational dynamics in various coordinate systems, including polar, cylindrical, and spherical coordinates.
GSC101.CO2	Analyze potential energy functions, central forces, and energy conservation to solve Kepler's problem and understand orbital mechanics and satellite maneuvers.
GSC101.CO3	Apply concepts of damped and forced harmonic motion to study oscillatory systems and predict resonance phenomena under different damping conditions.
GSC101.CO4	Evaluate the kinematics and dynamics of rigid body motion in two-dimensional systems using Euler's laws and angular momentum principles.
GSC101.CO5	Develop models to describe and differentiate between two-dimensional and three-dimensional rigid body motion, emphasizing angular velocity vectors and the moment of inertia tensor.

Unit I. Introduction and Transformation

Transformation of scalars and vectors under Rotation transformation; Forces in Nature. Newton's laws and its completeness in describing particle motion; Form in variance of Newton's Second Law; Solving Newton's equations of motion in polar coordinates; Problems including constraints and friction; Extension to cylindrical and spherical Coordinates.

Unit II. Potential Energy, Acceleration and Oscillator

Potential energy function; $F = -\text{Grad } V$, Equi-potential surfaces and meaning of gradient; Conservative and non-conservative forces, curl of a force field; Central forces; Conservation of Angular Momentum; Energy equation and energy diagrams; Elliptical, parabolic and hyperbolic orbits; Kepler problem; Application: Satellite maneuvers. Non-inertial frames of reference; Rotating coordinate system: Five-term acceleration formula. Centripetal and Coriolis accelerations; Applications: Weather systems, Foucault pendulum.

Unit III. Harmonic oscillator

Damped harmonic motion – over-damped, critically damped and lightly-damped oscillators; Forced oscillations and resonance.

Unit IV. Motion

Definition and motion of a rigid body in the plane; Rotation in the plane; Kinematics in a coordinate system rotating and translating in the plane; Angular momentum about a point of a rigid body in planar motion; Euler's laws of motion, their independence from Newton's laws, and their necessity in describing rigid body motion.

Unit V. Three-Dimensional Motion

Introduction to three-dimensional rigid body motion — only need to highlight the distinction from two-dimensional motion in terms of (a) Angular velocity vector, and its rate of change and (b) Moment of inertia tensor; Three-dimensional motion of a rigid body wherein all points move in a co-planar manner: e.g. Rod executing conical motion with center of mass fixed — only need to show that this motion looks two-dimensional but is three-dimensional, and two-dimensional formulation fails.

Text Books and References

1. Engineering Mechanics, 2nd ed. — MK Harbola
2. Introduction to Mechanics — MK Verma
3. An Introduction to Mechanics — D Kleppner & R Kolenkow
4. Principles of Mechanics — JL Synge & BA Griffiths
5. Mechanics — JP Den Hartog
6. Engineering Mechanics - Dynamics, 7th ed. - JL Meriam
7. Mechanical Vibrations — JP Den Hartog
8. Theory of Vibrations with Applications — WT Thomson

Faculty: Faculty of Computing and Information Technology

Program: B.Tech(CSE)

Semester: I

Session: ODD

Subject Code & Name: GSC102 Mathematics-I

Credits: 4

Course Outcome: At the end of the course, the students will be able to

GSC102.CO1	Understand the convergence of sequences and series and apply tests like the ratio, root, and Leibnitz tests to determine convergence.
GSC102.CO2	Solve linear systems using matrix operations, eigenvalues, eigenvectors, and the Cayley-Hamilton theorem, while evaluating linear independence.
GSC102.CO3	Analyze multivariable functions using partial derivatives, Jacobians, Taylor series, and Lagrange's method to determine extrema.
GSC102.CO4	Evaluate double and triple integrals in various coordinate systems to calculate areas and volumes.
GSC102.CO5	Formulate and solve physics and engineering problems using vector calculus concepts like gradient, divergence, curl, and related theorems.

Unit I. Series and Sequences

Sequences and Series Sequences, Convergence of Sequence. Series, Convergence of Series, Tests for onvergence: Comparison tests, Ratio test, Cauchy's root test, Raabe's test, Gauss test, Cauchy's Integral test, Alternating series, Leibnitz test, Absolute and Conditional Convergence.

Unit II. Matrices

Matrices Rank of a Matrix, elementary transformations, Row - reduced Echelon form. Vectors, Linear Independence and Dependence of Vectors. Consistency of system of linear equations. Eigenvalues, Eigenvectors, Cayley - Hamilton theorem.

Unit III. Advance Differential Calculus

Advance Differential Calculus Function of several variables, Limit, Continuity, Partial derivatives, Euler's theorem for homogeneous functions, Total derivatives, Chain rules, Jacobians and its properties, Taylor series for function of two variables, Maxima – Minima, Lagrange's method of multipliers.

Unit IV. Integral Calculus

Advance Integral Calculus Beta and Gamma functions: definition and properties. Double integrals, double integrals in polar coordinates, Change of order of integration, Triple Integrals, cylindrical and spherical coordinate systems, transformation of coordinates, Applications of double and triple integrals in areas and volumes.

Unit V. Vector Calculus

Vector Calculus Scalar and vector point functions, gradient, directional derivative, divergence, curl, vector equations and identities. Line Integral, Work done, Conservative field, Green's theorem in a plane, Surface and volume integrals, Gauss – divergence theorem, Stoke 's theorem.

Text Books & References:

1. E. Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
3. S. Narayan and P. K. Mittal, Differential Calculus, S. Chand Publication, 2016.
4. S. Narayan and P. K. Mittal, Integral Calculus, S. Chand Publication, 2018.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: I

Session: ODD

Subject Code & Name: GEC101 Basic Electrical Engineering

Credits: 4

Course Outcome: At the end of the course, the students will be able to

GEC101.CO1	Explain the fundamental concepts of electrical circuits, including electric fields, current, potential difference, Ohm's law, and energy storage in inductors and capacitors.
GEC101.CO2	Analyze DC electrical circuits using laws and theorems such as KCL, KVL, Thevenin's, and Norton's theorem, and solve problems involving transients in R-L and R-C circuits.
GEC101.CO3	Evaluate single-phase and three-phase AC systems by analyzing EMF generation, power factor, and complex power to assess their performance.
GEC101.CO4	Apply the principles of AC circuits to solve problems involving resistance, inductance, capacitance, and their combinations in series circuits.
GEC101.CO5	Design and analyze parallel AC circuits, including resonance and quality factors, using admittance methods and phasor diagrams for optimization.

Unit I. Electrical Engineering Fundamentals

Introduction to electrical circuits: Electric field, electric current, potential and potential difference, electric power, basic circuit components, ohm's law. Sources and its types, Ideal and practical sources, Source Conversion, independent and dependent sources, Energy Stored in Inductor and Capacitor, series, parallel and series and parallel circuit.

Unit II. DC Networks & Theorems

Laws and Theorems applicable to DC networks (KCL & KVL, Node voltage & Mesh current analysis, Star-Delta and Delta-Star conversion, Superposition principle, Thevenin & Norton theorem), Transients in R-L and R-C circuits with DC excitation, Simple problems

Unit III. AC Fundamentals

Single-Phase AC Circuits: Single-phase EMF Generation, Average and Effective value of periodic ac signals, Peak factor & Form factor, Phasor and Complex representation of sinusoids, Power factor, complex power. Three-Phase AC Circuits: Comparison between single-phase and three-phase systems, three phase EMF Generation, Line.

Unit IV. Single Phase A.C. Circuits

A.C. Circuits containing resistance only, A.C. Circuits containing inductance only, A.C. Circuits containing Capacitance only, R-L Series circuits, R-C series circuits, R-L-C Series circuits, Simple Numerical problems

Unit V. Parallel A.C. Circuits

R-L,R-C and R-L-C circuits. Admittance, susceptance, solution by admittance method, phasor diagram, Parallel resonance, quality factor. Comparison of Series and parallel Circuits.

Text Books & References:

1. J.B. Gupta , “Basic Electrical Engineering” , S K Kataria and sons, 2012 edition
2. B.L. Theraja , A.K. Theraja, “ A Text Book of Electrical Technology”, S.Chand 23 edition
3. D.P. Kothari, I.J. Nagrath, McGraw Hill Education India, 2019 edition
4. V. K. Mehta, RohitMehta , “Principles of electrical Machines” 2014 edition

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: I

Session: ODD

Subject Code & Name: GEC102 Engineering Graphics and Design

Credits: 1

Course Outcome: At the end of the course, the students will be able to

GEC102.CO1	Understand the significance, standards, conventions of engineering graphics, and the use of drafting tools for design and communication.
GEC 102.CO2	Apply drawing instruments, freehand sketching techniques, and drafting aids to create geometric shapes, shading, and renderings.
GEC 102.CO3	Analyze engineering drawings using standards like ASME Y14.5 and ISO 128 to interpret features such as threads, fasteners, and surface finishes.
GEC 102.CO4	Evaluate and construct orthographic and isometric projections, including sectional and auxiliary views, to represent 3D objects in 2D.
GEC 102.CO5	Create 2D sketches and 3D models using CAD software with advanced drawing and editing tools for engineering component design.

Unit I. Introduction to Engineering Graphics

Definition and significance of engineering graphics; Role of engineering graphics in design and communication; Overview of drafting instruments and materials; Introduction to drafting standards and conventions.

Unit II. Drawing Instruments and Techniques

Types of drawing instruments: Drawing board, T-square, set squares, compass, dividers, etc; Freehand sketching techniques: Line types, geometric shapes, shading, and rendering; Use of drafting aids: French curves, templates, ellipse guides, etc; Introduction to computer-aided drafting (CAD) software: Basic features and tools.

Unit III. Engineering Drawing Standards and Practices

Overview of engineering drawing standards: ASME Y14.5, ISO 128, DIN standards, etc; Drawing layout and organization: Title block, revision block, and notes; Conventional representations of common features: Holes, threads, fasteners, and surface finishes; Drawing interpretation and reading: Understanding engineering drawings.

Unit IV. Orthographic Projection

Principles of orthographic projection: Projection planes, reference planes, and viewing direction; First and third angle projection methods; Multi view drawings: Front view, top view, side view, and auxiliary views; Sectional views: Full section, half section, and offset section; Isometric and Oblique Projection.

Isometric projection:

Principles, axis alignment, and drawing techniques; Isometric scale and isometric grid; Oblique projection: Cabinet and cavalier projections; Drawing oblique pictorials and their applications.

Unit V. Introduction to Computer-Aided Design (CAD)

Basics of computer-aided design (CAD): History and evolution; Overview of CAD software packages: Auto CAD, etc; Drawing and editing tools in CAD software; Creating 2D sketches and 3D models using CAD software.

Text Books & References

1. Engineering Drawing and Design by David A. Madsen, David P. Madsen, and David P. Madsen.
2. Technical Drawing with Engineering Graphics by Frederick E. Giesecke, Alva Mitchell, Henry C. Spencer, and Ivan L. Hill.
3. Engineering Graphics with AutoCAD by James D. Bethune.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: I

Session: ODD

Subject Code & Name: GSC111 Physics Lab

Credits: 1.5

Course Outcome: At the end of the course, the students will be able to:

GSC111.CO1	Explain the principles of coupled oscillations, including normal modes and energy transfer, based on experimental observations.
GSC111.CO2	Apply air-track experiments to study near-frictionless motion dynamics and validate Newton's laws of motion.
GSC111.CO3	Analyze rotational motion by calculating the moment of inertia of rigid bodies using experimental setups.
GSC111.CO4	Evaluate gyroscopic motion, angular momentum, and torque to examine the stability of rotating systems.
GSC111.CO5	Investigate resonance phenomena by designing experiments to determine natural frequencies and understand damping effects in mechanical oscillators.

List of Experiment:

1. Coupled oscillators
2. Experiments on an air-track
3. Experiment on moment of inertia measurement,
4. Experiments with gyroscope
5. Resonance phenomena in mechanical oscillators

Text Books and References

1. Engineering Mechanics, 2nd ed. — MK Harbola
2. Introduction to Mechanics — MK Verma
3. An Introduction to Mechanics — D Kleppner & R Kolenkow
4. Principles of Mechanics — JL Synge & BA Griffiths
5. Mechanics — JP Den Hartog
6. Engineering Mechanics - Dynamics, 7th ed. - JL Meriam
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Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: I

Session: ODD

Subject Code & Name: GEC111 Basic Electrical Engineering Lab

Credits: 1

Course Outcome: At the end of the course, the students will be able to

GEC111.CO1	Understand and verify fundamental electrical principles such as Ohm's Law, Thevenin's theorem, Norton's theorem, and the superposition theorem through experiments.
GEC111.CO2	Apply concepts to design and analyze RL and RLC circuits, determine current and voltage characteristics, and measure power in three-phase star-connected loads.
GEC111.CO3	Evaluate the voltage transformation ratio of a single-phase transformer and analyze its performance under different conditions.
GEC111.CO4	Analyze the construction and operation of DC machines and three-phase induction motors to understand their functionality and applications.
GEC111.CO5	Design and perform transient analysis of series RL and RC circuits to observe time-dependent voltage and current characteristics.

List of Experiment:

1. Verification of Ohm's Law
2. Verification of Thevenin's & Norton's Theorem
3. Verification of Superposition Theorem
4. To design series RL circuit and find out the current flowing through each component
5. To study RLC series circuit
6. Determination of the voltage transformation ratio (K) of a single phase transformer
7. To measure the power of 3 load star
8. Study of different parts of DC machine.
9. Study of three phase induction motor
10. Study and use of Megger
11. Transient analysis of series RL and RC circuit

Text Books & References

1. J.B. Gupta , "Basic Electrical Engineering" , S K Kataria and sons, 2012 edition
2. B.L. Theraja , A.K. Theraja, " A Text Book of Electrical Technology", S.Chand 23 edition
3. D.P. Kothari, I.J. Nagrath, McGraw Hill Education India, 2019 edition
4. V. K. Mehta, Rohit Mehta , "Principles of electrical Machines" 2014 edition

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: I

Session: ODD

Subject Code & Name: GEC112 Engineering Graphics and Design Lab

Credits: 2

Course Outcome: At the end of the course, the students will be able to

GEC 112.CO1	Understand the layout of drawing sheets, types of lines, lettering, dimensioning methods, scales, and orthographic projection principles in engineering drawing.
GEC 112.CO2	Apply concepts of orthographic projections to represent points, lines, and planes in various quadrants and orientations relative to reference planes.
GEC 112.CO3	Analyze and create projections of 3D solids like cubes, prisms, pyramids, cylinders, and cones, including their lateral surfaces and intersections.
GEC 112.CO4	Create isometric projections and visualizations of engineering solids using isometric scales and construction techniques.
GEC 112.CO5	Utilize CAD software to draft, modify, and print 2D and 3D drawings using advanced drafting commands and editing tools.

List of Experiments:

a. **Experiment 1: Introduction and Orthographic Projections**

Introduction to Engineering Drawing: Layout of drawing sheets, sizes of drawing sheets, different types of lines used in drawing practice, Lettering, Dimensioning – linear, angular, aligned system, unidirectional system, parallel dimensioning, chain dimensioning, different type of scale (Plain and Diagonal scale).

Curves used in Engineering Practice: Explain conic sections with illustration, Explain terms like focus, vertex, directrix and eccentricity, Draw conics sections by eccentricity method- Ellipse, Parabola and Hyperbola.

Orthographic Projections: Reference planes, pictorial projection, orthographic projections – 1st, 2nd, 3rd, 4th angle projections, symbol of first and third angle projection

b. **Experiment 2: Points, Lines and Planes:**

Projections of Points and Lines: Projection of Points in different quadrants (1st, 2nd, 3rd, 4th) ,projections of lines (parallel to both reference planes, parallel to one and perpendicular to other, parallel to one and inclined to the other and inclined to both the reference planes).

Projection of planes: Projection of planes (Perpendicular to one plane and parallel to the other, perpendicular to the one plane and inclined to the other, perpendicular to both the reference planes).

c. **Experiment 3: Solids and Section of Solids**

Projections of Solids: Introduction of solids, projections of solids such as cube, tetrahedron, prism, pyramid, cylinder, cone & Sphere (with axes parallel to both the reference plane, axis perpendicular to one reference plane).

d. **Experiment 4: Development and Intersection of Solid Surfaces**

Development of Lateral Surfaces of Solids and Intersection of Surfaces: Introduction to Development of surfaces of various solids (Prism, Cylinder, Pyramid, Cone & Sphere). Development of Lateral Surface-concept, methods, applications, Interpenetration of solids such as prism-prism, prism- pyramid.

Isometric Projections: Draw isometric view & isometric projection of prism, pyramid, cone & cylinder with axis vertical with construction of isometric scale

e. **Experiment 5: Computer-Aided Drafting and Solid Modeling**

Introduction to Computer-Aided Drafting (CAD): Advantages of using Computer Aided Drafting (CAD) packages, applications of CAD, Cartesian and Polar Coordinate system, Absolute and Relative Coordinates systems; basic operation of drafting packages, use of various Basic Commands: Line, Point, Rectangle, Polygon, Circle, Arc, Ellipse, Poly-line, Basic editing Commands: Basic Object Selection Methods, Window and Crossing Window, Erase, Move, Copy, Offset, Fillet, Chamfer, Trim, Extend, Mirror; Display Commands: Zoom, Pan, Redraw, and Regenerate for drawing, dimensioning, editing, modifying, saving and printing/plotting the drawings.

Text Books & References:

1. Engineering Drawing by N.D.Bhatt&V.M.Panchal, Charotar publishing House, 50th Edition, 2012
2. Text Book on Engineering Drawing by Narayana / Kannaiah, Scitech publications, 9th Edition, 2012
3. Engineering Drawing with an Introduction to AutoCAD by Dhanjay A. Johle, Tata McGraw Hill publications, 1st Edition 2007
4. Machine Drawing (Includes AutoCAD) by Ajeet Singh, Tata McGraw Hill publications, 1st Edition, 2010
5. Engineering Drawing by Shah and Rana, Pearson Education, 2nd Edition, 2009
6. Engineering Drawing and Graphics by K.Venugopal, New Age International publications, 5th Edition, 2004

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Semester: II

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: II

Session: EVEN

Subject Code & Name: GSC201 Chemistry

Credits: 4

Course Outcomes: At the end of the course, the students will be able to

GSC 201.CO1	Understand atomic and molecular structures using quantum mechanics principles, including wave functions, molecular orbitals, and band structures, and their significance in material properties.
GSC 201.CO2	Apply spectroscopic techniques like electronic, vibrational, rotational, and NMR spectroscopy to analyze molecular structures and their applications in fields like medicine and materials.
GSC 201.CO3	Analyze intermolecular forces, potential energy surfaces, and free energy changes in real gases, chemical equilibria, and electrochemical processes such as corrosion and cell potentials.
GSC 201.CO4	Evaluate periodic trends, such as ionization energy, electronegativity, oxidation states, and molecular geometries, to predict chemical behavior and bonding in compounds.
GSC 201.CO5	Create 3D molecular models, analyze stereochemistry and isomerism, and design reaction pathways for organic synthesis, including developing commonly used drug molecules.

Unit I. Atomic and molecular structure

Schrodinger equation. Particle in a box solutions and their applications for conjugated molecules and nanoparticles. Forms of the hydrogen atom wave functions and the plots of these functions to explore their spatial variations. Molecular orbitals of diatomic molecules and plots of the multicenter orbitals. Equations for atomic and molecular orbitals. Energy level diagrams of diatomic. Pi-molecular orbitals of butadiene and benzene and aromaticity. Crystal field theory and the energy level diagrams for transition metal ions and their magnetic properties. Band structure of solids and the role of doping on band structures.

Unit II. Spectroscopic techniques and applications

Principles of spectroscopy and selection rules. Electronic spectroscopy. Fluorescence and its applications in medicine. Vibrational and rotational spectroscopy of diatomic molecules. Applications. Nuclear magnetic resonance and magnetic resonance imaging, surface characterisation techniques. Diffraction and scattering

Unit III. Intermolecular forces, potential energy surfaces and Use of free energy in chemical equilibria

Ionic, dipolar and van Der Waals interactions. Equations of state of real gases and critical phenomena. Potential energy surfaces of H_2 , H_2F and HCN and trajectories on these surfaces. energy, entropy and free energy. Estimations of entropy and free energies. Free energy and emf. Cell potentials, the Nernst equation and applications. Acid base, oxidation reduction and solubility equilibria. Water chemistry. Corrosion. Use of free energy considerations in metallurgy through Ellingham diagrams.

Unit IV. Periodic properties

Effective nuclear charge, penetration of orbitals, variations of s, p, d and f orbital energies of atoms in the periodic table, electronic configurations, atomic and ionic sizes, ionization energies, electron affinity and electronegativity, polarizability, oxidation states, coordination numbers and geometries, hard soft acids and bases, molecular geometries.

Unit V. Stereochemistry, Organic reactions and synthesis of a drug molecule

Representations of 3 dimensional structures, structural isomers and stereoisomers, configurations and symmetry and chirality, enantiomers, diastereomers, optical activity, absolute configurations and conformational analysis. Isomerism in transitional metal compounds.

Introduction to reactions involving substitution, addition, elimination, oxidation, reduction, cyclization and ring openings. Synthesis of a commonly used drug molecule.

Text Books & References:

1. Physical Chemistry, by P.W. Atkins
2. Engineering Chemistry by P.C.Jain&M.Jain; Dhanpat Rai Publishing Company (P) Ltd., New Delhi.
3. Fundamentals of Molecular Spectroscopy, by C.N. Banwell
3. Organic Chemistry: Structure and Function by K.P.C. Volhardt and N.E.Schore, 5th Edition.
4. University Chemistry, by B.M. Mahan, Pearson IV Edition.
5. Engineering Chemistry (NPTEL Web-book), by B.L. Tembe, Kamaluddin and M.S. Krishnan

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: II

Session: EVEN

Subject Code & Name: GSC202 Mathematics-II

Credits: 4

Course Outcome: At the end of the course, the students will be able to

GSC202.CO1	Understand and solve second- and higher-order linear differential equations using concepts like linear independence, the Wronskian, and methods such as operator techniques, Legendre's form, Euler-Cauchy form, and variation of parameters.
GSC202.CO2	Analyze ordinary and singular points of differential equations, and solve Bessel's and Legendre's differential equations, focusing on their properties and series solutions.
GSC202.CO3	Apply Fourier series to represent periodic functions and solve partial differential equations using methods like separation of variables to model wave and heat equations.
GSC202.CO4	Use complex variable techniques, such as Cauchy-Riemann equations, Cauchy's theorem, and residue theorem, to solve problems involving analytic functions, singularities, and series expansions.
GSC202.CO5	Evaluate and apply probability distributions, including Binomial, Poisson, and Normal distributions, in solving real-world problems involving random variables and probability functions.

Unit I. Functions of Several Variables

Linear differential equations, Wronskian, Linear independence and dependence of solutions, Linear differential equations of second and higher order, Operator method, Legendre's and Euler – Cauchy's form of linear differential equation, Method of variation of parameters.

Unit II. Ordinary Differential Equations

Ordinary and singular points of differential equation, Power and Frobenius series solutions. Bessel's differential equation, Bessel function of first kind and its properties. Legendre's differential equation, Legendre's polynomial and its properties.

Unit III. Fourier series and Partial Differential Equations

Fourier series: Euler formula for Fourier series, Dirichlet conditions, Half range Fourier series. Partial Differential Equations: Linear partial differential equations, Lagrange's method. Method of separation of variables and its application in solving one dimensional wave and heat equations.

Unit IV. Functions of Complex Variables

Function of a complex variable, Limit, Continuity, Differentiability, Analyticity, Analytic functions, Cauchy – Riemann equations. Harmonic functions, Harmonic Conjugate. Cauchy's theorem, Cauchy's Integral formula, Taylor and Laurent series expansions. Singularities and its types, Residues, Residue theorem

Unit V. Random Variables and Distribution Functions

Discrete and continuous random variables, cumulative distribution function, probability mass and density functions, expectation, variance, moment generating function. Introduction to Binomial, Poisson and Normal Distribution

Text Books & References

1. E. Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. R.K. Jain and S.R.K. Iyengar, Advanced Engineering Mathematics, 3rd Edition, Narosa Publishing, 2009.
3. D. G. Zill and W.S. Wright, Advanced Engineering Mathematics, 4th Edition, 2011.
4. J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Edition, McGraw Hill, 2004.
5. R. A. Johnson, I. Miller and J. Freund: Probability and Statistics for Engineers.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: II

Session: EVEN

Subject Code & Name: GEC201 Programming for Problem Solving

Credits: 3

Course Outcome: At the end of the course, the students will be able to

GEC201.CO1	Explain the fundamental components of computer systems, algorithms, flowcharts, and the C programming language structure.
GEC201.CO2	Use decision-making, loops, and unconditional control structures to solve real-world computational problems using the C language.
GEC201.CO3	Analyze the functionality of various user-defined and standard functions, including parameter passing and recursion, to design structured and modular programs.
GEC201.CO4	Implement arrays and strings for data manipulation and apply basic algorithms such as searching and sorting to solve practical problems.
GEC201.CO5	Design and implement programs using pointers, structures, and file-handling techniques to handle complex data processing and storage requirements.

Unit I. Introduction to Programming

Computer Systems-Hardware and Software, Computer Languages, Algorithm, Flowchart, Representation of Algorithm and Flowchart with examples. Introduction to C:History of C, Features of C, Structure of C Program, Character Set, C Tokens-Keywords, Identifiers, Constants, Variables, Data types, Operators, Variables and memory locations, Storage classes.

Unit II. Conditional Branching and Loops

Selection statements (Decision Making)- if and switch statements with examples, Repetition statements (loops)- while, for, do-while statements with examples, Unconditional statements-break, continue, goto statements with examples.

Unit III. Functions

Designing Structured Programs, Types of Functions-User defined functions, Standard, Categories of functions, Parameter Passing techniques, Storage classes, Recursion.

Unit IV. Arrays and String

Declaration and Initialization, One dimensional Arrays, Two dimensional Arrays.

Strings: Declaration and Initialization, String Input / Output functions, String manipulation functions.

Basic Algorithms: Searching, Basic Sorting Algorithms (Bubble, Insertion and Selection), Finding roots of equations

Unit V. Pointers and File Handling

Introduction, Definition and Declaration of pointers, address operator, Pointer variables, Pointers with Arrays. Structures: Introduction, Declaration and Initialization, Array of Structures, Unions. **File handling:** File I/O functions, Standard C pre processors, defining and calling macros, command-line arguments.

Text Books & References

1. Computer Programming with C, Special Edition-MRCET, Mc Graw Hill Publishers 2017.
2. Computer Science: A Structured Programming Approach Using C, B.A.Forouzan and R.F. Gilberg, Third Edition, Cengage Learning.
3. The C Programming Language, B.W. Kernighan and Dennis M.Ritchie, PHI.
4. Computer Programming, E.Balagurusamy, First Edition, TMH.
5. C and Data structures – P. Padmanabham, Third Edition, B.S. Publications.
6. Programming in C, *Ashok Kamthane*. Pearson Education India.
7. Let us C ,YashwanthKanethkar, 13th Edition, BPB Publications.

Faculty: Faculty of Computing and Information Technology**Program:** B.Tech (CSE)**Semester:** II**Session:** EVEN**Subject Code & Name:** GEC202 Workshop/ Manufacturing Practices**Credits:** 1**GEC 202 Workshop/Manufacturing Practices Credits 1****Course Outcome:** At the end of the course, the students will be able to:

GEC 202.CO1	Perform basic operations in various trades such as carpentry, fitting, tin-smithy, blacksmithy, plumbing, and house wiring to develop foundational mechanical engineering skills.
GEC 202.CO2	Explain the use and working principles of power tools in construction, woodwork, and mechanical engineering applications.
GEC 202.CO3	Analyze the mechanics of metal cutting, including cutting tool geometry, tool wear, machinability, and the economics of machining processes.
GEC 202.CO4	Operate lathes and milling machines for turning, indexing, gear cutting, and other machining operations, understanding the various components and mechanisms involved.
GEC 202.CO5	Design and execute drilling and boring operations by selecting appropriate tools, techniques, and machining parameters to achieve desired specifications and tolerances.

Unit I. Experiments in Mechanical Engineering

Experiments covering, the following trades for experiments (with a minimum of two exercises under each trade) - Carpentry, Fitting, Tin-Smithy and Development of jobs carried out and soldering, Black Smithy, House Wiring, Foundry (Moulding only) and Plumbing.

Unit II. Power Tools

Mechanical Engineering covering, the following trades for demonstration for exposure—Powertools in Construction, Wood working, Electrical and Mechanical Engineering practices

Unit III. Metal Cutting and Cutting Tools & Lathe Introduction

Types of cutting tools, orthogonal and oblique cutting, types of chips, chip breakers, cutting tool nomenclature, cutting action of hand tools, cutting feed and speed, friction and heat sources in cutting, tools life and wear, machinability, cutting tool materials, cutting fluids, economics of machining. Lathe- Turning on lathes, traditional lathes, capstan and turret lathes, automatic lathes

Unit IV. Milling Machine

Types of milling machines, milling cutters, indexing and dividing heads, indexing methods, calculations of indexing, multi-axis milling, Gear cutting methods, gear hobbing.

Unit V. Drilling

Introduction, types of drill, twist drill nomenclature, types of drilling machines, work holding devices, tool holding devices, drilling machine operation, speed, feed and machine time, Boring- Introduction, classification of boring machines, boring bars, boring heads, boring defects

Text Books and References

1. Kalpakjian, S., & Schmid, S. R. Manufacturing processes for engineering materials: Pearson Education.
2. DeGarmo, E. P., Black, J. T., & Kohser, R. A. Materials and processes in manufacturing: Wiley.
3. Lindberg, R. A. Processes and materials of manufacture: Allyn and Bacon.
4. Chapman, W. Workshop Technology: Edward Arnold.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: II

Session: EVEN

Subject Code & Name: HSC201 English

Credits: 2

Course Outcome: At the end of the course, the students will be able to

HSC201.1	Understand the concepts of word formation, root words, prefixes, suffixes, and the application of synonyms, antonyms, and abbreviations to expand vocabulary and improve language proficiency.
HSC201.2	Apply knowledge of sentence structures, phrases, clauses, punctuation, and paragraph organization to create clear, coherent, and well-structured documents that communicate ideas effectively.
HSC201.3	Analyze common writing errors, including subject-verb and noun-pronoun agreement, misplaced modifiers, articles, prepositions, redundancies, and clichés, and apply corrective techniques to enhance writing accuracy.
HSC201.4	Demonstrate sensible writing by using appropriate methods for describing, defining, classifying, and providing examples or evidence, while ensuring the clarity and conciseness of writing in academic and professional contexts.
HSC201.5	Practice writing skills such as comprehension, précis writing, and essay writing, developing the ability to convey complex ideas effectively in both structured and concise formats.

Unit I. Vocabulary Building

The concept of Word Formation, Root words from foreign languages and their use in English, Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives., Synonyms, antonyms, and standard abbreviations.

Unit II. Basic Writing Skills

Sentence Structures, Use of phrases and clauses in sentences, Importance of proper punctuation, Creating coherence, Organizing principles of paragraphs in documents, Techniques for writing precisely

Unit III. Identification of Errors

Identifying Common Errors in Writing, Subject-verb agreement, Noun-pronoun agreement, Misplaced modifiers, Articles, Prepositions, Redundancies, Clichés

Unit IV. Sensible Writing

Nature and Style of sensible Writing, Describing, Defining, Classifying, Providing examples or evidence, Writing introduction and conclusion

Unit V. Writing Practices

Writing Practices, Comprehension, Précis Writing, Essay Writing

Text Books & References

1. "Listening and Speaking Skills for English Language Learners" by Deborah Gordon and Cynthia Herd - Recommended Edition: 2nd Edition (2015)
2. "English Phonetics and Phonology: A Practical Course" by Peter Roach - Recommended Edition: 4th Edition (2009)
3. "Effective Communication Skills" by Dale Carnegie - Recommended Edition: Revised Edition (2009)
4. "Nonverbal Communication in Human Interaction" by Mark L. Knapp and Judith A. Hall - Recommended Edition: 8th Edition (2009)
5. "Business Communication: Building Critical Skills" by Kitty O. Locker and Stephen KyoKaczmarek - Recommended Edition: 6th Edition (2019)
6. "English for Careers: Business, Professional, and Technical" by Leila R. Smith - Recommended Edition: 11th Edition (2015)
7. "Public Speaking: Strategies for Success" by David Zarefsky - Recommended Edition: 8th Edition (2014)
8. "The Art of Storytelling: Easy Steps to Presenting an Unforgettable Story" by John Walsh - Recommended Edition: 2nd Edition (2014)
9. "Pronunciation Pairs: An Introduction to the Sounds of English" by Ann Baker and Sharon Goldstein - Recommended Edition: 3rd Edition (2012)
10. "Communicating at Work: Strategies for Success in Business and the Professions" by Ronald B. Adler and Jeanne Marquardt Elmhorst - Recommended Edition: 12th Edition (2018)

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: II

Session: EVEN

Subject Code & Name: GSC211 Chemistry Lab

Credits: 1.5

Course Outcome: At the end of the course, the students will be able to:

GSC211.CO1	Understand the procedures for determining chemical oxygen demand (COD) in industrial wastewater and analyze the results to assess the quality and environmental impact of wastewater.
GSC211.CO2	Apply iodometric and potentiometric methods to determine the percentage of copper in brass and the iron content in rust solution, demonstrating the ability to work with standard chemical techniques for quantitative analysis.
GSC211.CO3	Evaluate the total hardness of water samples using the rapid EDTA method, interpreting the results to understand water quality and suitability for different applications.
GSC211.CO4	Demonstrate the use of instruments such as the Bomb Calorimeter, Pensky-Martens apparatus, and Engler's viscometer to measure the calorific value, flash point, fire point, and viscosity of fuel and lubricants, ensuring proper handling of equipment for accurate results.
GSC211.CO5	Synthesize aspirin (acetylsalicylic acid) and apply Thin Layer Chromatography (TLC) to separate and identify amino acids, demonstrating advanced laboratory skills in organic synthesis and separation techniques.

List of Experiment:

(Choice of 10-12 experiments from the followings)

1. Determination of Chemical oxygen demand of industrial waste water
2. Determination of percentage of copper in brass by Iodometric method
3. Determination of percentage of iron in the given rust solution using standard Potassium Dichromate solution (External indicator method)
4. Determination of amount of HCl and CH₃COOH present in a mixture by conductometry
5. Determination of Nickel using EDTA by complexometric method
6. Determination of Total hardness of given water sample by rapid EDTA method
7. Estimation of FAS present in the given solution by potentiometric method
8. determine the calorific value and water equivalent of a supplied sample by using Bomb calorimeter (manually)
9. To determine the calorific value and water equivalent of a supplied sample by using Bomb calorimeter (Automatically)
10. To determine the proximate analysis of coal sample
11. To determine the ultimate analysis of the coal sample

12. To determine the viscosity of the given oil sample using Engler's viscometer
13. To determine the Cloud point and Pour point of a given fuel / lubricant / oil using Cloud and Pour point apparatus
14. To determine the Flash and Fire point of the given sample of oil using Pensky : Marten's apparatus
15. To determine the Acid value of a given oil
16. SYNTHESIS OF ASPIRIN (acetylsalicylic acid)
17. To separate a mixture of amino acids by Thin Layer Chromatography (TLC) and identify the test amino acids by measuring their R_f values.

Text Books and References

1. Vogel's A.I. A text book of quantitative analysis, 35th edition, 2012.
2. Willard, Merit, Dean and Settle, A text book of Instrumental analysis, 6th edition 2012.
3. Dr. Jyotsna Cherukuri (2012) Laboratory Manual of engineering chemistry-II, VGS Techno Series.
4. Chemistry Practical Manual, Lorven Publications.
5. K. Mukkanti (2009) Practical Engineering Chemistry, B.S. Publication.
6. G.H Jeffery, J Bassett, J Mendham and R.C. Denney Vogel's A.I. A text book of quantitative analysis, Dorling Kindersley (India) Pvt., Ltd. 35th edition, 2012.
7. Gary D Christian, Analytical Chemistry, Wiley India, 6th edition, 2015.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: II

Session: EVEN

Subject Code & Name: GEC211 Programming for Problem Solving Lab

Credits: 2

Course Outcome: At the end of the course, the students will be able to:

GEC201.CO1	Develop simple C programs to display formatted output, perform arithmetic computations, and implement decision-making and looping constructs.
GEC201.CO2	Analyze and implement problem-solving techniques using arrays, strings, and matrix operations, including sorting, searching, and string manipulations without built-in functions.
GEC201.CO3	Design modular programs by defining and using user-defined functions, recursive methods, and dynamic memory allocation for numerical and computational tasks.
GEC201.CO4	Use pointers and structures to manage and manipulate complex data efficiently and handle real-world scenarios such as student or employee data management.
GEC201.CO5	Test and debug programs to validate logic and ensure correctness while adhering to optimal coding practices for solving numerical, logical, and structural problems.

List of Experiment:

Lab1: Familiarization with programming environment

P1. Write a program in C to print the following:

UMU

Name- Vishal Gupta

Roll No.- 20B.TechCSE007

Branch- B.Tech (CS)

Sem-2nd

P2. Write a program in C to print the following (without using loop)

```
*  
  
*      *  
  
*      *      *  
  
*      *      *      *
```

Lab 2: Simple computational problems using arithmetic expressions

P3. Write a program to declare two integer variables and assign predefined values and then find the sum of the two variables.

P4. Write a program to take input from user two integer values and find the sum.

P5. Write a program to take input basic salary of an employee and find the gross salary according to the formula:

Gross salary = Basic + DA + TA + HRA

(DA=50% of Basic, TA=40% of Basic, HRA=30% of Basic)

Lab 3: Problems involving if-then-else structures

P6. Write a program to take input average marks obtained by a student and display the grade.

P7. Write a program to take input 5 positive integers and find out the sum using goto.

P8. Write a program to take input a number and determine whether the given number is odd or even.

P9. Write a program to display the number of integers greater than 100 and less than 200 which are divisible by 7 and also find their sum.

P10. Write a program to check whether an input number is prime or not. (without using loop)

P11. Write a program to perform the mathematical operations done by a calculator using switch statement.

P12. Write a program to check a triangle is equilateral or scalene or isosceles.

P13. Write a program to find the largest among three numbers.

P14. Write a program to check whether a number is positive, negative or zero.

P15. Write a program to check whether a year is leap year or not.

P16. Write a program to take input three angles of a triangle and check whether the triangle is valid or not.

P17. Write a program to take input Cost Price and Selling Price of a product and check whether it is Profit or Loss.

Lab 4: Iterative problems e.g., sum of series

P18. Write a program to find the sum of first 10 natural numbers (using loop).

- P19. Write a program to find the value of $y=k^n$, where k and n are user input.
- P20. Write a program to find the sum of even numbers between 10 & 40.
- P21. Write a program to display the multiplication table of 5 (using loop).
- P22. Write a program to display the multiplication table from 1 to 10.
- P23. Write a program to display 1 to 10 as well as 20 to 11 using single loop.
- P24. Write a program to print the Fibonacci series using nth term.
- P25. Write a program to check whether an input number is prime or not. (using loop)
- P26. Write a program to print prime numbers from 1 to 50.
- P27. Write a program to take input marks of 5 different subjects obtained by a student and display the average mark.
- P28. Write a program to compute the sum of digits of a given input number.
- P29. Write a program to print the following patterns (using loop)

- a. *
- * *
- * * *
- * * * *
- b. 1
- 1 2
- 1 2 3
- 1 2 3 4
- c. 1
- 2 2
- 3 3 3
- 4 4 4 4
- d. * * * *
- * * *
- * *

f. 

g.	1	0	1	0	1
	0	1	0	1	0
	1	0	1	0	1
	0	1	0	1	0

P30. Write a program to take input a number. If the number is odd find the factorial of the number and if the number is even find the factors of the number.

P31. Write a program to count the number of digits in a number.

P32. Write a program to display first and last digit of a number.

P33. Write a program to find the reverse of a number.

P34. Write a program to check whether a number is Pallindrome number or not.

Write a program to print all the factors of a number.

Write a program to print the following series:

$$1+4+7+10+13+15=50$$

Lab 5: 1D Array manipulation

P35. Write a program to declare an integer array of size 10. Take input the values from the user and display them.

P36. Write a program to take input an array (size of the array is 10). Find the maximum value from the array.

P37. Write a program to take input an array (size of the array is 10). Separate all the even numbers in one array and all the odd numbers in another array.

P38. Write a program to take input an array (size of the array is 10). Sort the elements of the array in ascending order.

P39. Write a program to declare an array of size 10. Take input the values from the user. Add 10 to the even location of the array and 5 to the odd locations of the array.

P40. Write a program to declare an array of size 10. Take input the values from the user. The values which are even multiply those values by 5 and display the array.

P41. Write a program to declare an array of size 10. Put value 0 to the odd locations of the array and 1 to the even locations of the array.

P42. Write a program to find the sum of the integer values that are stored in the array.

Lab 6: Matrix problems, String operations

P43. Write a program to initialize a 2-Dimensional array and display them.

P44. Write a program to perform matrix addition and matrix multiplication.

P45. Write a program take input a 2-Dimensional matrix and sort the matrix row-wise.

P46. Write a program to copy one string into another without using built-in function.

P47. Write a program to compare two strings without using built-in function.

P48. Write a program to take input a string and check whether the string is Pallindrome or not.

P49. Write a program to read a string from user using getchar().

P50. Write a program to find the length of a string without using built-in function.

P51. Write a program to take input a string and count the number of vowels and consonants present in a string.

P52. Write a program to print the characters and their ASCII values from A-Z

P53. Write a program to take input two strings and concatenate them.

Lab 7: Simple functions

P54. Write a program to perform four arithmetic operations using simple 4 functions.

P55. Write a program to find the factorial of a number using function.

P56. Write a program to check whether a number is odd or even using function.

P57. Write a program to swap two numbers using- Call by Value method.

P58. Write a program to find the value of $y=k^n$ using function.

Lab 8 and 9: Programming for solving Numerical methods problems

P59. Write a program using dynamic memory allocation (malloc) to input values in an array and display them.

P60. Write a program using dynamic memory allocation to input values in an array and find the sum of the elements.

P61. Write a program using dynamic memory allocation (calloc) to input values in an array and display them.

P62. Write a program using dynamic memory allocation (realloc) to reallocate already allocated array and display them.

P63. Write a program to take input an array and sort the array using Bubble Sort.

P64. Write a program to take input an array and sort the array using Selection Sort.

P65. Write a program to take input an array and sort the array using Insertion Sort.

P66. Write a program to take input values in an array and search an element from the array using linear search.

P67. Write a program to find the roots of an equation.

Lab 10: Recursive functions

P68. Write a program to print the sum of 'n' numbers along with the numbers using Recursion.

P69. Write a program to find the factorial of a number using Recursion.

P70. Write a program to print Fibonacci series using Recursion.

P71. Write a program that receives two non-negative integers and print $a*b$ recursively. [Hint: $a*b$ not multiplication operator. It will be working as $a+a+\dots+a$ for b times]

P72. Write a program to print the sum of the digits of a number using Recursion.

Lab 11: Pointers and structures

P73. Write a program to take input values in an array using pointer.

P74. Write a program to swap two numbers using Call by Reference method.

P75. Write a program to find the sum of two variables using pointer.

P76. Write a program to create a structure 'STUDENT' with the attributes Roll, Name, Gender and Marks in 5 different subjects. Take input the values of a student and display them.

P77. Write a program to create a structure 'STUDENT' with the attributes Roll, Name, Gender and Marks in 5 different subjects. Take input the values for 5 students and display them.

P78. Write a program to define a structure 'EMPLOYEE' having data members name, address, age and salary. Take the data for n employees in an array and find the average salary.

Text Books & References

- 1.Computer Programming with C, Special Edition-MRCET, Mc Graw Hill Publishers 2017.
- 2.Computer Science: A Structured Programming Approach Using C, B.A.Forouzan and R.F. Gilberg, Third Edition, Cengage Learning.
- 3.The C Programming Language, B.W. Kernighan and Dennis M.Ritchie, PHI.
- 4.Computer Programming, E.Balagurusamy, First Edition, TMH.
- 5.C and Data structures – P. Padmanabham, Third Edition, B.S. Publications.
- 6.Programming in C, *Ashok Kamthane*. Pearson Education India.
- 7.Let us C ,YashwanthKanethkar, 13th Edition, BPB Publications.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: II

Session: EVEN

Subject Code & Name: GEC212 Workshop/ Manufacturing Process Lab

Credits: 2

Course Outcome: At the end of the course, the students will be able to:

GEC 212.CO1	Explain the principles and applications of power tools, woodcutting, and electrical and mechanical engineering practices through practical demonstrations.
GEC 212.CO2	Demonstrate proficiency in performing basic manufacturing operations such as carpentry, fitting, tin-smithy, blacksmithy, house wiring, foundry molding, and plumbing to develop hands-on engineering skills.
GEC 212.CO3	Operate lathe machines and perform turning operations while understanding tool geometry, cutting mechanics, and the use of traditional, capstan, turret, and automatic lathes.
GEC 212.CO4	Analyze and perform machining operations such as milling, indexing, gear cutting, and hobbing on milling machines, applying knowledge of cutters and multi-axis milling techniques.
GEC 212.CO5	Evaluate grinding machine operations, including grinding tool nomenclature, work holding, and tool holding devices, to ensure optimal machining parameters and surface finishes.

List of Experiments:

1. Manufacturing process lab covering, the following trades for experiments Carpentry, Fitting, Tin-Smithy and Development of jobs carried out and soldering, Black Smithy, House Wiring, Foundry (Moulding only) and Plumbing.
2. A Manufacturing process lab covering, the following for demonstration - Power cutting Wood working, Electrical and Mechanical Engineering practices.
3. Lathe Introduction, types of lathe machine, orthogonal and oblique cutting, types of chips, chip breakers, cutting tool nomenclature, cutting action of hand tools, cutting feed and speed, friction and heat sources in cutting, tools life and wear, machinability, cutting tool materials, cutting fluids, economics of machining. Lathe- Turning on lathes, traditional lathes, capstan and turret lathes, automatic lathes.
4. Milling Machine Types of milling machines, milling cutters, indexing and dividing heads, indexing methods, calculations of indexing, multi-axis milling, Gear cutting methods, gear hobbing.
5. Grinding m/c tools: Introduction, types of grinding, grinding nomenclature, types of grinding machines, work holding devices, tool holding devices, grinding machine operation, speed, feed and machine time.

Text Books and References

1. Kalpakjian, S., & Schmid, S. R. Manufacturing processes for engineering materials: Pearson Education.
2. DeGarmo, E. P., Black, J. T., & Kohser, R. A. Materials and processes in manufacturing: Wiley.
3. Lindberg, R. A. Processes and materials of manufacture: Allyn and Bacon.
4. Chapman, W. Workshop Technology: Edward Arnold.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: II

Session: EVEN

Subject Code & Name: HSC211 English Lab

Credits: 1

Course Outcome: At the end of the course, the students will be able to:

HSC211.CO1	Enhance listening skills by identifying barriers and recognizing speech sounds, vowels, consonants, and clusters.
HSC211.CO2	Practice various types of listening, including general, specific, and intensive listening through audio clips.
HSC211.CO3	Understand word stress, syllable structure, and accent rules in different contexts.
HSC211.CO4	Develop oral communication skills through Just A Minute (JAM) sessions, role playing, and situational dialogues.
HSC211.CO5	Build effective communication skills in the workplace, including public speaking, interviews, and telephone etiquette.

List of Exercise:

- 1.Understanding the importance of listening skills. Identifying barriers to effective listening. Introduction to Phonetics: Speech Sounds, Vowels, Consonants, and Consonant Clusters.
- 2.Exploring the purpose and process of listening. Different types of listeningPractice: Listening to various audio clips to identify different types of listening.
- 3.Structure of Syllables and Word Stress Recognizing Weak Forms and Strong Forms in Context Basic Rules of Word Accent and Stress Shift
- 4.Listening practice with minimal pairs Listening for general content, specific information, and intensive listening Listening to different accents and variations in pronunciation.
- 5.Understanding the difference between spoken and written language Features of Good Conversation and Non-verbal Communication.
- 6.Oral practice through Just A Minute (JAM) Sessions Describing objects, situations, and people Role play activities in individual and group settings.
- 7.Situational Dialogues: Greetings, Taking Leave, Introducing Oneself and OthersIce-Breaking Activity and JAM Session.
- 8.Communication at Workplace: Effective communication skills Telephone Etiquette and Giving Directions.
- 9.Making Requests and Seeking Permissions. Public speaking and Oral Presentation skills.
- 10.Extempore speaking. Mock interviews for Resume Writing and Interview Skills.

11. Listening for General Details and Specific Details. Critical Analysis of Audio and Video materials.
12. Listening Comprehension Tests. Speech/Audio and Video Recording for Self-Analysis.
13. Developing storytelling skills through descriptions, narrations, and fictional writing. Storytelling through translation.

Text Books & References

1. "Listening and Speaking Skills for English Language Learners" by Deborah Gordon and Cynthia Herd - Recommended Edition: 2nd Edition (2015)
2. "English Phonetics and Phonology: A Practical Course" by Peter Roach - Recommended Edition: 4th Edition (2009)
3. "Effective Communication Skills" by Dale Carnegie - Recommended Edition: Revised Edition (2009)
4. "Nonverbal Communication in Human Interaction" by Mark L. Knapp and Judith A. Hall - Recommended Edition: 8th Edition (2009)
5. "Business Communication: Building Critical Skills" by Kitty O. Locker and Stephen KyoKaczmarek - Recommended Edition: 6th Edition (2019)
6. "English for Careers: Business, Professional, and Technical" by Leila R. Smith - Recommended Edition: 11th Edition (2015)
7. "Public Speaking: Strategies for Success" by David Zarefsky - Recommended Edition: 8th Edition (2014)
8. "The Art of Storytelling: Easy Steps to Presenting an Unforgettable Story" by John Walsh - Recommended Edition: 2nd Edition (2014)
9. "Pronunciation Pairs: An Introduction to the Sounds of English" by Ann Baker and Sharon Goldstein - Recommended Edition: 3rd Edition (2012).
10. "Communicating at Work: Strategies for Success in Business and the Professions" by Ronald B. Adler and Jeanne Marquardt Elmhorst - Recommended Edition: 12th Edition (2018)

USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Semester: III

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: GSC301 Mathematics-III

Credits: 2

Course Outcome: At the end of the course, the students will be able to

GSC301.CO1	Apply statistical techniques to analyze data using moments, skewness, kurtosis, and regression analysis.
GSC301.CO2	Demonstrate understanding of probability theory through discrete and continuous random variables and distribution functions.
GSC301.CO3	Solve transcendental and polynomial equations using numerical methods such as Bisection, Regula-Falsi, and Newton-Raphson.
GSC301.CO4	Implement numerical methods for solving systems of linear equations and integrate them with methods like Jacobi and Gauss-Seidel.
GSC301.CO5	Apply integral transformations, including Fourier and Z-transforms, to solve boundary value problems and partial differential equations.

Unit I. Statistical Techniques

Moments, Skewness, Kurtosis, Curve fitting, Method of least squares, Fitting of straight lines, Polynomials, Exponential curves, Correlation, regression analysis.

Unit II. Probability Theory

Discrete and continuous random variables, cumulative distribution function, expectation, variance, Introduction to Binomial, Poisson distribution.

Unit III. Numerical Techniques

Zeros of transcendental and polynomial equations using Bisection method, Regula-falsi method and Newton-Raphson method. Interpolation: Finite differences, Newton's forward and backward interpolation.

Unit IV. Numerical Techniques and Integration

Solution of system of linear equations, Matrix Decomposition methods, Jacobi method, Gauss-Seidel method. Numerical differentiation, Numerical integration: Trapezoidal rule, Simpson's one third and three-eighths rules, Solution of ordinary differential equations (first order, second order and simultaneous) by Euler's, Picard's and fourth-order Runge-Kutta methods.

Unit V. Integral Transformations

Boundary Value Problem, Finite Difference Method, Eigen Value Problems, Numerical solution of partial differential equations. Elliptic, parabolic and Hyperbolic equations. Integral Transforms: Fourier transform, Fourier sine and cosine transform, Applications of Fourier transform to simple one-dimensional heat transfer equations, wave equations and Laplace equations, Z- transform and its application to solve difference equations.

Text Books and References

1. B.S Grewal, Higher Engineering Mathematics, Khanna Publishers.
2. R.K. Jain & S. R. K. Iyenger, Advance Engineering Mathematics, Narosa Publication House.
3. S. S. Sastry, Introductory Methods of Numerical Analysis, PHI Learning Pvt. Limited, New Delhi.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: CSC301 Data Structure and Algorithms

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

CSC301.CO1	Explain the fundamental concepts of data structures, including arrays, linked lists, stacks, queues, trees, and graphs, along with their operations and applications.
CSC301.CO2	Implement data structures such as arrays, linked lists, stacks, queues, binary trees, and graphs in C for solving computational problems.
CSC301.CO3	Analyze algorithms for sorting, searching, and graph traversals in terms of time and space complexity to determine their efficiency and scalability.
CSC301.CO4	Evaluate the suitability of advanced data structures like AVL Trees, B-Trees, and hashing techniques for specific real-world problems.
CSC301.CO5	Design and develop algorithms using appropriate data structures to solve complex problems, optimizing for performance and resource usage.

Unit I. Introduction

Structures and Unions, Pointers, Dynamic Memory Allocation, Algorithm Specification, Space and Time Complexity.

Unit II. Arrays and Linked List

Arrays, Abstract Data Type, Dynamically Allocated Arrays, Polynomials, Two-dimensional Array, Address Calculation, Matrix Addition and Multiplication, Sparse Matrix . Singly Linked Lists and Chains, Representing Chains in C, Polynomials, Sparse Matrix, Doubly Linked Lists, Circular & Header Linked lists.

Unit III. Stacks and Queues

Stacks, Stacks using Dynamic Arrays and Linked List, Queues, Queue using Linked List, Circular Queues using Dynamic Arrays, Evaluation of Expressions, Priority Queue, Dequeue.

Unit IV. Trees and Graphs

Introduction, Binary Trees, Binary Tree Traversals, Threaded Binary Trees, Binary Search Trees, AVL Trees, mway Search Trees, B-Trees, B+-Trees, Tree Operation, Forests. The Graph ADT, Graph Representation, Graph Operation-DFS, BFS.

Unit V. Sorting and Searching

Insertion Sort, Quick Sort, Merge Sort, Heap Sort, Bubble Sort, Selection sort, Radix sort. Linear Search, Binary Search, Static Hashing, Dynamic Hashing

Text Book:

1. Data Structures, Schaum's OutLines, Seymour Lipschutz, TATA McGRAW HILL.

References:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahani, Anderson-Freed, Universities Press.
2. Data Structures A Pseudocode Approach with C, 2nd Edition, Richard F. Gilberg, Behrouz A. Forouzan, CENGAGE Learning, India Edition
3. Data Structures and Algorithm Analysis in C, Mark Allen Weiss, Pearson Education, 2nd Edition.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: GEC301 Analog Electronics Circuit

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

GEC301.CO1	Analyze and apply diode circuits, including rectifiers and Zener diodes, for various applications in electronics.
GEC301. CO 2	Examine and design transistor-based circuits, including BJT and FET configurations, for amplification and switching applications.
GEC301. CO 3	Design and analyze small-signal transistor amplifiers and feedback amplifiers for improving signal quality and gain.
GEC301. CO 4	Understand and apply operational amplifier configurations, including differential amplifiers and virtual ground concepts, for signal processing.
GEC301. CO 5	Design and analyze oscillators, including RC, LC, and Wien Bridge types, for generating periodic waveforms in electronic systems.

Unit I. Diodes and Applications

Semiconductor Diode - Ideal versus Practical, Resistance Levels, Diode Equivalent Circuits, Load Line Analysis; Diode as a Switch, Diode as a Rectifier, Half Wave and Full Wave Rectifiers with and without Filters; Breakdown Mechanisms, Zener Diode – Operation and Applications; Opto-Electronic Devices – LEDs, Photo Diode and Applications; Silicon Controlled Rectifier (SCR) – Operation, Construction, Characteristics, Ratings, Applications;

Unit II. Transistor Characteristics

Bipolar Junction Transistor (BJT) – Construction, Operation, Amplifying Action, Common Base, Common Emitter and Common Collector Configurations, Operating Point, Voltage Divider Bias Configuration; Field Effect Transistor (FET) – Construction, Characteristics of Junction FET, Depletion and Enhancement type Metal Oxide Semiconductor (MOS) FETs, Introduction to CMOS circuits.

Unit III. Transistor Amplifiers

Classification, Small Signal Amplifiers – Basic Features, Common Emitter Amplifier, Coupling and Bypass Capacitors, Distortion, AC Equivalent Circuit; Feedback Amplifiers – Principle,

Advantages of Negative Feedback, Topologies, Current Series and Voltage Series Feedback Amplifiers;

Unit IV. Operational Amplifiers and Applications

Introduction to Op-Amp, Differential Amplifier Configurations, CMRR, PSRR, Slew Rate; Block Diagram, Pin Configuration of 741 OpAmp, Characteristics of Ideal OpAmp, Concept of Virtual Ground;

Unit V. Oscillators:

Oscillators Classification, RC Phase Shift, Wien Bridge, High Frequency LC and Non-Sinusoidal type Oscillators.

Text Books & References:

1. Basic Electronics by BL Theraja, S Chand Publishing 5th Edition 2006.
2. Analog Circuits by A.K. Maini, Khanna Publishing House Ed. 2018 (ISBN: 978-93-86173-584)
3. Electronic Devices and Circuits by S. Salivahanan and N. Suresh Kumar, McGraw Hill Education; Fourth edition (1 July 2017) ISBN: 978-9339219505
4. Electronics Devices and circuit theory by Boyestad&Nashelsky, Pearson Education India; 11 edition (2015) ISBN: 978-9332542600.

Faculty: Faculty of Computing and Information Technology**Program:** B.Tech (CSE)**Semester:** III**Session:** ODD**Subject Code & Name:** GEC302 Digital Electronics**Credits:** 3**Course Outcome:** At the end of the course, the students will be able to:

GEC302. CO 1	Understand and apply number systems, binary arithmetic, and logic families in digital electronics design.
GEC302. CO 2	Simplify Boolean functions using algebraic methods and Karnaugh maps for efficient circuit design.
GEC302. CO 3	Analyze and design combinational circuits such as adders, multiplexers, and ALUs for logic and arithmetic operations.
GEC302. CO 4	Design and implement sequential circuits using flip-flops, latches, and triggering mechanisms for memory storage.
GEC302. CO 5	Examine and design memory systems, including counters, registers, and programmable logic devices for efficient digital storage.

Unit I. Number System and logic families

Digital signals, Number representation, Binary number system, Number base conversion, Octal, Hexadecimal and BCD codes, binary Arithmetic, one's and two's complements arithmetic, Logic Gates, Introduction to VHDL and Verilog, VHDL Models, Logic Families: TTL, ECL, and CMOS Logic Circuits, Logic levels, voltages and currents, fan-in, fan-out, speed, power dissipation. Comparison of logic families.

Unit II. Simplification of Boolean functions

Boolean Algebra, Basic theorems and Properties, De Morgan's theorem, Canonical & Standard forms, Simplification of Boolean function using Karnaugh map, POS & SOP simplification, Prime implicant, NAND and NOR implementation.

Unit III. Combinational Digital Circuits

Analysis and design procedure, Adders, Subtractors, 4-bit BCD adder/subtractor, Parity Generators and Checkers, look ahead carry, Adder, Magnitude comparator, Decoders, Encoders, Multiplexers, De-multiplexers, Design of 1 bit ALU for basic logic and arithmetic operations.

Unit IV. Design of Sequential Circuits

Basic Latch, Flip-Flops (SR, D, JK, T and Master-Slave), Triggering of Flip Flops,

Unit V. Memories

Synchronous and asynchronous counters, Registers, Shift Registers, Memories and Programmable Logic design, Types of memories, Memory Expansion and its decoding, Programmable Logic Arrays (PLA), Programmable Array Logic (PAL).

Text Books & References

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.
3. A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: CSC302 IT Workshop

Credits: 1

Course Outcome: At the end of the course, the students will be able to:

CSC302.CO1	Explain the MATLAB environment, data types, variables, operators, and fundamental programming constructs, including decision-making and loops.
CSC302.CO2	Utilize MATLAB for performing operations on matrices and arrays, creating basic and specialized 2D plots, and generating graphical representations for data visualization.
CSC302.CO3	Demonstrate file handling, debugging techniques, and MATLAB I/O to read, write, and manage data efficiently, including working with external file formats such as Excel.
CSC302.CO4	Analyze and solve mathematical problems such as linear algebraic equations, statistical computations, and numerical solutions to differential equations using MATLAB.
CSC302.CO5	Design and implement Simulink models and MATLAB-based applications for image processing, including segmentation, classification, and object identification.

Unit I. Matlab Environment

Data types and variables: Introduction to MATLAB, Data Types, and Inter-conversion of Data types, MATLAB Variables, Keywords and Constant, Session Command. MATLAB Operators and Operations: Operators (Arithmetic, Relational, Logical, Bitwise), Set Operations, Operator Precedence, Mathematical Functions.

Unit II. Programming in Matlab

MATLAB Programming: Script and Function, Decision Making, Loops, branches, Functions, Working on Script File (Creating, Saving and executing), MATLAB I/O, and Formatted I/O Method.

Unit III. Arrays and Graphics

Arrays: Introduction to Matrices, Operations on Arrays/Matrices, Manipulations of Arrays/Matrices, Expansion of Matrix Size, Reduction of Matrices/Arrays order. Graphics: Introduction to plot, Basic 2-D Plots (Style options, Labels, Axis control, etc.), specialized 2-D Plots, drawing multiple plots. Using MATLAB for fractals and chaos and Conway game of life.

Unit IV. File Handling

Introduction to file handling, working on files, accessing of Text File, Saving/ Loading MATLAB Variables, reading data without opening file, reading and writing Excel. Debugging: Introduction

to debugging, Break points, debugger, stepping, watching variable values, and debugging commands.

Unit V. Application Of Matlab

Linear algebraic equations, elementary solution method, matrix method for linear equation, Cramer's method. Statistics: Histogram and probability, normal distribution, random number generation. Interpolation: Analytical solution to differential equations – Numerical methods for differential equations. Introduction about Simulink – Simulink model. Image Segmentation, Classification and object identification.

Text Books and References

1. Delores M. Etter, David C. Kuncicky, Holly Moore, "Introduction to MATLAB 7.0", Pearson, 2013.
2. RudraPratap, "Getting Started with MATLAB", OXFORD University Press, 2010.
3. Agam Kumar Tyagi, "MATLAB and Simulink for Engineers", University Press, 2012.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: HSC301 Development of Life Skills- I

Credits: 3

Course Outcome: At the end of the course the students will be able to:

HSC301.CO1	Understand the importance and applications of Development of Life Skills (DLS) in the current context.
HSC301.CO2	Explore various information sources and methods for effective data collection, including internet research, interviews, and observations.
HSC301.CO3	Develop skills in written communication, including report writing, note-taking, and self-analysis for personal and professional growth.
HSC301.CO4	Enhance self-development through stress management, health management, time management, emotional intelligence, and goal setting.
HSC301.CO5	Enhance self-development through stress management, health management, time management, emotional intelligence, and goal setting.

Unit I. Importance of DLS,

Introduction to subject, importance in present context, application

Unit II. Information Search

Information source –Primary, secondary, tertiary Print and non – print, documentary, Electronic Information center, Library, exhibition, Government Departments. Internet Information search– Process of searching, collection of data –questionnaire, taking Interview, observation method.

Unit III. Written communication and Self Analysis

METHOD OF NOTE TAKING. Report writing –Concept, types and format. Understanding self— Attitude, aptitude, assertiveness, self esteem, Confidence buildings. Concept of motivation.

Unit IV. Self-Development

Stress Management –Concept, causes, effects, remedies to Avoid /minimize stress. Health Management – Importance, dietary guidelines and exercises. Time management- Importance, Process of time planning, Urgent Vs importance, Factors leading to time loss and ways to handle it, Tips for effective time management. Emotion-Concept, Types, Controlling, Emotional Intelligence. Creativity-Concept, Factors Enhancing Creativity. Goal Setting – Concept, Setting Smart Goal.

Unit V. Study habits

Ways to enhance memory and concentration. Developing reading skill. Organisation of knowledge, Model and methods of learning.

Text Books and References

1. Adams Time management, Marshall Cooks and Viva Books.
2. Basic Managerial Skills for All, E.H. Mc Grath , S.J. and Pretice Hall of India, Pvt Ltd.
3. Body Language, Allen Pease and Sudha Publications.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: GEC311 Analog Electronics Circuit Lab

Credits: 2

Course Outcome: At the end of the course, the students will be able to:

GEC311.CO 1	Demonstrate proficiency in the use of analog and digital instruments like multimeters, signal generators, and oscilloscopes for measurement and analysis.
GEC311.CO 2	Analyze the characteristics and behavior of diodes, including PN junction and Zener diodes, through practical experiments.
GEC311.CO 3	Evaluate and compare the performance of rectifiers with and without filters by constructing half-wave and full-wave rectifiers.
GEC311.CO 4	Examine and analyze the input-output characteristics of BJTs and FETs to determine their operational parameters.
GEC311.CO 5	Design and test various amplifiers, including unity gain, non-inverting, and inverting amplifiers, to observe and calculate gain and frequency response.

List of Experiments

- 1.Study the following devices: (a) Analog & digital multimeters (b) Function/ Signal generators (c) Regulated d. c. power supplies (constant voltage and constant current operations) (d) Study of analog CRO, measurement of time period, amplitude, frequency & phase angle using Lissajous figures.
- 2.Construct the circuit and plot the VI characteristics of the PN Junction Diode, find the cut in voltage, reverse Saturation current and static & dynamic resistances.
- 3.Construct the circuit and plot the characteristics of a Zener Diode. Find the breakdown voltage
- 4.Construct a Half Wave Rectifier and obtain regulation characteristics – Without Filters and with Filters Compare the results.
- 5.Construct a Full Wave Rectifier and obtain regulation characteristics – Without Filters and with Filters Compare the results
- 6.Plot input and output characteristics of BJT in CB, CC and CE configurations. Find their h-parameters.
- 7.Plot frequency response curve for single stage amplifier and to determine gain bandwidth product.
- 8.Plot drain current - drain voltage and drain current – gate bias characteristics of field effect transistor and measure of I_{dss} & V_p .

9.To Design and test the characteristics of the following amplifiers:

- a. A unity gain amplifier.
- b. A non-inverting amplifier with a gain of “A”.
- c. An inverting amplifier with a gain of “A”.

10.Study Wein bridge oscillator and observe the effect of variation in R & C on oscillator frequency.

11.Study transistor phase shift oscillator and observe the effect of variation in R & C on oscillator frequency and compare with theoretical value.

Text Books & References:

- 1.Basic Electronics byBL Theraja, S Chand Publishing 5th Edition 2006.
- 2.Analog Circuits by A.K. Maini, Khanna Publishing House Ed. 2018 (ISBN: 978-93-86173-584)
- 3.Electronic Devices and Circuits by S. Salivahanan and N. Suresh Kumar, McGraw Hill Education; Fourth edition (1 July 2017) ISBN: 978-9339219505
- 4.Electronics Devices and circuit theory by Boyestad&Nashelsky, Pearson Education India; 11 edition (2015) ISBN: 978-9332542600

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: GEC312 Digital Electronics Lab

Credits: 2

Course Outcome: At the end of the course, the students will be able to:

GEC312.CO1	Design and implement CMOS inverters and verify their functionality in digital circuits.
GEC312.CO2	Verify the truth tables of NAND and EX-OR gates using IC 7400 and apply digital logic principles.
GEC312.CO3	Design and implement a seven-segment display unit for visual representation of binary data.
GEC312.CO4	Construct and verify half adder and full adder circuits using logic gates and IC 7483.
GEC312.CO5	Design and implement a 3:8 decoder and an 8:3 priority encoder for data encoding and decoding operations.

List of Experiments

- 1.Design and implement a controlled CMOS Inverter.
- 2.To study and verify the truth table of NAND and EX-OR gate using IC 7400.
- 3.Design and implement SEVEN segment display unit.
- 4.Design and verify half adder and full Adder circuits using gates and IC 7483.
- 5.Design and implement a 3:8 Decoder.
- 6.Design and implement 8:3 priority encoder.
- 7.Design a 4 bit magnitude comparator using combinational circuits.
- 8.Design and implement 8:1 multiplexer and 1:4 Demultiplexer.
- 9.Design ALU with functions of ADD, SUB, INVERT, OR, AND. XOR, INC, DEC and CMP.
- 10.Design and verify decade Counter.
- 11.Design a ROM (8X4) using decoder, gates and diodes.
- 12.Design of pre settable up/down counter.

Text Books & References

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.
3. A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: CSC311 Data Structure and Algorithms Lab

Credits: 2

Course Outcome: At the end of the course, the students will be able to:

CSC311.C01	Implement and perform operations like creation, insertion, deletion, and traversal on singly, doubly, and circular linked lists.
CSC311.C02	Develop programs to implement stack and queue operations using arrays and linked lists (pointers) to understand fundamental data structures.
CSC311.C03	Demonstrate searching algorithms such as linear and binary search using both recursive and non-recursive methods.
CSC311.C04	Design and implement sorting algorithms including bubble sort, selection sort, quick sort, insertion sort, merge sort, and heap sort.
CSC311.C05	Construct programs to implement a dictionary ADT and tree traversal methods including binary search trees and AVL trees.

List of Experiments

- (1) Write a program that uses functions to perform the following operations on singly linked list
i) Creation ii) Insertion iii) Deletion iv) Traversal.
- (1) Write a program that uses functions to perform the following operations on doubly linked list
i) Creation ii) Insertion iii) Deletion iv) Traversal.
- (2) Write a program that uses functions to perform the following operations on circular linked List
i) Creation ii) Insertion iii) Deletion iv) Traversal.
- (3) Write a program that implement stack (its operations) using i) Arrays ii) Linked list (Pointers).
- (4) Write a program that implement Queue (its operations) using i) Arrays ii) Linked list (Pointers).
- (5) Write a program that implement Circular Queue using arrays. ii) Write a program that uses both recursive and non-recursive functions to perform the following searching operations for a Key value in a given list of integers: a) Linear search b) Binary search.
- (6) Write a program that implements the following sorting i) Bubble sort ii) Selection sort iii) Quick sort.
- (7) Write a program that implements the following i) Insertion sort ii) Merge sort iii) Heap sort.
- (8) Write a program to implement all the functions of a dictionary (ADT) using Linked List.
- (9) Write a program to perform the following operations: a) Insert an element into a binary search tree. b) Delete an element from a binary search tree. c) Search for a key element in a binary search tree.
- (10) Write a program to implement the tree traversal methods
- (11) Write a program to perform the following operations: a) Insert an element into a AVL tree. b) Delete an element from a AVL tree. c) Search for a key element in a AVL tree.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: III

Session: ODD

Subject Code & Name: CSC312 IT Workshop Lab

Credits: 2

Course Outcome: At the end of the course, the students will be able to:

CSC312.CO1	Demonstrate proficiency in basic MATLAB operations, including syntax, variables, and matrix manipulations.
CSC312.CO2	Develop programs using MATLAB to implement control structures, loops, and operations on arrays, vectors, and matrices, enabling efficient handling of data.
CSC312.CO3	Apply functions, input/output operations, and matrix algebra for solving real-world problems using MATLAB.
CSC312.CO4	Utilize MATLAB for graphical plotting, algebraic and calculus operations, including solving equations and performing transformations.
CSC312.CO5	Develop applications in MATLAB for image processing, including image conversion, filtering, and segmentation.

Unit I. Basic features and Variables

Lab 1: BASIC FEATURES

P1. Study the history and features of MATLAB.

P2. To study the local environment and features of MATLAB

Current Folder

Command Window

Command History

P3. Write a program in MATLAB to perform basic syntax operations.

Lab 2: Variables

P4. Write a program in MATLAB to perform basic variables operations:

who

whos

clear

P5. Write a program in MATLAB to perform basic Format Command:

format long

format short

format bank

format short e

format long e

format rat

P6. Write a program in MATLAB to create Row Vector, Column Vector.

P7. Write a program in MATLAB to create the 1D and 2-D matrices.

Unit II. M-Files and Scripts, Operations and Control Structures, Loops and Vectors

Lab 1: M-Files and Scripts

P8. Write a program to create a script file and perform the following operation:

```
No Of Students = 32000;  
Teaching Staff = 1150;  
Non Teaching Staff = 1920;  
Total = No Of Students + Teaching Staff + Non Teaching Staff;  
Disp (Total);
```

P9. Write a program to create a script file and perform the following operation:

```
a = 5; b = 7;  
c = a + b  
d = c + sin (b)  
e = 5 * d
```

P10. Write a program to perform various Data Type operation.

P11. Write a program to perform various Data Type Conversion.

```
Convert to character array to string.  
Convert integer data to string  
Convert matrix to string  
Convert number to string  
Convert string to number  
Convert binary number string  
Convert decimal to base N number  
Convert decimal to binary number  
Convert hexadecimal number string
```

P12. Write a program to create a script file with the following code:

```
isinteger()  
isfloat()  
isvector()  
isscalar()  
isnumeric(),
```

P13. Write a program to determine if the complex number is integer type.

P14. Write a program to determine if the array containing integer numbers is an integer type.

P15. Write a program to create a cell array that contains the array A and other integer numbers then use the class to identify the type of the cell array.

Lab 2: Operations and Control Structures

P16. Write a program to perform Matrix Arithmetic Operations, Bitwise operations, and Relational operations.

P17. Write a program to perform Logical operations and Set operations on matrix.

P18. Write a program to demonstrate the control structures If-Else, If-elseif-else.

P19. Write a program to demonstrate the Nested If statement.

P20. Write a program to demonstrate the switch statement, and Nested Switch Statements.

LAB 3: Loops and Vectors

P21. Write a program to demonstrate (For loop, While loop, Break and Continue statements.)

- P22. Write a program to perform the Vector operations on Matrix:
- Addition and Subtraction of Vectors
 - Scalar Multiplication of Vectors
 - Transpose of a Vector
 - Appending Vectors
 - Magnitude of a Vector
 - Vector Dot Product

Unit III. Matrix Operations, Array, Number and String

LAB 1: Matrix Operations

- P23. Write a program to perform the Addition & Subtraction of Matrices.
- P24. Write a program to perform the Inverse of a Matrix.
- P25. Write a program to perform the Determinant of a Matrix.
- P26. Write a program to perform the 2-D Matrix Multiplication.
- P27. Write a program to perform the Concatenating of a Matrices.
- P28. Write a program to perform the transpose of a Matrix.
- P29. Write a program to demonstrate the Scalar Operations of Matrices.
- P30. Write a program to perform the Left and Right Division of Matrices.

Lab 2: Array

- P31. Write a program to perform the special array operations:
- zeros() function
 - ones() function
 - eye() function
 - rand() function
 - magic() function
- P32. Write a program to create multidimensional arrays using the ones(), zeros() or the rand() functions.
- P33. Write a program to find the Length, Dimension and Number of elements in the array.
- P34. Write a program to perform the circular Shift of the Array Elements.
- P35. Write a program to perform the Sorting of Arrays.

Lab 3: Number and String

- P36. Write a program to convert the String to ASCII.
- P37. Write a program to convert the ASCII to String.
- P38. Write a program to combine the strings into a cell Array.

Unit IV. Function, Input and Output Operation

Lab 1: Function

- P39. Write a program to find the largest number in the array using the function.
- P40. Write a program to perform the inline function in the Matlab.
- P41. Write a program to calculate the roots of a quadratic equation. The function would take three inputs, the quadratic co-efficient, the linear co-efficient and the constant term and returns the roots.
- P42. Write a program to demonstrate the functionality of Global and Private function.

Lab 2: Input and Output Operation

P43. Write a program to import the text file and specify the Delimiter and Column Header.

P44. Write a program to describe the data in the file with format specifiers, such as '%s' for a string, '%d' for an integer, or '%f' for a floating-point number.

P45. Write a program to read the data and create the structured array.

Unit V. GRAPHICS, Algebra, Calculus, Application of Matlab

Lab 1: GRAPHICS - 2D PLOTS

P46. Write a program to single plot on the graph.

P47. Write a program to multiple plot on the same graph.

P49. Write a program to generate the plots:

$$y = e^{-1.5x} \sin(10x)$$

$$y = e^{-2x} \sin(10x)$$

Lab 2: Algebra and Calculus

P50. Write a program to solve the equation using solve function:

$$X-5=0$$

$$3x-4=0$$

$$5x-4=0$$

P51. Write a program to solve the quadratic equation:

$$x^2 - 7x + 12 = 0.$$

$$5x^2 + 10x + 22 = 0$$

P52. Write a program to solve the higher order equations

$$(x - 3)^2(x - 7) = 0$$

$$x^4 - 7x^3 + 3x^2 - 5x + 9 = 0.$$

P53. Write a program to solve the equations involving more than one variable

$$5x + 9y = 5$$

$$3x - 6y = 4$$

$$6x + 19y = 11$$

$$32x - 16y = 23$$

P54. Write a program to solve the equations involving more than one variables

$$x + 3y - 2z = 5$$

$$3x + 5y + 6z = 7$$

$$2x + 4y + 3z = 8$$

P55. Write a program to solve the higher order derivate having function

$$y = f(x) = 3 \sin(x) + 7 \cos(5x).$$

We will have to find out whether the equation $f'' + f = -5\cos(2x)$ holds true.

P56. Write a program to perform

Laplace Transform,

Inverse Laplace Transform,

Fourier Transforms, and

Inverse Fourier Transforms.

P57. Write a program to perform the Polynomial Curve Fitting.

- P58. Write a program to solve the Definite Integral Using MATLAB.
P59. Write a program to solve the Indefinite Integral Using MATLAB.

Lab 3: Application of Matlab

- P60. Write a program to read and store the image file.
P61. Write a program to convert the colour image to gray scale.
P62. Write a program to convert the gray scale image to binary.
P63. Write a program to plot the original, gray scale and binary image.
P64. Write a program to perform the mathematical operations:
 imadd(),
 imsubtract(),
 imabsdiff (),
 immultiply(),
 Fspecial (),
 imfilter()
P65. Write a program to separate the colour channel from colour image.
P66. Write a program to perform the colour Space Conversion.
P67. Write a program to perform the Edge Detection on gray scale image.
P68. Write a program to add noise and perform filtering in the image.
P69. Write a program to perform the threshold and image segmentation.
P70. Write a program to perform the Histogram Equalization and contrast Enhancement.
P71. Write a program to count the no of objects in the Image.

USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Semester: IV

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: CSC401 Discrete Mathematics

Credits: 4

Course Outcome: At the end of the course, the students will be able to:

CSC401.CO1	Explain foundational concepts of sets, relations, functions, recurrence relations, and their applications in problem-solving.
CSC401.CO2	Apply counting principles, such as permutations, combinations, and the pigeonhole principle, to solve real-world discrete problems.
CSC401.CO3	Analyze and evaluate logical statements using propositional logic, logical equivalence, and proof techniques to validate mathematical arguments.
CSC401.CO4	Explore and demonstrate Boolean algebra operations and algebraic structures like groups, monoids, and semigroups in various contexts.
CSC401.CO5	Model and solve problems related to graphs and trees, including spanning trees, graph coloring, and Euler/Hamiltonian circuits.

Unit I. Sets, Relation and Function

Operations and Laws of Sets, Cartesian Products, Binary Relation, Partial Ordering Relation, Equivalence Relation, Image of a Set, Sum and Product of Functions, Bijective functions, Inverse and Composite Function, **Size of a Set, Finite and infinite Sets**, Recurrence Relation, Linear Recurrence Relation

Countable and uncountable Sets, Cantor's diagonal argument and The Power Set theorem, Schroeder-Bernstein theorem.

Principles of Mathematical Induction: The Well-Ordering Principle, Recursive definition, The Division algorithm: Prime Numbers, The Greatest Common Divisor: Euclidean Algorithm, The Fundamental Theorem of Arithmetic.

Unit II. Counting Techniques

Basic counting techniques-inclusion and exclusion, pigeon-hole principle, permutation and combination.

Unit III. Propositional Logic

Syntax, Semantics, Validity and Satisfiability, Basic Connectives and Truth Tables, Logical Equivalence: The Laws of Logic, Logical Implication, Rules of Inference, The use of Quantifiers.

Proof Techniques: Some Terminology, Proof Methods and Strategies, Forward Proof, Proof by

Contradiction, Proof by Contraposition.

Unit IV. Boolean Algebra and Algebraic Structures

Boolean Algebra and Boolean Ring, Identities of Boolean Algebra, Duality, Representation of Boolean Function, Disjunctive and Conjunctive Normal, Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups

Unit V. Graphs and Trees

Graphs and their properties, Degree, Connectivity, Path, Cycle, Sub Graph, Isomorphism, Euler and Hamiltonian path and circuit, Graph Colouring, Planar Graphs, Colouring Vertices, Colouring Edges, Introduction to Spanning Trees, Minimum Spanning Trees.

Text Books and References

1. Kenneth H. Rosen, Discrete Mathematics and its Applications, Tata McGraw – Hill, 7th Edition, 2017.
2. C L Liu and D P Mohapatra, Elements of Discrete Mathematics A Computer Oriented Approach, Tata McGraw – Hill, 4th Edition, 2017.
3. Susanna S. Epp, Discrete Mathematics with Applications, Wadsworth Publishing Co. Inc., 4th edition, 2011.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: CSC402 Computer Organization and Architecture

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

CSC402.CO1	Understand and explain the fundamental concepts of computer organization, including the fetch-decode-execute cycle, instruction sets, and addressing modes.
CSC402.CO2	Analyze and evaluate the operation and organization of basic processing units, including the differences between hardwired and microprogrammed control.
CSC402.CO3	Design and implement arithmetic and logic unit (ALU) operations, including addition, multiplication, and floating-point arithmetic, using integer and floating-point representations.
CSC402.CO4	Evaluate and design memory systems, focusing on semiconductor memories, cache memory, virtual memory, and CPU-memory interfacing.
CSC402.CO5	Apply input/output organization concepts to solve problems involving interrupts, DMA, and interface circuits, and distinguish between various I/O bus systems.

Unit I. Fundamental of Computer Organization

Basic organization of the stored program computer and operation sequence for execution of a program. Role of operating systems and compiler/assembler. Fetch, decode and execute cycle, Concept of operator, operand, registers and storage, Instruction format. Instruction sets and addressing modes.

Unit II. Basic Processing Unit

Some Fundamental Concepts, Execution of a Complete Instruction, Single and Multiple Bus Organization, Hardwired Control, Micro programmed Control unit.

Unit III. ALU Design

The ALU: ALU organization, Integer Representation , Serial and parallel Adders, 1's and 2's Complement Arithmetic, Multiplication of Signed binary numbers, Overflow detection , Status flags. Floating point - IEEE 754 standard. Fixed and floating point representation of numbers. Floating point number arithmetic, Design of ALU.

Arithmetic

Design of fast adders, Multiplication of Positive Numbers, Signed Operand Multiplication, Fast Multiplication, Integer Division, Floating-point Numbers and Operations.

Unit IV. Memory System

Basic Concepts, Semiconductor RAM Memories, Read Only Memories, Speed, Size, and Cost, Memory unit design with special emphasis on implementation of CPU-memory interfacing. Memory organization, static and dynamic memory, memory hierarchy, associative memory. Cache memory, Virtual memory, Data path design for read/write access.

Unit V. Input/ Output Organization

Basic Input and Output Operations, Accessing I/O Devices, Interrupts – Interrupt Hardware, Enabling and Disabling Interrupts, Handling Multiple Devices, Controlling Device Requests, Exceptions, Direct Memory Access. Interface Circuits, Standard I/O Interfaces – PCI Bus, SCSI Bus, USB, Flynn's Classification, RISC vs CISC.

Text Book

1. Computer Organization and Design: The Hardware/Software Interface, David A. Patterson and John L. Hennessy, Elsevier, 5th Edition, 2013
2. Computer Organization, Carl Hamacher, Zvonko Vranesic and Safwat Zaky, McGraw Hill.
3. Computer Architecture and Organization, John P. Hayes, McGraw Hill, 3rd Edition, 2017

Reference Books:

4. Computer Organization and Architecture: Designing for Performance, William Stallings, Pearson Education, Ninth Edition, 2012.
5. Computer Systems Design and Architecture, Vincent P. Heuring and Harry F. Jordan, Pearson Education.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: CSC403 Operating Systems

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

CSC403.CO1	Define and describe the functions, evolution, and types of operating systems, including batch, interactive, time-sharing, real-time, and multi-threading systems.
CSC403.CO2	Analyze and solve concurrency issues by applying process concepts, critical section problems, semaphores, and inter-process communication mechanisms
CSC403.CO3	Evaluate various CPU scheduling algorithms and apply concepts of deadlock prevention, avoidance, detection, and recovery.
CSC403.CO4	Design and manage memory systems, focusing on techniques such as paging, segmentation, and virtual memory, and apply page replacement algorithms to optimize performance.
CSC403.CO5	Analyze and implement I/O management techniques, including disk scheduling, file organization, and distributed operating system concepts, with case studies like UNIX.

Unit I. Introduction

Operating system and functions, Evolution of operating system, Batch, Interactive, Time Sharing, Real Time System, Multi-Threading System, Operating structure and Services

Unit II. Concurrent Processes

Process concept, Principle of Concurrency, Critical Section problem, Semaphores, Classical problems in Concurrency, Inter Process Communication, Introduction to monitor, Process Generation, Process Scheduling.

Unit III. CPU Scheduling and Deadlock

Scheduling Concept, Performance Criteria Scheduling Algorithm, Evolution, System Model, Deadlock Characterization, Prevention, Avoidance and Detection, Recovery from deadlock combined approach.

Unit IV. Memory Management:

Resident monitor, Multiprogramming with fixed partition, Multiprogramming with variable partition, Multiple base register, Paging, Segmentation, Virtual memory concept, Demand paging, Performance, Page replacement algorithms, Allocation of frames, Thrashing.

Unit V. I/O Management & Disk Scheduling

I/O devices and organization of I/O function, I/O Buffering, DISK I/O, File Concept, File Organization and Access Mechanism, File Directories, File Sharing, Introduction to distributed operating system, Case Studies - The UNIX operating system

Text Book Reference Books

1. Operating System Concepts, A. Silverschwatz, P. Galvin & G. Gange, Willey
2. Operating System Concepts, Milenkovic, McGraw Hill
3. An introduction to operating system, Dietel, Addison Wesley
4. Operating system design and implementation, Tannenbaum, PHI

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: CSC404 Design & Analysis of Algorithms

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

CSC404. CO1	Describe the key concepts of algorithms, algorithm specification, time and space complexity, and asymptotic notations for performance analysis.
CSC404. CO2	Analyze the efficiency of sorting and searching algorithms, including bubble sort, insertion sort, merge sort, quick sort, and linear search, using complexity analysis techniques.
CSC404. CO3	Apply divide-and-conquer strategies in problems like binary search, merge sort, quick sort, and Strassen's matrix multiplication, while performing complexity analysis.
CSC404. CO4	Design greedy and dynamic programming algorithms for optimization problems like the knapsack problem, traveling salesman problem, and job sequencing, and analyze their complexities.
CSC404. CO5	Evaluate NP-hard and NP-complete problems, including the satisfiability problem and traveling salesman problem, by understanding the classes of P and NP.

Unit I. Basic Tools on Designing Algorithms

What is an algorithm? Algorithm specification, Pseudocode for expressing algorithms, performance analysis-space complexity, time complexity, asymptotic notation- big (O) notation, omega notation, theta notation and little (o) notation, recurrences, Masters Theorem.

Complexity Analysis of Sorting and Searching Techniques

Complexity analysis of –Bubble Sort, Insertion Sort, Merge Sort, Advanced Sorting techniques - Heap Sort, Quick Sort, Sorting in Linear Time - Bucket Sort, Radix Sort and Count Sort, Searching Techniques- Linear search and Binary search

Unit II. Divide-and-Conquer and Complexity Analysis

The general method, application to binary search, finding the maximum and minimum, merge sort, quick sort, the problem of selection and Strassen's matrix multiplication.

Unit III. The Greedy Method and Complexity Analysis

The general method, application to optimal storage on tapes, job sequencing with deadlines, optimal merge patterns and minimum weight spanning trees.

Dynamic Programming and Complexity Analysis

The general method, application to multistage graphs, all pairs shortest paths, optimal binary search trees, 0/1-Knapsack and traveling salesman problem, Flow shop scheduling.

Unit IV. Unit Backtracking and Complexity Analysis

The general method, application to 8- puzzle problem, 8- queen problem and sum of subsets.

Branch and Bound and Complexity Analysis: The method, application to 0/1 Knapsack traveling salesman problems, and efficiency considerations.

Unit V. NP- Hard and NP-Complete Problems

Introduction and basic concepts, non-deterministic Turing machine, the classes of P and NP, NP-hard graph problems, NP- completeness of the satisfiability problem, and polynomial-space-bounded problem.

Text Books and References

1. Sahani Horowitz, Fundamentals of Computer Algorithms, Universities Press, 2nd Edition, 2008
2. J.Kleinberg & E. Tardos – Algorithm Design, Pearson Education India, 1st Edition New Delhi, 2013
3. G.Brassard & P. Bratley – Fundamentals of Algorithmics, PHI, New Delhi, 2005
4. Thomas H. Cormen , Charles E. Leiserson , Ronald L. Rivest , Clifford Stein – Introduction to Algorithms – PHI, 3rd Edition, New Delhi, 2010

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: HSC401 Management-I

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

HSC401.CO1	Define and explain the fundamental concepts of management, its characteristics, and its roles as a science, art, and profession.
HSC401.CO2	Compare and contrast classical, neo-classical, and modern management theories, and evaluate the contributions of key management thinkers like Taylor, Fayol, and Mayo.
HSC401.CO3	Apply the principles of planning and decision-making, including strategic planning, MBO, and forecasting techniques, to real-world management scenarios.
HSC401.CO4	Analyze and design organizational structures, considering different forms of organization, delegation, authority, and the principles of centralization and decentralization.
HSC401.CO5	Evaluate and implement directing and controlling principles, including the human factor, leadership theories, and management models, to improve organizational performance.

Unit I. Concept and Nature of Management

Meaning, characteristics- management as a science or an art, management as a profession, university of management, management as a process, Management and Administration, levels of Management, Skills of a manager, Roles of a manager, Significance of management, Limitations of management, Business environment and its interaction with management.

Unit II. Management Theory Approaches

To management-Classical, Neo-classical and modern, contributions to management thoughts– Taylor and Scientific theory, Fayol and Administrative theory Mayo and Hawthorne Experiments

Unit III. Planning and Decision Making

Planning–Meaning, Process, Types, Principles, Limitations, Strategic Planning- meaning and process, MBO–Meaning, process and requirements for implementation, Planning premises– Meaning and Types, Forecasting– Meaning and Techniques.

Unit IV. Organization Design and Structure

Organization–Meaning, Process, Principles, Organization structure- Determinates and forms: line, functional, line and staff project, matrix and committees, Formal and Informal Organization, Departmentation–Meaning and Bases, Span of Control–Meaning and factors influencing, Authority, Responsible and Accountability, Delegation: Meaning, Process, Principles, Centralization and Decentralization–

Unit V. Fundamentals of Directing Principles of Direction.

Human Factor and Directing, Theory X&Y, Management Process with Theory X&Y, Immaturity & Maturity Theory, Management Models, Theory Z, Control–Meaning ,steps, Types, Techniques, Significance, Limitations

Text Books & References:

1. Principles of Management–M.Govindarajan&S.Natrajan
2. Essentials of Management–Koontz &Odonil
3. Principles of Management–Prakash Chandra Tripathi

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: CSC411 Computer Organization and Architecture Lab

Credits:2

Course Outcome: At the end of the course, the students will be able to:

CSC411.CO1	Write and implement programs using the 8085 Microprocessor for arithmetic operations (addition, subtraction, multiplication, division) on decimal, hexadecimal, and BCD numbers.
CSC411.CO2	Analyze and develop programs to find the largest and smallest number in an array using the 8085 instruction set.
CSC411.CO3	Demonstrate the ability to sort and arrange data in both ascending and descending order using the 8085 Microprocessor.
CSC411.CO4	Design and implement a simple interrupt service routine to understand and simulate interrupt handling in microprocessor systems.
CSC411.CO5	Interface and test communication between the 8085 Microprocessor and the 8251, including transmission and reception of characters.

List of Experiments

1. Write a program using 8085 Microprocessor for Decimal, Hexadecimal addition and subtraction of two Numbers.
2. Write a program using 8085 Microprocessor for addition and subtraction of two BCD numbers.
3. Write a program to perform multiplication and division of two 8 bit numbers using 8085.
4. Write a program to find the largest and smallest number in an array of data using 8085 instruction set.
5. Write a program to arrange an array of data in ascending and descending order.
6. Write a simple Interrupt service routine to understand interrupt.
7. Interfacing a program to initiate 8251 and to check transmission and reception of character.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: CSC412 Operating Systems Lab

Credits:2

Course Outcome: At the end of the course, the students will be able to:

CSC412.CO1	Simulate and analyze CPU scheduling algorithms such as FCFS, SJF, Round Robin, and Priority to understand their performance.
CSC412.CO2	Develop and implement C programs to solve synchronization problems such as the producer-consumer and dining-philosophers problems using semaphores.
CSC412.CO3	Simulate memory management techniques including MVT and MFT, and analyze their efficiency in managing system resources.
CSC412.CO4	Implement and evaluate contiguous memory allocation techniques (Worst fit, Best fit, and First fit) to understand memory utilization strategies.
CSC412.CO5	Simulate and compare page replacement algorithms (FIFO, LRU, and OPTIMAL) and disk scheduling algorithms (FCFS, SCAN, and C-SCAN) to analyze system performance in handling memory and I/O operations.

List of Experiments:

1. Simulate the following CPU scheduling algorithms.
a) FCFS b) SJF c) Round Robin d) Priority.
2. Write a C program to simulate producer-consumer problem using Semaphores
3. Write a C program to simulate the concept of Dining-philosophers problem.
4. Simulate MVT and MFT.
5. Write a C program to simulate the following contiguous memory allocation Techniques
a) Worst fit b) Best fit c) First fit.
6. Simulate all page replacement algorithms
a)FIFO b) LRU c) OPTIMAL
7. Simulate all File Organization Techniques
a) Single level directory b) Two level directory
8. Simulate all file allocation strategies
a) Sequential b) Indexed c) Linked.
9. Simulate Bankers Algorithm for Dead Lock Avoidance.
10. Simulate Bankers Algorithm for Dead Lock Prevention.
11. Write a C program to simulate disk scheduling algorithms.
a) FCFS b) SCAN c) C-SCAN

Text Books and References

1. An Introduction to Operating Systems, P.C.P Bhatt, 2nd edition, PHI.
2. Modern Operating Systems, Andrew S Tanenbaum, 3rd Edition, PHI

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: CSC413 Design and Analysis of Algorithms Lab

Credits:2

Course Outcome: At the end of the course, the students will be able to:

CSC413.CO1	Implement and analyze sorting (Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort) and searching algorithms (Linear Search, Binary Search) to understand algorithmic performance.
CSC413.CO2	Demonstrate the application of Warshall's algorithm for computing transitive closure in a directed graph.
CSC413.CO3	Solve and optimize the Knapsack problem using appropriate algorithmic techniques and evaluate the solution's efficiency.
CSC413.CO4	Implement and compare the Shortest Paths algorithm and Minimum Cost Spanning Tree algorithm to determine optimal solutions for graph-based problems.
CSC413.CO5	Design and implement algorithms for solving the N Queens Problem, Traveling Salesperson Problem, and All Pairs Shortest Paths, and analyze their computational complexity.

List of Experiments

1. Write a program to implement Sorting Algorithms (Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort) and Searching Algorithms (Linear search, Binary search)
2. Write a program to implement Warshall's algorithm
3. Write a program to implement Knapsack Problem
4. Write a program to implement Shortest Paths algorithm
5. Write a program to implement Minimum Cost Spanning Tree
6. Write a program to implement Tree Traversals
7. Write a program to implement Travelling Sales Person Problem
8. Write a program to implement All Pairs Shortest Paths
9. Write a program to implement N Queens Problem.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: IV

Session: EVEN

Subject Code & Name: MDC401 Environmental Sciences

Credits: NC

Course Outcome: At the end of the course, the students will be able to:

MDC401.CO1	Explain the scope, importance, and multidisciplinary nature of environmental studies and discuss the need for public awareness.
MDC401.CO2	Analyze the structure, function, and energy flow of ecosystems, and evaluate case studies of various ecosystems from Jharkhand.
MDC401.CO3	Evaluate the impact of human activities on natural resources, such as land, water, forests, and energy, through case studies related to Jharkhand.
MDC401.CO4	Discuss the significance of biodiversity, including its levels and conservation strategies, and identify threats to biodiversity, particularly in India.
MDC401.CO5	Identify the causes and effects of environmental pollution, and propose control measures for various types of pollution, including case studies.

Unit I. Introduction to environmental studies

Multidisciplinary nature of environmental studies. Scope and importance. Need for public awareness

Unit II. Ecosystems - Jharkhand perspective

Concept of an ecosystem. Structure and function of an ecosystem. Energy flow in an ecosystem: food chains, food

webs and ecological pyramids. Ecological succession. Case studies of the following ecosystems :

- a) Forest ecosystem
- b) Grassland ecosystem
- c) Desert ecosystem
- d) Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

Unit III. Natural Resources

Renewable and Non-renewable Resources - Jharkhand

Land resources and land use change: Land as a resource, land degradation, landslides (natural & man-induced), soil erosion and desertification. Forests & forest resources: Use and over-exploitation, deforestation, case studies. Impacts of deforestation, mining, dam building on environment, forests, biodiversity and tribal populations. Resettlement and rehabilitation of

project affected persons; problems and concerns, case studies. Water resources: Use and over-exploitation of surface and ground water, floods, drought, conflicts over water (international & inter-state). Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. Energy resources: Renewable and non-renewable energy sources, use of alternate energy sources, growing energy needs, case studies.

Unit IV. Biodiversity and Conservation

Levels of biological diversity: genetic, species and ecosystem diversity. Biogeographic zones of India. Ecosystem and biodiversity services: Ecological, economic, social, ethical, aesthetic and Informational values. Biodiversity patterns and global biodiversity hot spots. India as a mega-biodiversity nation; Endangered and endemic species of India. Threats to biodiversity: Habitat loss, poaching of wildlife, man-wildlife conflicts, biological invasions. Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

Unit V. Environmental Pollution

- What is environmental pollution and its types?
- Causes, effects and control measures of :

Air pollution, Water pollution – freshwater and marine, Soil pollution, Noise pollution, Thermal pollution, Nuclear hazards and human health risks, Solid waste management: Control measures of urban and industrial waste., Role of an individual in prevention of pollution., Pollution case studies.

Text Books and References

- Tiwari, G.N. & Ghosal. M. K. 2005. Renewable Energy Resources: Basic Principles and Application. Narosa Publishing House.
- Bharucha, E. 2003, Textbook for Environmental Studies, University Grants Commission, New Delhi and Bharati Vidyapeeth Institute of Environmental Education and Research, Pune. 361
- Purohit, S.S. & Ranjan, R. 2007. Ecology, Environment & Pollution. Agrobios Publications.
- Singh, J.S., Singh, S.P. & Gupta, S.R. 2006. Ecology, Environment and Resource Conservation. Anamaya Publications.
- Mitra, A.P., Sharma, S., Bhattacharya, S., Garg, A., Devotta, S. & Sen, K. 2004. Climate Change and India. Universities Press, India.

- Krishnamurthy, K.V. 2004. An Advanced Text Book of Biodiversity - Principles and Practices. Oxford and IBH Publications Co. Pvt. Ltd. New Delhi.
- Asnani, P. U. 2006. Solid waste management. India Infrastructure Report 570.

USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Semester: V

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V

Session: ODD

Subject Code & Name: GEC501 Signals and Systems

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

GEC501.CO1	Identify and classify various types of signals and systems, and explain their properties such as linearity, causality, and stability.
GEC502. CO2	Analyze and characterize the behavior of continuous and discrete-time LTI systems, including convolution, stability, and system representation through state-space and differential equations.
GEC503. CO3	Apply Fourier series and Fourier transforms to analyze periodic and non-periodic signals in the time and frequency domains, and understand their effects on signal processing.
GEC504. CO4	Evaluate system behavior using Laplace and z-transforms for continuous and discrete-time systems, including poles, zeros, and solving differential equations..
GEC505. CO5	Examine the implications of the Sampling Theorem, and understand the effects of aliasing, signal reconstruction, and the relationship between continuous and discrete-time systems.

Unit I. Introduction to Signals and Systems (6 hours)

Signals and systems as seen in everyday life, and in various branches of engineering and science. Energy and power signals, continuous and discrete time signals, continuous and discrete amplitude signals. System properties: linearity: additivity and homogeneity, shift-invariance, causality, stability, reliability.

Unit II. Behavior of continuous and discrete-time LTI systems

Linear shift-invariant (LSI) systems, impulse response and step response, convolution, input-output behavior with a periodic convergent inputs. Characterization of causality and stability of linear shift invariant systems. System representation through differential equations and difference equations. State-space Representation of systems. State-Space Analysis, Multi-input, multi-output representation. State Transition Matrix and its Role. Periodic inputs to an LTI system, the notion of a frequency response and its relation to the impulse response.

Unit III. Fourier series

Fourier series representation of periodic signals, Waveform Symmetries, Calculation of Fourier Coefficients. Fourier Transform, convolution/multiplication and their effect in the frequency domain, magnitude and phase response, Fourier domain duality. The Discrete-Time Fourier Transform (DTFT) and the Discrete Fourier Transform (DFT). Parseval's Theorem.

Unit IV. Fourier, Laplace and z- Transforms (10 hours)

Review of the Laplace Transform for continuous time signals and systems, system functions, poles and zeros of system functions and signals, Laplace domain analysis, solution to differential equations and system behavior. The z-Transform for discrete time signals and systems, system functions, poles and zeros of systems and sequences, z-domain analysis.

Unit V. Sampling and Reconstruction (6 hours)

The Sampling Theorem and its implications- Spectra of sampled signals. Reconstruction: ideal interpolator, zero-order hold, first-order hold, and so on. Aliasing and its effects. Relation between continuous and discrete time systems.

Text Books & References:

- (1) A. V. Oppenheim, A. S. Willsky and S. H. Nawab, "Signals and systems", Prentice Hall India, 1997.
 - (2) J. G. Proakis and D. G. Manolakis, "Digital Signal Processing: Principles, Algorithms, and Applications", Pearson, 2006.
 - (3) H. P. Hsu, "Signals and systems", Schaum's series, McGraw Hill Education, 2010.
 - (4) S. Haykin and B. V. Veen, "Signals and Systems", John Wiley and Sons, 2007.
 - (5) A. V. Oppenheim and R. W. Schaffer, "Discrete-Time Signal Processing", Prentice Hall, 2009.
- B. P. Lathi, "Linear Systems and Signals", Oxford University Press, 2009.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V

Session: ODD

Subject Code & Name: CSC501 Database Management Systems

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

CSC501.CO1	Explain the fundamentals of database systems, including the definition of DBMS, data models, and the three-schema architecture.
CSC501.CO2	Design and implement a relational database using the E/R model, and convert database specifications into relational schemas while understanding indexing techniques.
CSC501.CO3	Develop and execute queries using Relational Algebra and SQL, including the use of nested queries, aggregation, and data manipulation.
CSC501.CO4	Analyze and apply normalization techniques to optimize relational database design, including the use of functional dependencies and normalization to 3NF, BCNF, and beyond.
CSC501.CO5	Evaluate and implement transaction processing concepts, including ACID properties, concurrency control, and recovery techniques in a DBMS.

Unit 1. Introduction

General introduction to database systems; Database - DBMS Definition, approaches to building a database, data models, three-schema architecture of a database, challenges in building a DBMS, various components of a DBMS.

Unit 2. Relational Data Model

Concept of relations and its characteristics, schema-instance, integrity constraints, E/R Model - Conceptual data modelling - motivation, entities, entity types, various types of attributes, relationships, relationship types, E/R diagram notation, Extended E/R Model, Converting the database specification in E/R and Extended E/R notation to the relational schema. File organizations, primary, secondary index structures, hash-based indexing, dynamic hashing techniques, multi-level indexes, B+ trees.

Unit 3. Relational Query Language

Relational Algebra operators: selection, projection, cross product, various types of joins, division, example queries, tuple relation calculus, domain relational calculus. Introduction to SQL, Data definition in SQL, Table, Key and Foreign key definitions, Data manipulation in SQL. Nested queries, Notion of aggregation, PL/SQL.

Unit 4. Relational Database Design

Dependencies and Normal forms - Importance of a good schema design, problems encountered with bad schema designs, motivation for normal forms, dependency theory - functional dependencies, Armstrong's axioms for FD's, closure of a set of FD's, minimal covers, definitions of 1NF, 2NF, 3NF and BCNF, decompositions and desirable properties of them, algorithms for 3NF and BCNF normalization, multi-valued dependencies and 4NF, join dependencies and definition of 5NF.

Unit 5. Transaction Processing

Concepts of transaction processing, ACID properties, concurrency control, locking based protocols, recovery and logging methods.

Text Book and References

1. Database System Concepts by Silberschatz, Korth & Sudarshan (McGraw-Hill Education)
2. Fundamentals of Database System By Elmasari & Navathe- Pearson Education
3. Database Management Systems by Rama Krishna & Gehrke (McGraw-Hill Education)
4. Fundamentals of Relational Database management Systems by Sumathi & Esakkirajan, Springer.

Faculty: Faculty of Computing and Information Technology**Program:** B.Tech (CSE)**Semester:** V**Session:** ODD**Subject Code & Name:** CSC502 Formal Language and Automata Theory**Credits:** 3**Course Outcome:** At the end of the course, the students will be able to:

CSC502.CO1	Explain the fundamentals of database systems, including the definition of DBMS, data models, and the three-schema architecture.
CSC502.CO2	Design and implement a relational database using the E/R model, and convert database specifications into relational schemas while understanding indexing techniques.
CSC502.CO3	Write and execute queries using Relational Algebra and SQL, including the use of nested queries, aggregation, and data manipulation.
CSC502.CO4	Analyze and apply normalization techniques to optimize relational database design, including the use of functional dependencies and normalization to 3NF, BCNF, and beyond.
CSC502.CO5	Evaluate and implement transaction processing concepts, including ACID properties, concurrency control, and recovery techniques in a DBMS.

Unit I. Introduction

Fundamentals: Basic definition of sequential circuit, block diagram, mathematical representation, and concept of transition table and transition diagram (Relating of Automata concept to sequential circuit concept) Design of sequence detector, Introduction to finite state model Finite state machine: Definitions, capability & state equivalent, kth-equivalent concept, Finite memory definiteness, testing table & testing graph. Minimization of FSM-completely specified and incompletely specified (Merger

graph, Merger table, Compatibility graph) Equivalence between two FSM's, Limitations of FSM Application of finite automata, Finite Automata with output- Moore & Melay machine.

Unit II. Finite Automation

Deterministic finite automaton and non deterministic finite automaton. Transition diagrams and Language recognizers. Finite Automata: NFA with $\hat{I}$ transitions - Significance, acceptance of languages. Conversions and Equivalence: Equivalence between NFA with and without $\hat{I}$ transitions. NFA to DFA conversion. DFA minimization. Myhill-Nerode theorem Regular Languages: Regular sets. Regular expressions, identity rules. Arden's theorem state and prove Constructing finite Automata for a given regular expressions, Regular string accepted by NFA/DFA. Pumping lemma of regular sets. Closure properties of regular sets (proofs not required). Grammar Formalism: Regular grammars-right linear and left linear grammars. Equivalence between regular linear grammar and FA.

Unit III. Context Free Grammar

Derivation trees, sentential forms. Right most and leftmost derivation of strings. Context Free Grammars, Ambiguity in context free grammars. Minimization of Context Free Grammars.

Chomsky normal form and Greibach normal form. Pumping Lemma for Context Free Languages. Enumeration of properties of CFL. Closure property of CFL, Ogden's lemma & its applications.

Unit IV. Push Down Automata

Push Down Automata: Push down automata, definition. Acceptance of CFL, Acceptance by final state and acceptance by empty state and its equivalence. Equivalence of CFL and PDA, interconversion. Introduction to DCFL and DPDA.

Unit V. Turing Machine

Turing Machine, definition, model, Design of TM, Computable functions, Church's hypothesis, counter machine, Types of Turing machines Universal Turing Machine, Halting problem.

Text Books and References

1. "Theory of Computer Science-Automata Languages and Computation", Mishra and Chandrashekar, 2nd edition, PHI
2. "Switching & Finite Automata", ZVI Kohavi, 2nd Edn., Tata McGraw Hill
3. "An Introduction to Computing", Peter Linz, Narosa. "Introduction to Automata Theory Language and Computation", Hopcroft H.E. and Ullman J. D.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V

Session: ODD

Subject Code & Name: CSC503 Object Oriented Programming

Credits:2

Course Outcome: At the end of the course, the students will be able to:

CSC503.CO1	Understand the concept of abstract data types (ADT) and demonstrate their implementation with appropriate abstraction functions and invariants.
CSC503.CO2	Apply the principles of object-oriented programming (OOP), such as encapsulation, polymorphism, and inheritance, to solve real-world problems.
CSC503.CO3	Analyze and implement design patterns, particularly the Model-View-Controller (MVC) pattern, to structure software applications effectively.
CSC503.CO4	Evaluate memory management techniques and use generic types and collections in object-oriented programming to enhance performance and flexibility
CSC503.CO5	Design and implement graphical user interfaces (GUIs) using Scala and Swing, applying object-oriented concepts in the software development process.

Unit I. Introduction to Abstract Datatypes

Abstract data types and their specification. How to implement an ADT. Concrete state space, concrete invariant, abstraction function. Implementing operations, illustrated by the Text example.

Unit II.Features of OOPS

Features of object-oriented programming. Encapsulation, object identity, polymorphism – but not inheritance. Inheritance in OO design.

Unit III.MVC Pattern

Design patterns. Introduction and classification. The iterator pattern. Model-view-controller pattern. Commands as methods and as objects. Implementing OO language features

Unit IV. Memory management.

Memory management, Generic types and collections.

Unit V. GUIs and SDP

GUIs. Graphical programming with Scala and Swing. The software development process.

Text Book and References

1. Herbert Schildt Java Complete Reference TMH.
2. Mr Kotiyana JAVA. The Complete Core Reference ORACLE

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V

Session: ODD

Subject Code & Name: HSC501 Development of Life Skills-II

Credits: 3

Course Outcome: At the end of the course the students will be able to:

HSC501.CO1	Understand and demonstrate key life skills such as self-awareness, empathy, critical thinking, and decision-making, as identified by WHO.
HSC501.CO2	Apply stress management techniques and emotional coping strategies to enhance personal and professional well-being.
HSC501.CO3	Analyze and utilize 21st-century skills such as creativity, critical thinking, and problem-solving for effective decision-making and collaboration.
HSC501.CO4	Evaluate group and team dynamics, managing conflicts and fostering effective teamwork to achieve common objectives.
HSC501.CO5	Develop leadership skills, understanding different leadership styles and frameworks, and applying them to lead teams and manage diverse stakeholders.

Unit I.Outline of Life Skills

Meaning and significance of life skills, Life skills identified by WHO: Self- awareness, Empathy, Critical thinking, Creative thinking, Decision making, problem solving, Effective communication, interpersonal relationship, coping with stress, coping with emotion.

Life skills for professionals: positive thinking, right attitude, attention to detail, having the big picture, learning skills, research skills, perseverance, setting goals and achieving them, helping others, leadership, motivation, self-motivation, and motivating others, personality development, IQ, EQ, and SQ.

Unit II. Self-awareness

Definition, need for self-awareness; Coping With Stress and Emotions, Human Values, tools and techniques of SA: questionnaires, journaling, reflective questions, meditation, mindfulness, psychometric tests, feedback. 3 Stress Management: Stress, reasons and effects, identifying stress, stress diaries, the four As of stress management, techniques, Approaches: action-oriented, emotion oriented, acceptance- oriented, resilience, Gratitude Training, Coping with emotions: Identifying and managing emotions, harmful ways of dealing with emotions, PATH method and relaxation techniques. Morals, Values and Ethics: Integrity, Civic Virtue, Respect for Others, Living Peacefully. Caring, Sharing, Honesty, Courage, Valuing Time, Time management, Cooperation, Commitment, Empathy, Self-Confidence, Character, Spirituality, Avoiding Procrastination, Sense of Engineering Ethics.

Unit III. 21st century skills

Creativity, Critical Thinking, Collaboration, Problem Solving, Decision Making, Need for Creativity in the 21st century, Imagination, Intuition, Experience, Sources of Creativity, Lateral Thinking, Myths of creativity, Critical thinking Vs Creative thinking, Functions of Left Brain & Right brain, Convergent & Divergent Thinking, Critical reading & Multiple Intelligence. Steps in problem solving: Problem Solving Techniques, Six Thinking Hats, Mind Mapping, Forced Connections.

Analytical Thinking, Numeric, symbolic, and graphic reasoning. Scientific temperament and Logical thinking

Problem solving technique. (any one technique may be considered)

- 1) Trial and error
- 2) Brain storming
- 3) Lateral thinking.

Unit IV. Group and Team Dynamics

Introduction to Groups: Composition, formation, Cycle, thinking, clarifying expectations, Problem Solving, Consensus, Dynamics techniques, Group vs Team, Team Dynamics, Virtual Teams. Managing team performance and managing conflicts, Intrapreneurship. Establishing good rapport, interest with others and work effectively with them to meet common objectives.

Unit V. Leadership

Leadership framework, entrepreneurial and moral leadership, vision, cultural dimensions. Growing as a leader, turnaround leadership, managing diverse stakeholders, crisis management. Types of Leadership, Traits, Styles, VUCA Leadership, Levels of Leadership, Transactional vs Transformational Leaders, Leadership Grid, Effective Leaders.

Text Books and References

1. **"The 7 Habits of Highly Effective People"** by Stephen R. Covey, published by Simon & Schuster.
2. **"How to Win Friends and Influence People"** by Dale Carnegie, published by Gallery Books.
3. **"Mindset: The New Psychology of Success"** by Carol S. Dweck, published by Ballantine Books.
4. **"Grit: The Power of Passion and Perseverance"** by Angela Duckworth, published by Scribner.
5. **"Emotional Intelligence: Why It Can Matter More Than IQ"** by Daniel Goleman, published by Bantam

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V

Session: ODD

Subject Code & Name: CSC511 Database Management Systems Lab

Credits:2

Course Outcomes: At the end of the course students will be able to:

CSC511.CO1	Demonstrate proficiency in implementing DDL commands (Create, Alter, Drop) in SQL to define and manage database structures.
CSC511.CO2	Implement and manage data using DML commands (Insert, Update, Delete) to manipulate database records.
CSC511.CO3	Utilize SQL functions such as number, aggregate, character, conversion, and date functions to perform data operations.
CSC511.CO4	Apply SQL operators and clauses (Arithmetic, Logical, Comparison, Group By, Order By) to filter and manipulate data efficiently.
CSC511.CO5	Develop PL/SQL programs to automate complex database operations and enhance data processing.

List of Programs/ Experiments

Lab Assignment No: 1

Objective: Implementation of DDL commands of SQL with suitable examples

- Create table
- Alter table
- Drop Table

Lab Assignment No: 2

Objective: Implementation of DML commands of SQL with suitable examples

- Insert
- Update
- Delete

Lab Assignment No: 3

Objective: Implementation of different types of function with suitable examples

- Number function
- Aggregate Function

- Character Function
- Conversion Function
- Date Function

Lab Assignment No: 4

Objective Implementation of different types of operators in SQL

- Arithmetic Operators
- Logical Operators
- Comparison Operator
- Special Operator
- Set Operation

Lab Assignment No: 5

Objective: Study and Implementation of

- Group By & having clause
- Order by clause

Lab Assignment No: 6

Objective: Study & Implementation of different types of constraints.

Lab Assignment No: 7

Objective: Study & Implementation of

- Sub queries
- Views

Lab Assignment No: 8

Objective: Implementation of different types of Joins

- Inner Join
- Outer Join
- Natural Join etc.

Lab Assignment No: 9

Objective: Study & Implementation of PL/SQL.

Text books:

1. “Database System Concepts”, 6th Edition by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill.
2. Elmasri, S.B. Navathe, Fundamentals of Database Systems, 7th Edition, Pearson Education, 2016.

Reference Book:

1. Raghu Ramakrishnan, Database Management Systems, 3rd edition, TMH, 2014.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V

Session: ODD

Subject Code & Name: CSC512 Object Oriented Programming Lab

Credits:2

Course Outcome: At the end of the course, the students will be able to:

CSC512.CO1	Recall fundamental Java programming syntax and tools required for setting up and executing Java programs.
CSC512.CO2	Demonstrate the use of Object-Oriented Programming (OOP) concepts such as inheritance, polymorphism, and abstraction in a lab environment.
CSC512.CO3	Implement Java programs to perform tasks such as data processing, file handling, and basic algorithmic problem-solving.
CSC512.CO4	Analyze the efficiency of Java code by examining execution flow, identifying potential errors, and optimizing logic during debugging.
CSC512.CO5	Develop interactive and functional Java applications integrating graphical user interfaces (GUIs), multithreading, and database connectivity.

List of Experiments

1. To find the sum of any number of integers entered as command line arguments
2. To find the factorial of a given number
3. To learn use of single dimensional array by defining the array dynamically.
4. To learn use of .length in case of a two dimensional array
5. To convert a decimal to binary number
6. To check if a number is prime or not, by taking the number as input from the keyboard
7. To find the sum of any number of integers interactively, i.e., entering every number from the keyboard, whereas the total number of integers is given as a command line argument
8. Write a program that show working of different functions of String and StringBuffer classes like setCharAt(), setLength(), append(), insert(), concat() and equals().
9. Write a program to create a —distance class with methods where distance is computed in terms of feet and inches, how to create objects of a class and to see the use of this pointer
10. Modify the —distance class by creating constructor for assigning values (feet and inches) to the distance object. Create another object and assign second object as reference variable to another object reference variable. Further create a third object which is a clone of the first object.
11. Write a program to show that during function overloading, if no matching argument is found, then java will apply automatic type conversions (from lower to higher data type)
12. Write a program to show the difference between public and private access

specifiers. The program should also show that primitive data types are passed by value and objects are passed by reference and to learn use of final keyword

13. Write a program to show the use of static functions and to pass variable length arguments in a function.
14. Write a program to demonstrate the concept of boxing and unboxing.
15. Create a multi-file program where in one file a string message is taken as input from the user and the function to display the message on the screen is given in another file (make use of Scanner package in this program).
16. Write a program to create a multilevel package and also creates a reusable class to generate Fibonacci series, where the function to generate fibonacci series is given in a different file belonging to the same package.
17. Write a program that creates illustrates different levels of protection in classes/subclasses belonging to same package or different packages
18. Write a program —DivideByZero that takes two numbers a and b as input, computes a/b, and invokes Arithmetic Exception to generate a message when the denominator is zero.
19. Write a program to show the use of nested try statements that emphasizes the sequence of checking for catch handler statements.
20. Write a program to create your own exception types to handle situation specific to your application (Hint: Define a subclass of Exception which itself is a subclass of Throwable).
21. Write a program to demonstrate priorities among multiple threads.
22. Write a program to demonstrate multithread communication by implementing synchronization among threads (Hint: you can implement a simple producer and consumer problem).
23. Write a program to create URL object, create a URLConnection using the openConnection() method and then use it to examine the different components of the URL and content.
24. Write a program to implement a simple datagram client and server in which a message that is typed into the server window is sent to the client side where it is displayed.
25. Write a program that creates a Banner and then creates a thread to scroll the message in the banner from left to right across the applet's window.
26. Write a program to get the URL/location of code (i.e. java code) and document (i.e. html file).
27. Write a program to demonstrate different mouse handling events like mouseClicked(), mouseEntered(), mouseExited(), mousePressed, mouseReleased() and mouseDragged().
28. Write a program to demonstrate different keyboard handling events.
29. Write a program to generate a window without an applet window using main() function.
30. Write a program to demonstrate the use of push buttons.

Text Book and References

1. Herbert Schildt Java Complete Reference TMH.
2. Mr. Kottiyana JAVA. The Complete Core Reference ORACLE
3. Kathy Seira Head Forst Java Orielley

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V

Session: ODD

Subject Code & Name: MDC501 Constitution of India

Credits: NC

Course Outcome: At the end of the course, the students will be able to:

MDC501.CO1	Understand the historical evolution and foundational principles of the Indian Constitution and its significance in governance.
MDC501.CO2	Explain the framework of fundamental rights, including equality, freedom, and personal liberty, as enshrined in the Constitution.
MDC501.CO3	Analyze the role of Fundamental Duties and Directive Principles of State Policy in shaping responsible citizenship and governance.
MDC501.CO4	Evaluate the federal structure and distribution of legislative and financial powers between the Union and States in India.
MDC501.CO5	Assess the impact of constitutional amendments and emergency provisions on the governance and democratic process of India.

Unit I. Meaning of Constitution Law

Meaning of Constitution Law and Constitutionalism, Historical Perspective of Constitution of India.

Unit II. Characteristics of Constitution of India

Salient features and characteristics of Constitution of India Scheme of the fundamental rights

- Scheme of fundamental right to Equality
- Scheme of fundamental right to certain freedom under Article 19
- Scope of the Right to Life and Personal Liberty under Article 21

Unit III. Scheme of Fundamental Duties

Scheme of Fundamental Duties and its legal status, The Directive Principles of State Policy-Its importance and implementation

Unit IV. Federal structure and distribution of Legislative

Federal structure and distribution of Legislative and financial powers between the Union and the States, Parliamentary form of Government in India- The Constitution powers and status of the President of India, Amendment of the Constitutional Powers and Procedure.

Unit V. The historical perspective

The historical perspective of the constitutional amendments in India, Emergency Provisions: National Emergency, President rule, Financial Emergency, Local Self Government- Constitutional Scheme in India.

Text Books and References

1. JN Pandey, Constitutional Law of India, Central Law Agency.
2. AK Jain, Constitution of India, Ascent Publications
3. Constitution of India, Bare Act with Short Notes, Universal Publication.



USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Semester: VI

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI

Session: EVEN

Subject Code & Name: CSC601 Compiler Design

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

CSC601.CO1	Explain the phases of compilation and analyze the role of lexical and syntax analysis in transforming source code to intermediate representations.
CSC601.CO2	Design and implement finite state machines and parsing techniques for lexical and syntax analysis of programming languages.
CSC601.CO3	Generate intermediate representations of source programs using syntax-directed translation and perform type checking.
CSC601.CO4	Analyze and optimize code using data flow analysis, loop optimization, and other local/global optimization techniques..
CSC601.CO5	Develop object code generation algorithms by applying register allocation, instruction selection, and machine-dependent optimizations.

Unit I. Introduction, Lexical and Syntax Analysis

Overview of Compilation: Introduction to Compiler, Phases of Compilation, Grouping of Phases. Role of Lexical Analyzer, Input Buffering, Specification of Tokens, Finite state machines and regular expressions and their applications to lexical analysis. Context-free grammars, Top-down Parsing – Backtracking, LL(1), recursive descent parsing, Predictive parsing, Bottom-up parsing – Shift Reduce parsing, LR and LALR parsing, Error recovery in parsing, handling ambiguous grammar.

Unit II. Semantic analysis

Intermediate forms of source Programs – abstract syntax tree, polish notation and three address codes. Attributed grammars, Syntax directed translation, Conversion of popular Programming languages language Constructs into Intermediate code forms, Type checker.

Unit III. Symbol Tables

Symbol table format, organization for block structures languages, hashing, tree structures representation of scope information.

Unit IV. Code optimization, Data flow analysis and Object code generation

Consideration for Optimization, Scope of Optimization, local optimization, loop optimization, frequency reduction, folding, DAG representation.

Unit V. Data flow analysis and Object code generation

Flow graph, data flow equation, global optimization, redundant sub expression elimination, Induction variable elements, Live variable analysis, Copy propagation. Object code forms, machine dependent code optimization, register allocation and assignment generic code generation algorithms, DAG for register allocation.

Text Book and References

1. Compilers- Principles, Techniques and Tools, By A.V. Aho, M.S. Lam, R Sethi and J.D.Ullman, Pearson Education.
2. lex &yacc – John R. Levine, Tony Mason, Doug Brown, O'reilly
3. Engineering a Compiler, by Cooper & Linda, Elsevier.
4. Compiler Construction, K.C. Loudon, Thomson Brooks/Cole.

Faculty: Faculty of Computing and Information Technology**Program:** B.Tech (CSE)**Semester:** VI**Session:** EVEN**Subject Code & Name:** CSC602 Computer Networks**Credits:** 3**Course Outcome:** At the end of the course, the students will be able to:

CSC602.CO1	Describe network architectures, models, and transmission media while differentiating between OSI and TCP/IP reference models.
CSC602.CO2	Implement and evaluate data link layer protocols including error detection, correction techniques, and medium access control methods.
CSC602.CO3	Analyze network layer protocols and routing algorithms for efficient packet delivery, congestion control, and internetworking.
CSC602.CO4	Illustrate transport layer mechanisms for reliable and efficient process-to-process communication using UDP and TCP.
CSC602.CO5	Demonstrate application layer protocols and internetworking devices for real-world communication and data management.

Unit I. Introduction to Computer Network

Network hardware, Network software, OSI, TCP/IP Reference models, Example Networks: ARPANET, Internet. Physical Layer: Guided Transmission media: twisted pairs, coaxial cable, fiber optics, Wireless transmission.

Unit II. Data link layer

Data link layer: Design issues, framing, Error detection and correction. Elementary data link protocols: simplex protocol, A simplex stop and wait protocol for an error-free channel, A simplex stop and wait protocol for noisy channel. Sliding Window protocols: A one-bit sliding window protocol, A protocol using Go-Back-N, A protocol using Selective Repeat, Example data link protocols. Medium Access sub layer: The channel allocation problem, Multiple access protocols: ALOHA, Carrier sense multiple access protocols, collision free protocols. Wireless LANs.

Unit III. Network Layer

Network Layer: Design issues, Routing algorithms: shortest path routing, Flooding, Hierarchical routing, Broadcast, Multicast, distance vector routing, Congestion Control Algorithms, Quality of Service, Internetworking, The Network layer in the internet. Introduction to Internet protocol, IPv4 header, IPv4 Datagrams, Encapsulation, Fragmentation and Reassembly, IP routing, Subnet addressing, Subnet mask, Super netting- special case of IP addresses, Ipv6-Motivation, frame format and addressing. ICMP: Introduction, ICMP Header, ICMP message types.

Unit IV. Transport Layer

Process to Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP)s; Quality of Service, QoS improving techniques: Leaky Bucket and Token Bucket algorithm.

Unit V. Presentation Layer and Application Layer

Application Layer –Domain name system, SNMP, Electronic Mail; the World WEB, HTTP, Streaming audio and video, DNS: Distribution of name spaces, DNS in the internet, FTP:- Connection, Communication, command processing, TFTP, E-Mail: SMTP, POP, IMAP, SNMP. study of internetworking devices and their configuration– switches, hubs, Bridges, routers and Gateways.

Text Book

1. “Computer Networks: A top-down approach”, by Forouzan and Mosharraf, Pearson.

Reference Book

1. “Computer Networking: A top-down approach”, by Kurose and Ross, 5th Edition, Pearson.
2. “Computer Networks”, by Andrew S. Tanenbaum and David Wetherall, 5th Edition, Pearson.
3. “Computer Networks: A System Approach”, Larry L. Peterson and Bruce S. Davie, 5th Edition, Morgan Kaufmann Publishers

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI

Session: EVEN

Subject Code & Name: CSC611 Compiler Design Lab

Credits:2

Course Outcome: At the end of the course, the students will be able to:

CSC611.CO1	Develop a lexical analyzer and parser using tools like Lex and YACC to analyze and process syntactic structures of programming languages.
CSC611.CO2	Design and implement programs for automata-based transformations, such as NFA to DFA conversion, ϵ -closure computation, and DFA minimization.
CSC611.CO3	Simulate parsing techniques like recursive descent and operator precedence parsing to evaluate the correctness of given grammars.
CSC611.CO4	Generate intermediate code and perform basic code optimizations, including loop unrolling and constant propagation.
CSC611.CO5	Implement back-end code generation to translate intermediate representations into assembly instructions executable on an 8086 processor.

List of Programs / Applications

1. Design and implement a lexical analyzer for given language using C and the lexical analyzer should ignore redundant spaces, tabs and new lines.
2. Implementation of Lexical Analyzer using Lex Tool
3. Generate YACC specification for a few syntactic categories.
 - a) Program to recognize a valid arithmetic expression that uses operator +, −, * and /.
 - b) Program to recognize a valid variable which starts with a letter followed by any number of letters or digits.
 - c) Implementation of Calculator using LEX and YACC
 - d) Convert the BNF rules into YACC form and write code to generate abstract syntax tree
4. Write program to find ϵ – closure of all states of any given NFA with ϵ transition.
5. Write program to convert NFA with ϵ transition to NFA without ϵ transition.
6. Write program to convert NFA to DFA
7. Write program to minimize any given DFA.
8. Develop an operator precedence parser for a given language.
9. Write program to find Simulate First and Follow of any given grammar.
10. Construct a recursive descent parser for an expression.
11. Construct a Shift Reduce Parser for a given language.

12. Write a program to perform loop unrolling.
13. Write a program to perform constant propagation.
14. Implement Intermediate code generation for simple expressions.
15. Implement the back end of the compiler which takes the three address code and produces the 8086 assembly language instructions that can be assembled and run using an 8086 assembler. The target assembly instructions can be simple move, add, sub, jump etc

Text Books and References

1. Aho A Ravi Sethi and J D Ullman, Compilers Principles Techniques and Tools, Addison Wesley
2. Kenneth C Loudon, “Compiler Construction Principles and Practice”, Cenage Learning Indian Edition
3. D M Dhamdhare, System programming and operating system, Tata McGraw Hill & Company
4. Tremblay and Sorenson, The Theory and Practice of Compiler Writing - Tata McGraw Hill & Company.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI

Session: EVEN

Subject Code & Name: CSC612 Computer Networks Lab

Credits:2

Course Outcome: At the end of the course, the students will be able to:

CSC612.CO1	Set up and simulate network topologies using network simulation tools to analyze data traffic and routing protocols.
CSC612.CO2	Implement and evaluate core networking algorithms, including distance vector routing and congestion control using the leaky bucket algorithm.
CSC612.CO3	Configure and simulate various network protocols, such as DHCP, ARP, and VLAN, to understand their operations in real-world scenarios.
CSC612.CO4	Develop and test data link layer mechanisms, including sliding window protocol and CRC error detection for reliable data transfer.
CSC612.CO5	Apply encryption and decryption techniques to ensure secure communication and evaluate their effectiveness in protecting data..

List of Experiments:

- 1.Network Simulator installation
- 2.Network Simulator structure
- 3.How to start
- 4.Creating network topology
- 5.Sending data
- 6.Wired lab
- 7.Node creation
- 8.UDP data traffic
- 9.Traffic file analysis
- 10.TCP data traffic
- 11.Creating a LAN segment
- 12.Implementing switch
- 13.Implement distance vector routing algorithm for obtaining routing tables at each node.
14. Implement data encryption and data decryption
- 15.Write a program for congestion control using Leaky bucket algorithm.
- 16.Implementing VLAN
- 17.DHCP Protocol configuration
- 18.ARP Protocol configuration
- 19.Develop a simple data link layer that performs the flow control using the sliding window protocol, and loss recovery using the Go-Back-N mechanism.
- 20.Simulate Cyclic Redundancy Check (CRC) error detection algorithm for noisy channel.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI

Session: EVEN

Subject Code & Name: CSC613 Project-I

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

CSC613.CO1	Set up and simulate network topologies using network simulation tools to analyze data traffic and routing protocols.
CSC613.CO2	Implement and evaluate core networking algorithms, including distance vector routing and congestion control using the leaky bucket algorithm.
CSC613.CO3	Configure and simulate various network protocols, such as DHCP, ARP, and VLAN, to understand their operations in real-world scenarios.
CSC613.CO4	Develop and test data link layer mechanisms, including sliding window protocol and CRC error detection for reliable data transfer.
CSC613.CO5	Apply encryption and decryption techniques to ensure secure communication and evaluate their effectiveness in protecting data.

- i. Project work is an integral part of B.Tech in Computer Science and Engineering programme. B.Tech in Computer Science and Engineering students are required to undertake Project-I(Minor Project) for a period of one semester in the University, in semester VI of credits 3.
- ii. The topic/title of the Project and the name of the supervisor earmarked shall be approved by an Internal Project Guide from the University. Every B.Tech in Computer Science and Engineering student shall also be required to prepare a Project Report/ Dissertation under the supervision of a Project Guide.
- iii. Final Project Report shall be submitted to the Head of the Department duly signed by the Project Guide, minimum in triplicate on or before a specified date fixed for this purpose of evaluation by the Head of the Department. Students are expected to maintain academic integrity of the submitted project report.
- iv. During End-semester evaluation, every student shall be required to make a presentation of his/her project report, including execution of Project work undertaken by him/her during the semester before the duly constituted committee formed by the Head of the department.
- v. Each student may be asked to present Synopsis of the Project Study before the members of the Faculty for consideration and finalization of Project Study Title.
- vi. Once the project report is submitted to and approved by the project guide, the same shall be finally examined by project guide and internal faculty members. Project guide will evaluate out of 40 marks and internal faculty members will evaluate out of 60 marks in End Term Examination.



USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Semester: VII

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VII

Session: ODD

Subject Code & Name: GSC701 Biology

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

GCC701.CO1	Explain the fundamental principles and importance of biology as a scientific discipline in understanding natural phenomena and technological applications.
GCC701.CO2	Classify organisms based on cellular, structural, and ecological characteristics and analyze the hierarchical organization of life forms.
GCC701.CO3	Apply genetic principles, including Mendelian laws and molecular genetics, to explain inheritance patterns and genetic variations.
GCC701.CO4	Evaluate the role of enzymes in catalysis, reaction mechanisms, and biological processes using examples and quantitative parameters.
GCC701.CO5	Illustrate metabolic pathways such as glycolysis, Krebs cycle, and photosynthesis, and relate energy transformations to biological thermodynamics.

Unit I. Introduction & Classification

Purpose: To convey that Biology is as important a scientific discipline as Mathematics, Physics and Chemistry. Bring out the fundamental differences between science and engineering by drawing a comparison between eye and camera, Bird flying and aircraft. Mention the most exciting aspect of biology as an independent scientific discipline. Why we need to study biology? Discuss how biological observations of 18th Century that lead to major discoveries. Examples from Brownian motion and the origin of thermodynamics by referring to the original observation of Robert Brown and Julius Mayor. These examples will highlight the fundamental importance of observations in any scientific inquiry.

Classification

Purpose: To convey that classification per se is not what biology is all about. The underlying criterion, such as morphological, biochemical or ecological be highlighted.

Hierarchy of life forms at phenomenological level. A common thread weaves this hierarchy Classification. Discuss classification based on (a) cellularity- Unicellular or multicellular (b) ultrastructure- prokaryotes or eucaryotes. (c) energy and Carbon utilisation -Autotrophs, heterotrophs, lithotrophs (d) Ammonia excretion — aminotelic, uricotelic, ureotelic (e) Habitataaquatic or terrestrial (e) Molecular taxonomy- three major kingdoms of life. A given organism can come under different category based on classification. Model organisms for the study of biology come from different groups. E.coli, S.cerevisiae, D. Melanogaster, C. elegance, A. Thaliana, M. musculus

Unit II. Genetics & Biomolecules

Purpose: To convey that "Genetics IS to biology what Newton's laws are to Physical Sciences" Mendel's laws, Concept of segregation and independent assortment. Concept of allele. Gene mapping, Gene interaction, Epistasis. Meiosis and Mitosis be taught as a part of genetics. Emphasis to be give not to the mechanics of cell division nor the phases but how genetic material passes from parent to offspring. Concepts of recessiveness and dominance. Concept of mapping of phenotype to genes. Discuss about the single gene disorders in humans. Discuss the concept of complementation using human genetics. building blocks and yet the manifestations are as diverse as one can imagine Molecules of life. In this context discuss monomeric units and polymeric structures. Discuss about sugars, starch and cellulose. Amino acids and proteins. Nucleotides and DNA/RNA. Two carbon units and lipids.

Unit III. Enzymes

Purpose: To convey that without catalysis life would not have existed on earth Enzymology: How to monitor enzyme catalysed reactions. How does an enzyme catalyse reactions? Enzyme classification. Mechanism of enzyme action. Discuss at least two examples. Enzyme kinetics and kinetic parameters. Why should we know these parameters to understand biology? RNA catalysis.

Unit IV. Information Transfer

Purpose: The molecular basis of coding and decoding genetic information is universal Molecular basis of information transfer. DNA as a genetic material. Hierarchy of DNA structure from single stranded to double helix to nucleosides. Concept of genetic code. Universality and degeneracy of genetic code. Define gene in terms of complementation and recombination. Proteins- structure and function. Hierarch in protein structure. Primary secondary, tertiary and quaternary structure. Protoms as enzymes, transporters, receptors and structural elements.

Unit V. Metabolism

Thermodynamics as applied to biological systems. Exothermic and endothermic versus endergonic and exergoinc reactions. Concept of K_{eq} and its relation to standard free energy. Spontaneity. ATP as an energy currency. This should include the breakdown of glucose to $C_0_2 + H_2O$ (Glycolysis and Krebs cycle) and synthesis of glucose from C_0_2 and H_2O (Photosynthesis). Energy yielding and energy consuming reactions. Concept of Energy charge Module 9. (3 hours)- Microbiology Concept of single celled organisms. Concept of species and strains. Identification and classification of microorganisms. Microscopy. Ecological aspects of single celled organisms. Sterilization and media compositions. Growth kinetics.

Text Books & References

- 1) Biology: A global approach: Campbell, N. A.; Reece, J. B.; Urry, Lisa; Cain, M, L.; Wasserman, S. A. ; Minorsky, P. V. ; Jackson, R. B. Pearson Education Ltd
- 2) Outlmes of Biochemistry, Conn, E.E; Stumpf, P.K; Bruening, G; Doi, R.H. John Wiley and Sons
- 3) Principles of Biochemistry (V Edition), By Nelson, D. L.; and Cox, M. M. W.H. Freeman and Company
- 4) Molecular Genetics (Second edition), Stent, G. S.; and Calender, R. W.H. Freeman and company, Distributed by Satish Kumar Jain for CBS Publisher
- 5) Microbiology, Prescott, L.M J.P. Harley and C.A. Klein 1995. 2nd edition Wm, C. Brown Publishers

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VII

Session: ODD

Subject Code & Name: CSC711 Project-II

Credits:6

Course Outcome: At the end of the course, the students will be able to:

CSC711.CO1	Identify and formulate a research problem or engineering challenge within the domain of Computer Science and Engineering for in-depth study.
CSC711.CO2	Develop and implement innovative solutions or prototypes based on systematic research and technical methodologies.
CSC711.CO3	Evaluate project outcomes and technical performance through rigorous testing, validation, and iteration.
CSC711.CO4	Prepare and present a comprehensive project report or dissertation with proper documentation and adherence to academic integrity standards.
CSC711.CO5	Demonstrate effective communication and project management skills by delivering structured presentations and adhering to timelines in a collaborative environment.

- i. Project work is an integral part of B.Tech in Computer Science and Engineering programme. B.Tech in Computer Science and Engineering students are required to undertake Project-II for a period of one semester in the University in semester VII of credits 6.
- ii. The Project work for VII semester is Minor project
- iii. The topic/title of the Project and the name of the supervisor earmarked shall be approved by an Internal Project Guide from the University. Every B.Tech in Computer Science and Engineering student shall also be required to prepare a Project Report/ Dissertation under the supervision of a Project Guide.
- iv. Final Project Report shall be submitted to the Head of the Department duly signed by the Project Guide, minimum in triplicate on or before a specified date fixed for this purpose of evaluation by the Head of the Department. Students are expected to maintain academic integrity of the submitted project report.
- v. During End-semester evaluation, every student shall be required to make a presentation of his/her project report, including execution of Project work undertaken by him/her during the semester before the duly constituted committee formed by the Head of the department.
- vi. Each student may be asked to present Synopsis of the Project Study before the members of the Faculty for consideration and finalization of Project Study Title.
- vii. Once the project report is submitted to and approved by the project guide, the same shall be finally examined by an external examiner and internal examiner(s). Evaluation of VII semester project is done by project guide and internal faculty members. Project guide will evaluate out of 40 marks and internal faculty members will evaluate out of 60 marks in End Term Examination.



USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Semester: VIII

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VIII

Session: EVEN

Subject Code & Name: CSC811 Project- III

Credits: 6

Course Outcome: At the end of the course, the students will be able to:

CSC811.CO1	Identify and define a complex engineering problem in Computer Science and Engineering, framing it within a structured project scope.
CSC811.CO2	Design and develop an innovative system, application, or research solution using advanced engineering tools and methodologies.
CSC811.CO3	Critically analyze and evaluate project deliverables based on performance metrics, technical feasibility, and user requirements.
CSC811.CO4	Document a detailed and professional project report or dissertation reflecting the research process, development methodologies, and outcomes with academic integrity.
CSC811.CO5	Effectively communicate project findings and technical contributions through structured presentations and demonstrations to diverse audiences, including external evaluators.

- i. Project work is an integral part of B.Tech in Computer Science and Engineering programme. B.Tech in Computer Science and Engineering students are required to undertake Project-III for a period of one semester in the University, in semester VIII of credits 6.
- ii. The Project work for VIII semester is Major Project.
- iii. The topic/title of the Project and the name of the supervisor earmarked shall be approved by an Internal Project Guide from the University. Every B.Tech in Computer Science and Engineering student shall also be required to prepare a Project Report/ Dissertation under the supervision of a Project Guide.
- iv. Final Project Report shall be submitted to the Head of the Department duly signed by the Project Guide, minimum in triplicate on or before a specified date fixed for this purpose of evaluation by the Head of the Department. Students are expected to maintain academic integrity of the submitted project report.

- v. During End-semester evaluation, every student shall be required to make a presentation of his/her project report, including execution of Project work undertaken by him/her during the semester before the duly constituted committee formed by the Head of the department.
- vi. Each student may be asked to present Synopsis of the Project Study before the members of the Faculty for consideration and finalization of Project Study Title.
- vii. Once the project report is submitted to and approved by the project guide, the same shall be finally examined by an external examiner and internal examiner(s).
- viii. Evaluation of VIII semester project is done by project guide, external examiner and internal faculty members. Project guide will evaluate out of 40 marks and in End Term Examination both external examiner and internal faculty members will evaluate out of 60 marks.



USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Open Elective Courses

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC001 Soft Skills and Interpersonal Development

Credits:3

Course Outcome: At the end of the course, the students will be able to:

OEC001.CO1	Explain the importance of soft skills and demonstrate effective interpersonal communication in personal and professional contexts.
OEC001.CO2	Analyze personal strengths and weaknesses using SWOT techniques to set realistic goals and develop a positive mindset.
OEC001.CO3	Demonstrate proficiency in corporate communication, including public speaking, group discussions, and interview preparation, to effectively convey ideas.
OEC001.CO4	Interpret non-verbal cues and practice body language techniques to enhance personality and communication effectiveness.
OEC001.CO5	Develop teamwork and leadership abilities by engaging in collaborative activities and applying business etiquette in professional environments.

Unit I. Soft Skills & Interpersonal Communication

An Introduction – Definition and Significance of Soft Skills; Process, Importance and Measurement of Soft Skill Development. Interpersonal relations; communication models, process and barriers; team communication; developing interpersonal relationships through effective communication; listening skills; essential formal writing skills; corporate communication styles– assertion, persuasion, negotiation.

Unit II. SWOT & Creative Thinking

Discovering the Self; Setting Goals; Beliefs, Values, Attitude, Virtue. Developing Positive Thinking and Attitude; Driving out Negativity; Meaning and Theories of Motivation; Enhancing

Motivation Levels

Unit III. Corporate Communication

Public Speaking: Skills, Methods, Strategies and Essential tips for effective public speaking.

Group Discussion: Importance, Planning, Elements, Skills assessed; Effectively disagreeing, Initiating, Summarizing and Attaining the Objective.

Interview & Presentation Skills: Interviewer and Interviewee–in-depth perspectives. Before, During and After the Interview. Tips for Success: Types, Content, Audience Analysis, Essential Tips– Before, During and After, Overcoming Nervousness.

Unit IV. Non-Verbal

Non-Verbal Communication & Personality Development Importance and Elements; Body Language. Concept, Essentials, Tips Meaning, Nature, Features, Stages, Models; Learning Skills; Adaptability Skills.

Unit V. Business Etiquette & Team Work

Concept of Teams; Building effective teams; Concept to Leadership and honing Leadership skills. Meaning, Nature, Features, Stages, Models; Learning Skills; Adaptability Skills.

Text Books and References

- 1.ManagingSoftSkillsforPersonalityDevelopment edited by B. N. Ghosh, McGraw Hill India, 2012.
- 2.Effective Communication and Soft Skills, Nitin Bhatnagar, Pearson Education India, 2011
- 3.EnglishandSoftSkills– S.P. Dhanavel, Orient BlackswanIndia,2010.
- 4.EffectiveBusinessCommunication, Kulbhushan Kumar, Khanna Publishing House, 2021.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC002 Human Resource Development and Organizational Behavior

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

OEC002.CO1	Define key concepts of HRM and HRD, and differentiate between them while analyzing their impact on organizational development.
OEC002.CO2	Explain the HRD process and apply career and succession planning principles in designing and evaluating HRD systems.
OEC002.CO3	Examine individual behavior theories such as learning, attitude, and job satisfaction to understand their influence on organizational performance.
OEC002.CO4	Assess motivational theories and leadership styles, applying them to enhance employee engagement and organizational effectiveness.
OEC002.CO5	Evaluate personality traits and values using models like MBTI and Big Five, and integrate them into workplace dynamics for better person-job and person-organization fit.

Unit I. HRM & HRD

Concepts, Definition of Human Resources Management, Meaning of Human Resource Development, Difference between HRM &HRD, Theoretical Framework of Human Resource Development, Challenges in Human Resource Development, Human Resource Development Goals. Human Resource Development Climate and Practices in India, Factors contributing to Human Resource Development Climate.

Unit II. HRD Process:

Concepts and Definition of Career Planning & Succession Planning, Human Resource Development Systems, Human Resource Development Processes, (Designing Human Resource Development, Need assessment, Design, Implementation, Evaluation).

Unit III. Introduction OB

Concept of OB; Management roles, skills and activities; Disciplines that contribute to OB;

Individual Behavior: Learning, attitude and job satisfaction Concept of learning, conditioning, shaping and reinforcement, Concept of attitude, components, behavior and attitude, Job satisfaction.

Unit IV. Motivation & Leadership:

Motivation: Concept; Theories (Hierarchy of needs, X and Y, Two factor, McClelland, Goal setting, Self-efficacy, Equity theory); Job characteristics model; Redesigning job and work arrangements; Employee involvement; Flexible benefits, Intrinsic rewards.

Leadership: Concept; Trait theories, Behavioral theories

Unit V. Personality and Values:

Concept of personality; MBTI; Big Five model, Relevance of values; Indian values, Linking personality and values to the workplace (person-job fit, person- organization fit).

Group Behavior: Groups and Work Teams:

Concept; Five stage model of group development; Group think and shift; Indian perspective on group norms

Text Books & References:

1. Kandula, R. Srinivas, Strategic Human Resource Development, Prentice Hall of India
2. Ishwar Dayal, Designing HRD systems, Concept Publishing Company
3. T.V.Rao, Readings in Human Resource Development, Oxford and IBH publishing Co.Ltd.

Faculty: Faculty of Computing and Information Technology**Program:** B.Tech (CSE)**Semester:** VI/VII/VIII**Session:** ODD/EVEN**Subject Code & Name:** OEC003

Cyber Law and Ethics

Credits: 3**Course Outcome:** At the end of the course, the students will be able to:

OEC003.CO1	Comprehend the concepts of cyber jurisprudence and different approaches to law, such as the doctrinal, consensual, and real approaches.
OEC003.CO2	Grasp the overview of the IT Act, its amendments, and limitations, understanding its role in governing digital transactions and electronic records.
OEC003.CO3	Analyze the relevance of cyber law in civil and criminal procedural codes, evidence acts, and banking regulations.
OEC003.CO4	Analyze the legal implications of e-commerce models such as B2B and B2C, along with legal issues related to e-security, taxation, electronic payments, and supply chain management.
OEC003.CO5	Create ethical frameworks for emerging technologies such as artificial intelligence and blockchain, incorporating core ethical principles into their development and use.

UNIT 1: Introduction to Cyber Law

Evolution of computer Technology, emergence of cyber space. Cyber Jurisprudence, Jurisprudence and law, Doctrinal approach, Consensual approach, Real Approach, Cyber Ethics, Cyber Jurisdiction, Hierarchy of courts, Civil and criminal jurisdictions, CyberspaceWeb space, Web hosting and web Development agreement, Legal and Technological Significance of domain Names, Internet as a tool for global access.

UNIT 2: Information Technology Act

Overview of IT Act, 2000, Amendments and Limitations of IT Act, Digital Signatures, Cryptographic Algorithm, Public Cryptography, Private Cryptography, Electronic Governance, Legal Recognition of Electronic Records, Legal Recognition of Digital Signature, Certifying Authorities, Cyber Crime and Offences, Network Service Providers Liability, Cyber Regulations Appellate Tribunal, Penalties and Adjudication.

UNIT 3: Cyber law and Related Legislation

Patent Law, Trademark Law, Copyright, Software – Copyright or Patented, Domain Names and Copyright disputes, Electronic Data Base and its Protection, IT Act and Civil Procedure Code, IT Act and Criminal Procedural Code, Relevant Sections of Indian Evidence Act, Relevant Sections of Bankers Book Evidence Act, Relevant Sections of Indian Penal Code, Relevant Sections of Reserve Bank of India Act, Law Relating To Employees And Internet, Alternative Dispute Resolution , Online Dispute Resolution (ODR).

UNIT 4: Electronic Business and legal issues

Evolution and development in E-commerce, paper vs paper less contracts E-Commerce models- B2B, B2C, E security. Business, taxation, electronic payments, supply chain, EDI, E-markets,

Emerging Trends.

UNIT 5: Cyber Ethics

The Importance of Cyber Law, Significance of cyber Ethics, Need for Cyber regulations and Ethics.

Ethics in Information society, Introduction to Artificial Intelligence Ethics: Ethical Issues in AI and core Principles, Introduction to Block chain Ethics.

Text Books and References

1. Cyber Laws: Intellectual property & E Commerce, Security- Kumar K, dominant Publisher
2. Cyber Ethics 4.0, Christoph Stuckelberger, Pavan Duggal, by Globethic
3. Information Security policy & Implementation Issues, NIIT, PHI
4. Computers, Internet and New Technology Laws, Karnika Seth, Lexis Nexis Butterworths Wadhwa Nagpur.
5. Legal Dimensions of Cyber Space, Verma S, K, Mittal Raman, Indian Law Institute, New Delhi,
6. Cyber Law, Jonthan Rosenoer, Springer, New York, (1997).
7. The Information Technology Act, 2005: A Handbook, OUP Sudhir Naib,, New York, (2011)
8. Information Technology Act, 2000, S. R. Bhansali,, University Book House Pvt. Ltd., Jaipur (2003).
9. Cyber Crimes and Law Enforcement, Vasu Deva, Commonwealth Publishers, New Delhi, (2003).

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC004 Indian Music System

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

OEC004.CO1	Demonstrate an understanding of the basic elements of Indian classical music, including Swara, Raga, Tala, and Shruti, and their historical development.
OEC004.CO2	Analyze the structure and characteristics of Ragas and Talas, and classify them based on their types and rhythmic cycles.
OEC004.CO3	Apply theoretical knowledge in vocal or instrumental improvisation and rendition of Ragas, Talas, and traditional compositions such as Dhrupad, Khayal, and Varnam.
OEC004.CO4	Evaluate the emotional and thematic content of musical compositions and performances through critical listening and peer feedback.
OEC004.CO5	Compose and perform solo or group pieces that integrate elements of Raga, Tala, and traditional compositional forms, showcasing an appreciation for Indian classical music's cultural and social context.

Unit 1. Fundamentals of Indian Music

- Introduction to Indian classical music (Hindustani or Carnatic)
- Understanding the basic elements: Swara (notes), Raga (melody), Tala (rhythm)
- Overview of the historical development of Indian classical music
- Introduction to the concept of Shruti (microtones) and its significance
- Listening exercises and basic vocalization practice

Unit 2. Raga and Tala

- In-depth study of Ragas: classification, characteristics, and structure
- Understanding Tala: rhythm cycles, types of Tala, and their importance
- Exploration of Alap, Bandish, and Taan in Hindustani music or Alapana, Kriti, and Kalpanaswaram in Carnatic music
- Practical exercises in Raga improvisation and Tala rendition
- Listening sessions to analyze different Ragas and Talas

Unit 3. Compositional Forms

- Study of traditional compositions: Dhrupad, Khayal, Thumri, Bhajan in Hindustani music or Varnam, Kriti, Padam, Tillana in Carnatic music
- Analysis of the structure, lyrics, and thematic content of compositions
- Practical application through learning and performing select compositions

- Understanding the importance of Bhava (emotion) and expression in rendering compositions
- Comparative analysis of compositional forms across different regions and styles

Unit 4. Instruments in Indian Music

- Overview of traditional Indian musical instruments: string, wind, and percussion
- Study of instrument-specific techniques and playing styles
- Role of accompanying instruments in ensemble performances
- Exploration of solo and ensemble repertoire for various instruments
- Practical sessions for basic playing techniques on select instruments

Unit 5. Performance Practice and Appreciation

- Practical application of learned concepts through solo and group performances
- Peer evaluation and constructive feedback sessions
- Attendance of live concerts and recitals by renowned artists
- Critical listening and analysis of performances
- Understanding the cultural and social context of Indian classical music through historical and contemporary perspectives

Text Books and References

- 1.The Raga Guide: A Survey of 74 Hindustani Ragas" by Joep Bor, Suvarnalata.
- 2.Rao, and Wim van der Meer, published by Nimbus Records.
Fundamentals of Raga and Tala" by V. K. Krishna Prasad, published by Sapna Book House.
- 3.The Oxford Illustrated Companion to South Indian Classical Music" by Ludwig Pesch, published by Oxford University Press.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC005 History of Science

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

OEC005.CO1	Explain the evolution of scientific methodologies and their logical frameworks.
OEC005.CO2	Analyze the influence of ancient and medieval scientific traditions on the progression of science.
OEC005.CO3	Evaluate the contributions of the Scientific Revolution to modern scientific thought.
OEC005.CO4	Summarize major scientific advancements in biology and physics and their relevance to contemporary science.
OEC005.CO5	Assess the relationship between science, society, and public health in historical contexts.

Unit I. Introduction

Introduction and Overview, contributions to sciences, Paradigm Shifts and Science History, Black Box Activity, Logic and methodology of sciences.

Unit II. Ancient and Medieval Origins

Aristotelian Worldview, Classical Cosmology, Ancient Astronomy, Traditions of Learning in Medieval University, Aristotle in Medieval University.

Unit III. Scientific Revolution

Vesalius and Scientific Renaissance, Copernicus and Scientific Renaissance, Early Modern Big Science, Galileo Affair, Bacon and Empiricism, Cultures of Experimentation, Galileo vs. Boyle, Descartes and Mechanical Philosophy, Newtonianism, Chemical Revolution and Industrialism.

Unit IV. Biology and Physics

Darwinian Revolution, Formation of Modern Biology, Discovery of DNA, Big Biology, Atomic Model, Einstein and Relativity, Quantum Mechanics, Big Science and Cold War.

Unit V. Science and Public Health

World War I: Chemistry and Total War, World War II: Eugenics and Total War, Epidemics, Tuberculosis, Cholera, and Spanish Flu, Epidemiology and Germ Theory of Disease, Eugenics.

Text Books and References

1. "The Structure of Scientific Revolutions" by Thomas S. Kuhn, first published in 1962.
2. "Cosmos" by Carl Sagan, first published in 1980.
3. "A Brief History of Time" by Stephen Hawking, first published in 1988.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC006 Introduction to Arts and Aesthetics

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

OEC006.CO1	Explain and summarize the fundamental concepts of aesthetics, including Indian and Western philosophical approaches to art and judgment.
OEC006.CO2	Analyze the principles of Indian philosophy and religious thought, along with their influence on traditional and modern artistic expressions.
OEC006.CO3	Apply theoretical frameworks like Rasa, Bhava, and artistic representation to interpret and critique various forms of art, including literature and visual art.
OEC006.CO4	Evaluate the evolution of art through different eras, from classical traditions to contemporary digital aesthetics and media.
OEC006.CO5	Create a comprehensive understanding of art in the digital age, exploring its impact on cinema, conceptual art, and new media practices.

Unit I. Introduction

Introduction to Aesthetics and its Scope, Philosophy and Art, Introduction To Indian Aesthetics, Judgment, Painting and Literature, Appreciation, Aesthetic experience, qualities and attitude.

Unit II. Principles of Philosophy

Introduction to basic Principles of Indian Philosophy and Religious Thoughts, Vedic, Upanishadic, Greek Philosophy, Western Philosophy, Social and Political, Western Philosophy.

Unit III. Fundamentals of Art

Fundamentals of Indian Art, Principles of Painting and Shilpa Texts like Chitrasutram, Chitralakshanam, Defining Art, Theories of art, Art as Representation (Plato), Art as Communication (Tolstoy), Art as Symbolic Form (Susane Langer), Art as a Expression, Art as a Divine Intuition, Art as Play.

Unit IV. Rasa and Bhava

Concept of Rasa, Rasanubhuti, Ras Nishpatti, Bharat Lollat, Sankhya, Bharat Muni, Bhava, Types of Bhava, Concept of Rang Sangrah, Rasa- Internal or External, Concept of Shadanga, Alankar, Bhamak, Dhawani Sidhant, Anad Vardhan, Auchitya, Kasmendra, Riti, Vaman, Abhinavgupta on Rasa And Rasadhvani, Literary Concept- Guna, Alamkar, Varieties of Kavya - Drishya and Sravya, Relation Between Rasa And Kavya.

Unit V. Art in Digital Era

Visual art, Modern art, Abstract art, Abstract expression, Conceptual art, Dada, Formalism, Modernism, Postmodernism, Primitivism, Tachisme, Contemporary Aesthetics, New Media Art, New Aesthetic, Role of the Artist, Digital Era, Art in Digital Era, Art in Cinema,

Text Books and References

1. Chakravarty, Nilima (1992), Indian Philosophy: The Path Finder's and System Builders, New Delhi: Allied Publishers.
2. Chatterjee, S.C. & D.M. Datta (1984), An Introduction to Indian Philosophy, reprint, University of Calcutta.
3. Hiriyana, M: (1951), Outlines of Indian Philosophy, London: Allen & Unwin.
4. Mohanty, J.N. (1992), Reason and Tradition in Indian Thought, Oxford, Clarendon Press.
5. Organ, Troy Wilson. (1964), The Self in Indian Philosophy, London, Mouton & Co.
6. Coomaraswami, A. K.: The Transformation of Nature in Art
7. Ghosal, S.N.: Elements of Indian Aesthetics
8. Goswami, Trailokyanath(1980): Nandantatta: Prachya Aru Pachatya, Bani Prakashan

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC007 Economic Policies in India

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

OEC007.CO1	Explain the fundamental features, challenges, and historical evolution of the Indian economy, including poverty, unemployment, and income inequality.
OEC007.CO2	Analyze the sectoral composition of the Indian economy, focusing on agriculture, land reforms, Green Revolution, and related policies.
OEC007.CO3	Evaluate the role of industrial and service sectors in India's economic development and assess the impact of industrial policies.
OEC007.CO4	Compare the objectives and functions of economic planning in India, including the transition from the Planning Commission to NITI Aayog and key fiscal and monetary policies.
OEC007.CO5	Assess the impact of globalization, foreign trade, FDI, and India's participation in international frameworks like WTO on the country's economy.

Unit I. Basic features and problems of Indian Economy

Economic History of India; Nature of Indian Economy, demographic features and Human Development Index, Problems of Poverty, Unemployment, Inflation, income inequality, Black money in India.

Unit II. Sectorial composition of Indian Economy

Issues in Agriculture sector in India, land reforms Green Revolution and agriculture policies of India.

Unit III. Industrial development

Industrial development, small scale and cottage industries, industrial Policy, Public sector in India, service sector in India

Unit IV. Economic Policies

Economic Planning in India, Planning commission v/s NITI Aayog, Five Year Plans, monetary policy in India, Fiscal Policy in India, Centre state Finance Relations, Finance commission in India. LPG policy in India

Unit V. External sector in India

External sector in India India's foreign trade value composition and direction, India Balance of payment since 1991, FDI in India, Impact of Globalization on Indian Economy, WTO and India.

Text Books and references

1. Dutt Rudder and K.P.M Sunderam (2017). Indian Economy. S Chand & Co. Ltd. New Delhi.
2. Mishra S.K & V.K Puri (2017). Indian Economy and –Its Development Experience. Himalaya Publishing House.
3. Singh, Ramesh, (2016): Indian Economy, Tata-McGraw Hill Publications, New Delhi.
4. Dhingra, I.C., (2017): March of the Indian Economy, Heed Publications Pvt. Ltd.
5. Karam Singh Gill, (1978): Evolution of the Indian Economy, NCERT, New Delhi
6. Kaushik Basu (2007): The Oxford Companion to Economics of India, Oxford University Press.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC008 Environmental Engineering and Disaster Management

Credits:3

Course Outcome: At the end of the course, the students will be able to:

OEC008.CO1	Explain the fundamental concepts of environmental engineering, water supply systems, and their significance in urban and rural areas.
OEC008.CO2	Evaluate drinking water quality standards as per Indian norms and outline the processes involved in water treatment for safe consumption.
OEC008.CO3	Analyze the characteristics and disposal methods of domestic wastewater, including the basics of sewer design and wastewater treatment processes.
OEC008.CO4	Assess the composition, characteristics, and disposal techniques for solid waste and evaluate air pollution types, their effects, and mitigation strategies based on BIS standards.
OEC008.CO5	Develop disaster management strategies by identifying various types of disasters and understanding their impacts on environmental systems.

Unit I: Introduction: Objective, scope and outcome of the course.

Unit II: Importance of safe water supply system. Domestic water requirements for urban and rural areas. Sources of Water supply. Intakes and transportation of water.

Unit III: Drinking water quality. Indian Standards of drinking water. Introduction to water treatment for safe drinking, Importance of sanitation.

Unit IV: Domestic waste water: quantity, characteristics, disposal in urban and rural areas. Sewer: types, design discharge and hydraulic design. Introduction to domestic wastewater treatment.

Unit V: Solid waste: quantity, characteristics and disposal for urban and rural areas. Introduction to air pollution. Types of pollutants, properties and their effects on living beings. BIS standards for pollutants in air and their abetments. Introduction to various disaster, Importance of disaster management.

Text Books & References:

- Environment Engineering and Disaster Management- Dr. Sanjay K. Sharma, Laxmi Publications, First Edition, 2017

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC009 Sustainable Engineering

Credits:3

Course Outcome: At the end of the course, the students will be able to:

OEC009.CO1	Explain the principles of sustainable development, sustainable engineering models, and the significance of environmental ethics and legislations in promoting sustainability.
OEC009.CO2	Analyze local and global environmental issues such as solid waste, resource degradation, and climate change, and evaluate mitigation strategies like the zero-waste concept and carbon trading.
OEC009.CO3	Apply tools for sustainability, including Environmental Management Systems (EMS), Life Cycle Assessment (LCA), and Environmental Impact Assessment (EIA), to assess the environmental impact of engineering projects.
OEC009.CO4	Design sustainable habitats and industrial systems by integrating concepts like green building materials, energy-efficient systems, and sustainable urbanization models.
OEC009.CO5	Evaluate renewable energy technologies and green business practices to promote sustainable energy solutions and environmentally friendly innovations.

Unit I. Introduction to Sustainable Engineering

Sustainable development, concepts of sustainable development: three pillar model, egg of sustainability model, Atkisson's pyramid model, prism model, principles of sustainable development, sustainable engineering, threats for sustainability.

Environmental Ethics and Legislations – Environmental ethics and education, multilateral environmental agreements and protocols, enforcement of environmental laws in India – The Water Act, The Air Act, The Environment Act.

Unit II. Local Environmental Issues

Solid waste, impact of solid waste on natural resources, zero waste concept and three R concept, waste to energy technology: thermo-chemical conversion, biochemical conversion.

Global Environmental Issues- Resource degradation: deterioration of water resources, land degradation, air pollution, climate change and global warming, ozone layer depletion, carbon footprint, carbon trading.

Unit III. Tools for Sustainability

Environmental management System (EMS), concept of ISO14000, life cycle assessment (LCA): basic components, advantages, disadvantages, case study. Environmental impact assessment (EIA), environmental auditing, bio mimicking, case studies.

Unit IV. Sustainable Habitat

Concept of green building, green building materials, green building certification and rating: green

rating for integrated habitat assessment (GRIHA) , leadership in energy and environmental design (LEED) rating, energy efficient buildings, sustainable cities, sustainable transport, sustainable pavements, case studies in sustainability engineering: Green building, sustainable city, sustainable transport system.

Sustainable Industrialization and Urbanization – Sustainable urbanization, industrialization, material selection, pollution prevention, industrial ecology, industrial symbiosis, poverty reduction.

Unit V. Renewable energy resources

Conventional and non- conventional forms of energy, solar energy, fuel cells, wind energy, small hydro plants, biogas systems, biofuels, energy from ocean, geothermal energy, conservation of energy.

Green technology and Green Business: Sustainable business, green technology, green energy, green construction, green transportation, green chemistry, green computing

Text Book:

R. L. Rag and Lekshmi Dinachandran Remesh. *Introduction to Sustainable Engineering*. 2nd Edition, PHI Learning Pvt. Ltd., 2016.

Reference Books:

1. D. T. Allen and D. R. Shonnard. *Sustainability Engineering: Concepts, Design and Case Studies*, 1st Edition, Prentice Hall, 2011.
2. A.S. Bradley, A. O. Adebayo, P.Maria. *Engineering applications in sustainable design and development*, 1st Edition, Cengage learning, 2016.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC010 Water Pollution Control Engineering

Credits:3

Course Outcome: At the end of the course, the students will be able to:

OEC010.CO1	Describe the characterization, monitoring, and reuse strategies of industrial and municipal wastewater and the criteria for selecting water pollution treatment plants.
OEC010.CO2	Analyze physico-chemical treatment methods such as sedimentation, coagulation, and flocculation, and the fundamentals of biological treatment for nutrient removal.
OEC010.CO3	Explain the principles and applications of aerobic and anaerobic biological treatment processes, including activated sludge systems, trickling filters, and UASB reactors.
OEC010.CO4	Evaluate advanced wastewater treatment processes like membrane filtration, wet-air oxidation, and adsorption for effective pollution control and sludge management.
OEC010.CO5	Apply wastewater treatment strategies through case studies of industries such as petroleum, petrochemicals, and paper manufacturing to design sustainable disposal solutions.

Unit I. Introduction:

Characterization and monitoring of industrial and municipal waste water, recycling and reuse of wastewater. Basic philosophy and selection of water pollution treatment plants; Design criteria: hydraulic loading rate, organic loading rate, residence time, dilution rate.

Unit II. Physico-Chemical Treatment Methods:

Sedimentation, coagulation, flocculation, thickening, floatation. Biological Treatment Fundamentals: Microbial metabolism, bacterial growth kinetics; Biological nitrification, denitrification and phosphorus removal; Anerobic fermentation and aerobic treatment.

Unit III. Aerobic Suspended and Attached Growth Biological Treatment Processes:

Aerated lagoon, activated sludge systems, trickling filter, sequential batch reactor, fluidized bed bioreactors. Anaerobic Suspended and Attached Growth Biological Treatment Processes: UASB and hybrid UASB reactors, bio-towers.

Unit IV. Advanced Treatment Processes:

Membrane processes- reverse osmosis, ultrafiltration, nanofiltration and electrodialysis; Wet air oxidation, adsorption and ion-exchange; Wet-land and root-zone treatment of industrial and municipal wastes; Design of sludge drying beds, thermal and biological processes for sludge and land fillings.

Unit V. Case Studies:

Waste water treatment and disposal strategies in petroleum, petrochemical, fertilizer, distillery, pulp and paper industries.

Text Books and References:

1. Water Pollution: Assessment and Management by Arvind Kumar, Daya Publishing House
2. Fair. G. M., “Water and Waste water engineering Vol. I & II”, John Wiley and sons, New York, 2010
3. Water Management in India, Concept Publishing Company, New Delhi, 2004

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: OEC011 Intellectual Property Rights

Credits:3

Course Outcome: At the end of the course, the students will be able to:

OEC011.CO1	Describe the fundamental concepts and types of intellectual property rights, including patents, copyrights, trademarks, and geographical indications, and their role in innovation and commerce..
OEC011.CO2	Analyze the processes and criteria for patent and copyright registration, including infringement issues, remedies, and specific cases like software copyrights.
OEC011.CO3	Explain the nature and legal framework of trademarks, including registration, infringement, domain names, and remedies for protection in cyberspace.
OEC011.CO4	Evaluate the significance of design registration and international conventions on design protection, including procedures and cancellation.
OEC011.CO5	Apply the principles of the Information Technology Act, including e-commerce, digital signatures, and legal measures against cybercrimes, in the context of intellectual property protection.

Unit I:

Introduction to IPR: Meaning of property, Origin, Nature, Meaning of Intellectual Property Rights
Introduction to TRIPS and WTO.

Kinds of Intellectual property rights—Copy Right, Patent, Trade Mark, Trade Secret and trade dress, Design, Layout Design, Geographical Indication, Plant Varieties and Traditional Knowledge.

Unit II:

Patent Rights and Copy Rights— Origin, Meaning of Patent, Types, Inventions which are not patentable, Registration Procedure, Rights and Duties of Patentee, Assignment and licence , Restoration of lapsed Patents, Surrender and Revocation of Patents, Infringement, Remedies & Penalties.

Copy Right— Origin, Definition &Types of Copy Right, Registration procedure, Assignment & licence, Terms of Copy Right, Piracy, Infringement, Remedies, Copy rights with special reference to software.

Unit III:

Trade Marks— Origin, Meaning & Nature of Trade Marks, Types, Registration of Trade Marks, Infringement & Remedies, Offences relating to Trade Marks, Passing Off, Penalties. Domain Names on cyber space.

Unit IV:

Design- Meaning, Definition, Object, Registration of Design, Cancellation of Registration, International convention on design, functions of Design.

Unit V:

Basic Tenants of Information Technology Act-2000 – IT Act – Introduction, E-Commerce and legal provisions, E- Governance and legal provisions, Digital signature and Electronic Signature, Cybercrimes

Text Books:

1. Intellectual Property Rights and the Law, Gogia Law Agency, by Dr. G.B. Reddy
2. Law relating to Intellectual Property, Universal Law Publishing Co, by Dr. B.L.Wadehra
3. IPR by P. Narayanan
4. Law of Intellectual Property, Asian Law House, Dr.S.R. Myneni.



USHA MARTIN UNIVERSITY, JHARKHAND

(Established by Jharkhand Government under Sec.2(f) of UGC Act 1956)

Elective Courses

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECT001 Theory of Computation

Credits: 3

Course Outcomes: After the completion of this course, students will be able to:

ECT001.CO1	Define the concepts of finite automata, regular expressions, and grammars, establishing the foundation of automata theory for modeling computational systems.
ECT001.CO2	Apply regular expressions and finite automata to recognize and analyze regular and non-regular languages.
ECT001.CO3	Analyze context-free grammars (CFGs) and pushdown automata (PDAs) for parsing techniques, derivation processes, and ambiguity resolution in language processing.
ECT001.CO4	Evaluate the role and variations of Turing machines in determining language recognition, decidability, and computational problem classification.
ECT001.CO5	Explain the theoretical underpinnings of computational complexity classes (P, NP, NP-C, NP-Hard) and their relevance to solving real-world problems.

Unit 1. Automata

Introduction to Automata: (mathematical model of digital devices, including real computer), State Transition Graph, Finite Automaton (FA) and its types, Deterministic Finite Automaton (DFA), Non-deterministic Finite Automaton (NFA), Complement, Union, Intersection of FA's, Conversion Strategy from NFA to DFA, Minimization of FA, Finite Automaton with Output, Applications of FA.

Unit 2. Regular Expression

Introduction, R.E.'s and basic operations, Algebraic laws on Regular Expression, Finite and Infinite Languages, Equivalence of finite Automaton and regular expressions, Constructing NDFA from Regular Expression, Pumping Lemma for Regular Language, Closure properties of Regular Languages, Non-regular languages, Applications of Regular Expression.

Unit 3. Grammar

Introduction, Formal Definition of Grammar, The Chomsky Hierarchy of Grammar, Designing Regular grammar from DFA, Context Free Grammar, Closure properties of Context Free Languages, CFG and Normal form: Chomsky Normal Form, Greibach Normal Form, Non-Context Free Language, Applications of CFGs.

Unit 4. Push Down Automata

Introduction, Definition of PDA, Types of Pushdown Automata (DPDA and NPDA), Converting CFG to PDA, Derivation (Parsing), Parsing Techniques, Ambiguous and Unambiguous Grammar, and Demerits of Ambiguous Grammar.

Unit 5. Turing Machine

Turing Machine(TM): Single Tape TM, Variations of TM, Halting Problem, Turing Machine and Languages, Enumerable Languages, Decidable, Recognizable and Undecidable languages, Solvable and Unsolvable problems, Post Correspondence Problems(PCP), Classes of Problems: P, NP, NP-C and NP-Hard.

Text Books and References

1. Hopcroft J.E., Motwani R. and Ullman J.D, Introduction to Automata Theory, Languages and Computations, Second Edition, Pearson Education, 2008. (T1)
2. Mishra K.L.P. and Chandrasekaran N. , Theory of Computer Science: Automata, Languages and Computation, 3rd Edition, PHI.(R1)
3. Martin John C., Introduction to Languages and the Theory of Computation, 3rd Edition, Tata McGraw Hill Publishing Company, New Delhi, 2007. (R2)
4. Lewis Harry R. and Papadimitriou Christos H., Elements of the theory of Computation, 2nd Edition, Prentice-Hall of India Pvt. Ltd. (R3).

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECT002 Graph Theory

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECT002.CO1	Understand fundamental graph concepts, including paths, cycles, Eulerian circuits, and graph coloring, and apply proof techniques such as induction.
ECT002.CO2	Apply tree properties, spanning tree algorithms, and optimization techniques like Kruskal's and Dijkstra's to solve real-world problems.
ECT002.CO3	Analyze matching and covering problems using key theorems like Hall's, König-Egeváry, and Gallai's to deduce optimal solutions.
ECT002.CO4	Evaluate graph connectivity and flow networks using Menger's Theorem and the Ford-Fulkerson algorithm to determine max-flow and min-cut solutions.
ECT002.CO5	Create efficient graph coloring and planarity solutions using advanced concepts such as Turán's Theorem, Vizing's Theorem, and Kuratowski's Theorem.

Unit I. Fundamental concepts of graphs

Basic definitions of graphs and multigraphs; adjacency matrices, isomorphism, girth, decompositions, independent sets and cliques, graph complements, vertex coloring, chromatic number, important graph like cubes and the Petersen graph, Paths, cycles, and trails; Eulerian circuits, Vertex degrees and counting; large bipartite subgraphs, the handshake lemma, Havel-Hakimi Theorem, Directed graphs: weak connectivity, connectivity, strong components Induction and other fundamental proof techniques

Unit II. Trees:

Basics of equivalent characterizations of trees, forests, Spanning trees and 2-switches, Distance and center, Optimization: Kruskal's Theorem and Dijkstra's Theorem.

Unit III. Matching and covering

Bipartite matching, vertex cover, edge cover, independent set, M-alternating path, Hall's Theorem, König-Egeváry Theorem, Gallai's Theorem

Unit IV. Connectivity

Vertex cuts, separating sets, bonds; vertex and edge connectivity, block-cut-point tree, Menger's Theorem: undirected vertex and edge versions Network flow: Ford-Fulkerson Labeling algorithm, flow integrality, Max-flow/Min-cut Theorem, proof of Menger's Theorem

Unit V. Coloring

Chromatic number: lower bounds from clique number and maximum, independent set, upper bounds from greedy coloring (& Welsh-Powell), Szekeres-Wilf, and Brooks Theorem. Also k-critical graphs, Cartesian product of graphs, and interval graphs, k-Chromatic graphs: Mycielsk is construction, Turán's Theorem, Edge coloring, line graphs, Vizing's Theorem, Planarity Embeddings, dual graphs, Euler formula, Kuratowski Theorem, Coloring, including the 5-color theorem.

Text Books and References

1. Douglas B. West, Introduction to Graph Theory, Prentice Hall India Ltd., 2001
2. Narasingh Deo, Graph theory, PHI, 1979.
3. Robin J. Wilson, Introduction to Graph Theory, Longman Group Ltd., 2010
4. R. Diestel, Graph Theory, free online edition, 2016: diestel-graph-theory.com/basic.html.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECT003 Advanced Algorithms

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECT003.CO1	Understand the foundational concepts of algorithm analysis, including order notation, recurrences, and randomized algorithms, and apply them to sorting and order statistics.
ECT003.CO2	Apply advanced design techniques such as dynamic programming and greedy methods to solve problems like matrix chain multiplication, vertex cover, and Huffman coding.
ECT003.CO3	Analyze graph algorithms for tasks like topological sorting, minimum spanning trees, shortest paths, and maximum flow to evaluate their efficiency and correctness.
ECT003.CO4	Evaluate the performance of sorting networks and advanced matrix operations, including Strassen's matrix multiplication and solutions to linear equation systems.
ECT003.CO5	Design solutions for string matching problems using algorithms like Rabin-Karp and Knuth-Morris-Pratt, and classify problems based on computational complexity, including NP-completeness.

Unit I. Introduction

Role of Algorithms in computing, Order Notation, Recurrences, Probabilistic Analysis and Randomized Algorithms. Sorting and Order Statistics: Heap sort, Quick sort and Sorting in Linear Time.

Unit II. Advanced Design and Analysis Techniques

Dynamic Programming- Matrix chain Multiplication, Longest common Subsequence and optimal binary Search trees, Vertex cover Problem, Travelling Sales person problem

Unit III. Greedy and Graph Algorithms

Huffman Codes, Activity Selection Problem. Amortized Analysis, Topological Sorting, Minimum Spanning trees, Single Source Shortest Paths, Maximum Flow algorithms.

Unit IV. Sorting Networks and Matrix Operations: Comparison Networks, Zero-one principle, bitonic Sorting Networks, Merging Network, Sorting Network. Strassen's Matrix Multiplication, inverting matrices, solving system of linear Equations

Unit V. String Matching and NP- Completeness: Naive String Matching, Rabin-Karp algorithm, matching with finite Automata, Knuth- Morris - Pratt algorithm, Polynomial time, polynomial time verification, NP-Completeness and reducibility, NP-Complete problems. Approximation

Text Books and References

1. Introduction to Algorithms," T. H. Cormen, C.E. Leiserson, R. L. Rivest, and C. Stein, Third Edition, PHI.
- 2.Fundamentals of Computer Algorithms, Ellis Horowitz, SatrajSahni and Rajasekharam, Galgotia publicationspvt. Ltd.
- 3.Design and Analysis Algorithms - Parag Himanshu Dave, HimanshuBhalchandra Dave Publisher: Pearson
- 4.Algorithm Design: Foundations, Analysis and Internet examples, M.T. Goodrich and R. Tomassia, John Wiley and sons.
5. Data structures and Algorithm Analysis in C++, Allen Weiss, Second edition, Pearson education.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECT004 Parallel and Distributed Algorithms

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECT004.CO1	Understand the fundamental concepts of parallel and distributed computing, including their architectures and techniques for enhancing computation speed.
ECT004.CO2	Apply message-passing techniques and divide-and-conquer strategies to design and evaluate parallel programs effectively.
ECT004.CO3	Analyze pipelining techniques and their applications to optimize computational performance on parallel platforms.
ECT004.CO4	Evaluate resource management strategies, including load balancing and distributed termination in synchronous and shared-memory systems.
ECT004.CO5	Design efficient distributed algorithms for sorting and numerical computations, leveraging shared memory systems and distributed primitives.

Unit I. Introduction

Basic Techniques, Parallel Computers for increase Computation speed, Parallel & Cluster Computing.

Unit II. Message Passing Technique

Message Passing Technique- Evaluating Parallel programs and debugging, Portioning and Divide and Conquer strategies examples

Unit III. Pipelining

Pipelining- Techniques computing platform, pipeline programs examples

Unit IV. Resource management

Synchronous Computations, load balancing, distributed termination examples, programming with shared memory, shared memory multiprocessor constructs for specifying parallelism sharing data parallel programming languages and constructs, open MP.

Unit V. Memory and algorithms

Distributed shared memory systems and programming achieving constant memory, distributed shared memory programming primitives, algorithms – sorting and numerical algorithms.

Text books and References

1. Kai Hwang, Jack Dongarra& Geoffrey C. Fox, “Distributed and Cloud Computing: Clusters, Grids, Clouds, and the Future Internet (DCC)”, 2012.

2. Andrew S. Tanenbaum & Maarten van Steen, “Distributed Systems: Principles and Paradigms”, Prentice Hall, 2017.
3. Parallel Programming, Barry Wilkinson, Michael Allen, Pearson Education, 2nd Edition.
4. Introduction to Parallel algorithms by Jaja from Pearson, 1992.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECT005 Computational Complexity

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECT005.CO1	Explain the foundational models of computation, including Turing machines, and classify problems based on their computational complexity.
ECT005.CO2	Analyze undecidable problems like the Halting Problem and apply reductions and Rice's theorem to explore undecidability in various languages.
ECT005.CO3	Apply complexity class concepts, such as P, NP, and NP-completeness, and demonstrate polynomial-time reductions using problems like satisfiability and knapsack.
ECT005.CO4	Evaluate space complexity classes, hierarchy theorems, and randomized complexity classes like BPP, RP, and ZPP to determine computational limits and trade-offs
ECT005.CO5	Design approximation algorithms for optimization problems, including vertex cover and traveling salesman, with a focus on error analysis and polynomial-time solutions.

Unit I. Introduction

Easy and hard problems. Algorithms and complexity.

Turing machines: Models of computation. Multi-tape deterministic and non-deterministic Turing machines. Decision problems.

Unit II. The Halting Problem and Undecidable Languages

Counting and diagonalization. Tape reduction. Universal Turing machine. Undecidability of halting. Reductions. Rice's theorem.

Deterministic Complexity Classes: DTIME[t]. Linear Speed-up Theorem. P Time. Polynomial reducibility. Polytime algorithms: 2-satisfiability, 2-colourability.

Unit III. NP and NP-completeness

Non-deterministic Turing machines. NTIME[t]. NP. Polynomial time verification. NP-completeness. Cook-Levin Theorem.

Polynomial transformations: 3-satisfiability, clique, colourability, Hamilton cycle, partition problems. Pseudo-polynomial time. Strong NP-completeness, Knapsack. NP-hardness.

Unit IV. Space complexity and hierarchy theorems

DSPACE[s]. Linear Space Compression Theorem. PSPACE, NPSpace. PSPACE = NPSpace. PSPACE-completeness. Quantified Boolean Formula problem is PSPACE-complete. L, NL and NL-completeness. NL=coNL. Hierarchy theorems.

Randomized Complexity: The classes BPP, RP, ZPP. Interactive proof systems: IP = PSPACE.

Unit V. Optimization and approximation

Combinatorial optimization problems. Relative error. Bin-packing problem. Polynomial and fully polynomial approximation schemes. Vertex cover, traveling salesman problem, minimum partition.

Text Books and References

1. Michael Sipser, Introduction to the Theory of Computation, (First edition – PWS Publishing Company, January 1997, or second edition - Thomson Course Technology, 2005).
2. Sanjeev Arora and Boaz Barak, Computational Complexity: A Modern Approach, Cambridge University Press, 2009
3. Christos H Papadimitriou, Computational Complexity, Addison-Wesley, 1994.
4. M R Garey and D S Johnson, Computers and Intractability: A Guide to the Theory of Completeness, Freeman, 1979.
5. Oded Goldreich, Computational Complexity, Cambridge University press, 2008.
6. Vijay Vazirani, Approximation Algorithms, Springer--Verlag, 2001

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECS001 Advanced Computer Architecture

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECS001.CO1	Understand the principles of pipelining and vector processing, including RISC pipelines and array processors, to enhance computational performance.
ECS001.CO2	Apply computer arithmetic algorithms for addition, subtraction, multiplication, and division, along with floating-point operations, in hardware implementation.
ECS001.CO3	Analyze the performance attributes of parallel computer models, including shared and distributed memory systems, vector supercomputers, and SIMD architectures.
ECS001.CO4	Evaluate advanced processor technologies and memory hierarchy designs, such as CISC, RISC, and superscalar processors, to optimize processing capabilities.
ECS001.CO5	Design efficient pipelining and superscalar techniques by optimizing scheduling, dynamic instruction handling, and branch prediction mechanisms.

Unit I. Pipeline and vector processing

Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processors.

Unit II. Computer Arithmetic

Addition and Subtraction, Hardware Implementation, Multiplication Algorithms and Hardware Implementation, Division Algorithms and Hardware Implementation, Floating Point Arithmetic Operations.

Unit III. Parallel Computer Models

Evolution of Computer Architecture, System Attributes to Performance, Shared Memory Multiprocessors, Distributed Memory Multicomputer, Vector SuperComputers, SIMD Super Computers.

Unit IV. Processors and Memory Hierarchy

Advanced Processor Technology: Design Space of Processors, Instruction-Set Architectures, CISC scalar Processors, RISC scalar Processors, SuperScalar and Vector Processors: Super scalar Processors.

Unit V. Pipelining and Super scalar Techniques

Linear Pipeline Processors: Asynchronous and Synchronous models, Clocking and Timing Control, Speedup, Efficiency and Throughput, Pipeline Schedule Optimization, Instruction Pipeline Design: Instruction Execution Phases, Mechanisms for Instruction Pipelining, Dynamic Instruction Scheduling, Branch Handling Techniques.

Text Books and References

1. Kai Hwang and Naresh Jotwani, Advanced Computer Architecture (SIE): Parallelism, Scalability, Programmability, McGraw Hill Education 3/e. 2015.
2. Computer System Architecture, Morris M. Mano, 3rd edition, Pearson/Prentice Hall India.
3. Advanced Computer Architecture, Kai Hwang, McGraw-Hill, India.
4. John L. Hennessy and David A. Patterson, Computer Architecture: A quantitative approach, 5th edition, Morgan Kaufmann Elsevier, 2013.
5. Computer Organization and Architecture, William Stallings, 8th edition, PHI.
6. Computer Organization, Carl Hamacher, Vranesic, Zaky, 5th edition, McGraw Hill.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECS002 Software Engineering

Credits: 3

Course Outcome: After completing this course, the students will be able to:

ECS002.CO1	Identify key concepts and components of software engineering, including various software process models and metrics.
ECS002.CO2	Analyze software requirements, including functional and non-functional requirements, and apply the requirements engineering process.
ECS002.CO3	Design software architectures and models, applying design concepts, principles, and UML diagrams in both object-oriented and structured designs.
ECS002.CO4	Evaluate software testing strategies, quality management, and various testing techniques, including black-box and white-box testing.
ECS002.CO5	Develop risk management strategies, assess software risks, and apply quality management practices in software development to ensure reliability and quality.

Unit I. Introduction to SE

S/W myths, Definitions and components of SE, Software processes and S/W characteristics, S/W applications. Software Process Models: Waterfall, Spiral, Prototype, RAD, 4 GT, Incremental, Evolutionary development, Iterative enhancement, Selection of a model.

Software Metrics

Definitions, Types of S/W metrics, products and processes S/W size and effort estimations: LOC, KLOC etc. Function point estimations COCOMO: basic, intermediate, detailed modes for organic, semidetached and embedded S/W.

Unit II. Software Requirements

Software Requirements: Functional and non-functional requirements, user requirements, system requirements, interface specification, the software requirements document.

Requirements engineering process: Feasibility studies, requirements elicitation and analysis, requirements validation, requirements management.

System models: Context models, behavioral models, data models, object models, structured methods.

Unit III. Software Design

Design Engineering: Design process and design quality, design concepts, the design model.

Creating an architectural design: software architecture, data design, architectural styles and patterns, architectural design, conceptual model of UML, basic structural modeling, class diagrams,

sequence diagrams, collaboration diagrams, use case diagrams, component diagrams. Structured design: modularity, coupling, cohesion, OO design principles.

Unit IV. Software Testing

Testing Strategies and Quality Management: A strategic approach to software testing, test strategies for conventional software, black- Testing Strategies & Product Metrics

Objective, principles, test strategies for conventional software, types of S/W testing: black-box and white-box testing, validation testing, system testing, metrics for testing, metrics for maintenance, the art of debugging.

S/w quality assurance: concepts of S/W quality, S/W reliability, S/W safety.

Unit V. Risk management and Quality

Risk management: Reactive Vs proactive risk strategies, software risks, risk identification, risk projection, risk refinement, RMMM, RMMM plan.

Quality Management: Quality concepts, software quality assurance, metrics for software quality, software reviews, formal technical reviews, statistical software quality assurance, software reliability, the ISO 9000 quality standards.

Text Books:

1. Software Engineering, A practitioner's Approach- Roger S. Pressman, 6th edition, Mc Graw Hill International Edition.
2. Software Engineering- Sommerville, 7th edition, Pearson Education.
3. The unified modeling language user guide Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education.

Reference Books:

4. Software Engineering, an Engineering approach- James F. Peters, Witold Pedrycz, John Wiley.
5. Software Engineering principles and practice- Waman S Jawadekar, The Mc Graw-Hill Companies.
6. Fundamentals of object-oriented design using UML Meiler page-Jones: Pearson Education.

Subject Code & Name: ECS003 Distributed Systems

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECS003.CO1	Explain the characteristics, challenges, and fundamental models of distributed systems to understand their architecture and applications.
ECS003.CO2	Apply concepts of time synchronization, global states, and distributed coordination to solve problems like mutual exclusion and consensus in distributed environments.
ECS003.CO3	Analyze interprocess communication mechanisms, including remote procedure calls and distributed object communication, with relevant case studies like Java RMI..
ECS003.CO4	Evaluate distributed file systems, name services, and distributed shared memory models, focusing on their design, implementation, and consistency requirements.
ECS003.CO5	Design solutions for distributed transactions and concurrency control using techniques such as atomic commit protocols, deadlock resolution, and transaction recovery.

Unit I. Characterization of Distributed Systems

Introduction, Examples of Distributed Systems, Resource Sharing and the Web, Challenges.

System Models: Introduction, Architectural Models, Fundamental Models.

Unit II. Time and Global States

Introduction, Clocks Events and Process States, Synchronizing Physical Clocks, Logical Time and Logical Clocks, Global States, Distributed Debugging.

Coordination and Agreement

Introduction, Distributed Mutual Exclusion, Elections, Multicast Communication, Consensus and Related Problems.

Unit III. Inter Process Communication

Introduction, the API for the Internet Protocols, External Data Representation and Marshalling, Client-Server Communication, Group Communication, and Case Study: IPC in UNIX.

Distributed Objects and Remote Invocation

Introduction, Communication between Distributed Objects, Remote Procedure Call, Events and Notifications, Case Study: JAVA RMI.

Unit IV. Distributed File Systems

Introduction, File Service Architecture, and Case Study 1: Sun Network File System, Case Study 2: The Andrew File System.

Name Services: Introduction, Name Services and the Domain Name System, Directory Services, Case Study of the Global Name Services.

Distributed Shared Memory

Introduction, Design and Implementation Issues, Sequential Consistency and IVY case study, Release Consistency, Munin Case Study, Other Consistency Models.

Unit V. Transactions and Concurrency Control

Introduction, Transactions, Nested Transactions, Locks, Optimistic Concurrency Control, Timestamp Ordering, Comparison of Methods for Concurrency Control.

Distributed Transactions: Introduction, Flat and Nested Distributed Transactions, Atomic Commit Protocols, Concurrency Control in Distributed Transactions, Distributed Deadlocks, Transaction Recovery.

Text Book and References

1.Distributed Systems, Concepts and Design, George Coulouris, J Dollimore and Tim Kindberg, Pearson Education, Edition. 2009.

2.Distributed Systems, Principles and Paradigms, Andrew S. Tanenbaum, Maarten Van Steen, 2nd Edition, PHI.

3.Distributed Systems, An Algorithm Approach, Sukumar Ghosh, Chapman Hall/CRC, Taylor &Fransis Group, 2007.

Subject Code & Name: ECS004 Embedded Systems

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECS004.CO1	Explain the characteristics, classification, and quality attributes of embedded systems, differentiating them from general-purpose computing systems.
ECS004.CO2	Analyze the components of a typical embedded system, including processors, memory, sensors, actuators, and communication interfaces, to determine their application-specific roles.
ECS004.CO3	Apply embedded firmware design techniques using various development approaches and hardware circuits like reset, brown-out protection, and watchdog timers.
ECS004.CO4	Evaluate the role of RTOS in embedded system design, focusing on task scheduling, multitasking, and multiprocessing for efficient operation.
ECS004.CO5	Design task communication and synchronization mechanisms using shared memory, message passing, and device drivers to address real-time constraints in embedded systems.

Unit I. Introduction to Embedded Systems

Definition of Embedded System, Embedded Systems Vs General Computing Systems, History of Embedded Systems, Classification, Major Application Areas, Purpose of Embedded Systems, Characteristics and Quality Attributes of Embedded Systems.

Unit II. Typical Embedded System

Core of the Embedded System: General Purpose and Domain Specific Processors, ASICs, PLDs, Commercial Off-The-Shelf Components (COTS), Memory: ROM, RAM, Memory according to the type of Interface, Memory Shadowing, Memory selection for Embedded Systems, Sensors and Actuators, Communication Interface: Onboard and External Communication Interfaces.

Unit III. Embedded Firmware

Reset Circuit, Brown-out Protection Circuit, Oscillator Unit, Real Time Clock, Watchdog Timer, Embedded Firmware Design Approaches and Development Languages.

Unit IV. RTOS Based Embedded System Design

Operating System Basics, Types of Operating Systems, Tasks, Process and Threads, Multiprocessing and Multitasking, Task Scheduling.

Unit V. Task Communication

Shared Memory, Message Passing, Remote Procedure Call and Sockets, Task.

Synchronization: Task Communication/Synchronization Issues, Task Synchronization Techniques, Device Drivers, How to Choose an RTOS.

Text Books and References

- 1.Introduction to Embedded Systems - Shibu K.V, Mc Graw Hill.
- 2.Embedded Systems - Raj Kamal, TMH.
- 3.Embedded System Design - Frank Vahid, Tony Givargis, John Wiley.
- 4.Embedded Systems – Lyla, Pearson, 2013.
- 5.An Embedded Software Primer - David E. Simon, Pearson Education.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECS005 Advanced Operating System

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECS005.CO1	Understand the fundamental structures and components of advanced operating systems, including process scheduling and inter-process communication.
ECS005.CO2	Analyze memory management techniques, virtual memory mechanisms, and file system implementations to optimize resource allocation.
ECS005.CO3	Apply deadlock handling methods and process synchronization techniques to ensure system reliability and consistency.
ECS005.CO4	Evaluate security principles and access control mechanisms to identify and mitigate operating system vulnerabilities.
ECS005.CO5	Design fault-tolerant distributed systems using concepts like clock synchronization, consistency models, and secure communication protocols.

Unit I. Operating System Introduction, Structures

Simple Batch, Multi programmed, time-shared, Personal Computer, Parallel, Distributed Systems, Real-Time Systems, System components, Operating-System services, System Calls, Virtual Machines, System Design and Implementation. Process and CPU Scheduling - Process concepts and scheduling, Operation on processes, Cooperating Processes, Threads, and Interposes Communication Scheduling Criteria, Scheduling Algorithm, Multiple -Processor Scheduling, Real-Time Scheduling.

Unit II. Memory Management and Virtual Memory

Logical versus Physical Address Space, Swapping, Contiguous Allocation, Paging, Segmentation, Segmentation with Paging. Demand Paging, Performance of Demanding Paging, Page Replacement, Page Replacement Algorithm, Allocation of Frames, Thrashing. File System Interface and Implementation - Access methods, Directory Structure, Protection, File System Structure, Allocation methods, Free-space Management, Directory Management, Directory Implementation, Efficiency and Performance.

Unit III. Deadlocks

System Model, Dead locks Characterization, Methods for Handling Dead locks

Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, and Recovery from Deadlock.
Process Management and Synchronization - The Critical Section Problem, Synchronization Hardware, Semaphores, and Classical Problems of Synchronization, Critical Regions, Monitors.

Unit IV. Operating System Security Issues

Introduction to the topic of Security in Operating Systems, Principles of Information Security, Access Control Fundamentals, and Generalized Security Architectures. Introduction to Distributed systems: Goals of distributed system, hardware and software Concepts, design issues. Elementary introduction to the terminologies within Modern Oss: Parallel, Distributed, Embedded & Real Time, Mobile, Cloud and Other Operating System Models.

Unit V. Clock Synchronization

Clock Synchronization, Logical Clocks, Mutual Exclusion, Global Positioning Of Nodes, Data Centric Consistency Models, Client-Centric Consistency Models, Consistency Protocols. Introduction to Fault Tolerance, Process Resilience, Reliable Client-Server Communication, Reliable Group Communication, Distributed Commit, Recovery, Secure Channels, Access Control, Security Management

Case Study: Over View of UNIX, LINUX, Windows NT, Android and IOS Operating systems.

Text Books and References

- 1.Modern Operating Systems, Andrew S Tanenbaum 2nd edition Pearson/PHI
- 2.Operating System Principles- Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley
- 3.Operating Systems, Dhamdhere, TMH.
- 4.Distributed Operating System - Andrew. S. Tanenbaum, PHI.
- 5.Gary K. Yeap, "Practical Low Power Digital VLSI Design", KAP, 2002.
- 6.Rabaey, Pedram, "Low Power Design Methodologies" Kluwer Academic.
- 7.Kaushik Roy, Sharat Prasad, "Low-Power CMOS VLSI Circuit Design" Wiley, 2000

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECS008 Real Time Systems

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECS008.CO1	Understand the key concepts and challenges in real-time computing, including task scheduling, performance measures, and task assignment in real-time systems.
ECS008.CO2	Apply uniprocessor and multiprocessor scheduling algorithms to real-time tasks, utilizing techniques such as the RM algorithm, priority ceiling, and fault tolerance strategies.
ECS008.CO3	Analyze real-time communication protocols, including VTCSMA, PB CSMA, and deterministic collision resolution for multi-packet messages.
ECS008.CO4	Evaluate real-time database systems and compare them with general-purpose databases, focusing on transaction management, concurrency control, and disk scheduling algorithms.
ECS008.CO5	Design real-time models using Petri nets and apply them in real-world case studies like air traffic control systems and distributed air defense systems.

Unit I. Introduction to Task Scheduling

Introduction - Issues in Real Time Computing, Structure of a Real Time System, Task classes, Performance Measures for Real time Systems, Task Assignment and Scheduling – Classical uniprocessor scheduling algorithms, RM algorithm with different cases-Priority ceiling precedence constraints- using of primary and alternative tasks.

Unit II . UNI and multiprocessor scheduling

Uniprocessor scheduling of IRIS tasks, Task assignment, Utilization balancing – Next fit- Bin packing-Myopic off-line - Focused addressing and bidding- Buddy strategy- Fault Tolerant.

Scheduling: A periodic scheduling - spring algorithm, Horn algorithm- Bratley. – Sporadic scheduling.

Unit III. Real Time Communication

Introduction – VTCSMA – PB CSMA- Deterministic collision resolution protocol- DCR for multi packet messages - dynamic planning based- Communication with periodic and aperiodic messages.

Unit IV. Real Time Databases

Basic Definition, Real time Vs General purpose databases, Main Memory Databases, Transaction priorities, Transaction Aborts, Concurrency control issues, Disk Scheduling Algorithms, Two-phase Approach to improve Predictability, Maintaining Serialization Consistency, Databases for Hard Real Time System.

Unit V. Real-Time Modeling and Case Studies

Petrinets and applications in real-time modeling, Air traffic controller system – Distributed air defense system.

Text Books and References

- 1.C.M. Krishna, Kang G. Shin, “Real Time Systems”, Tata McGraw - Hil, 2010.
- 2.Giorgio C. Buttazzo, “Hard real-time computing systems: predictable scheduling algorithms and applications”, Springer, 2008.
- 3.C. Siva Ram Murthy, G. Manimaran, “Resource management in real-time systems and networks”, PHI, 2009.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECS011 Internet of Things

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECS011.CO1	Explain the fundamental concepts of IoT, including its architecture, functional blocks, and communication models.
ECS011.CO2	Analyze the transition from Machine-to-Machine (M2M) communication to IoT, and the value chains associated with IoT systems.
ECS011.CO3	Evaluate IoT reference architectures, considering technical design constraints and different architectural views such as functional and deployment perspectives.
ECS011.CO4	Apply IoT concepts to develop domain-specific applications in areas like home automation, industrial systems, and surveillance.
ECS011.CO5	Develop IoT solutions using tools such as Arduino, Raspberry Pi, and cloud/fog computing platforms, addressing privacy and security issues.

UNIT I

Introduction to IoT: Sensing, Actuation, Networking basics, Communication Protocols, Sensor Networks, Machine-to-Machine Communications, IoT Definition, Characteristics. IoT Functional Blocks, Physical design of IoT, Logical design of IoT, Communication models & APIs.

UNIT II

M2M to IoT - The Vision-Introduction, From M2M to IoT, M2M towards IoT-the global context, A use case example, Differing Characteristics. Definitions, M2M Value Chains, IoT Value Chains, An emerging industrial structure for IoT.

M2M vs IoT an Architectural Overview – Building architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. Reference Architecture and Reference Model of IoT.

UNIT III

IoT Reference Architecture - Getting Familiar with IoT Architecture, Various architectural views of IoT such as Functional, Information, Operational and Deployment. Constraints affecting design in IoT world- Introduction, Technical design Constraints.

UNIT IV

Domain specific applications of IoT: Home automation, Industry applications, Surveillance applications, Other IoT application.

UNIT V

Developing IoT solutions: Introduction to Python, Introduction to different IoT tools, Introduction to Arduino and Raspberry Pi Implementation of IoT with Arduino and Raspberry, Cloud Computing, Fog Computing, Connected Vehicles, Data Aggregation for the IoT in Smart Cities, Privacy and Security Issues in IoT.

Text Book:

- 1) Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, “From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence”, 1st Edition, Academic Press, 2014.

References Books:

- 1) Vijay Madisetti and ArshdeepBahga, “Internet of Things (A Hands-onApproach)”, 1st Edition, VPT, 2014.
- 2) Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013.
- 3) CunoPfister, Getting Started with the Internet of Things, O’Reilly Media, 2011, ISBN: 978-1-4493- 9357-1.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECD001 Artificial Intelligence

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECD001.CO1	Describe the fundamental concepts, objectives, and search strategies in artificial intelligence, including heuristic search and genetic algorithms.
ECD001.CO2	Apply knowledge representation techniques such as propositional and predicate logic for logical reasoning and problem-solving in AI systems.
ECD001.CO3	Analyze network-based representation methods like semantic networks, frames, and conceptual dependencies for reasoning and natural language processing.
ECD001.CO4	Evaluate game strategies and adversarial search methods, including the minimax procedure and alpha-beta pruning, for decision-making in competitive environments.
ECD001.CO5	Develop reasoning models for uncertain environments using fuzzy logic, Bayesian networks, and Hidden Markov Models to address real-world problems.

Unit I. Introduction

Objective, scope and outcome of the course

Meaning and definition of artificial intelligence, Physical Symbol System Hypothesis, production systems, Characteristics of production systems; Breadth first search and depth first search techniques. Heuristic search Techniques: Hill Climbing, Iterative deepening DFS, bidirectional search. Analysis of search methods. A* algorithm, and their analysis. Introduction to Genetic Algorithms.

Unit II. Knowledge Representation

Knowledge Representation, Problems in representing knowledge, knowledge representation using propositional and predicate logic, logical consequences, syntax and semantics of an expression, semantic Tableau. Forward and backward reasoning. Proof methods, substitution and unification, conversion to clausal form, normal forms, resolution, refutation, deduction, theorem proving, inferencing, monotonic and non-monotonic reasoning. Introduction to prolog.

Unit III. Network-based representation and reasoning

Network-based representation and reasoning, Semantic networks, Conceptual Graphs, frames. Description logic (DL), concept language, reasoning using DL. Conceptual dependencies (CD), scripts, reasoning using CD. Introduction to natural language processing.

Unit IV. Adversarial search and Game theory

Adversarial search and Game theory, classification of games, game playing strategies, prisoner's Dilemma. Game playing techniques, minimax procedure, alpha-beta cut-offs. Complexity of alpha-beta search. Automated planning, classical planning problem, forward planning, partial order planning, planning with proposal logic, hierarchical task planning, multiagent planning.

Unit V. Reasoning in uncertain environments

Reasoning in uncertain environments, Fuzzy logic, fuzzy composition relation, operations on fuzzy sets. Probabilistic reasoning, Bayes theorem, construction of Bayesian networks, belief propagation. Markov processes and Hidden Markov models

Text Books and References

1. “Artificial Intelligence” -By Elaine Rich and Kevin Knight (2nd Edition) Tata McGraw-Hill.
2. Artificial Intelligence: A Modern Approach, Stuart Russel, Peter Norvig, PHI
3. Introduction to Prolog Programming By Carl Townsend.
4. “PROLOG Programming For Artificial Intelligence” -By Ivan Bratko (Addison-Wesley).
5. “Programming with PROLOG” –By Klocksin and Mellish.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECD002 Machine Learning

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECD002.CO1	Understand various learning problems, decision tree learning, and inductive bias concepts for effective machine learning model development.
ECD002.CO2	Apply neural network models and genetic algorithms, including backpropagation and evolutionary strategies, to solve complex learning tasks.
ECD002.CO3	Analyze and implement Bayesian techniques, such as Bayes' Theorem and Naïve Bayes classifiers, to build probabilistic models in machine learning.
ECD002.CO4	Evaluate and apply instance-based learning techniques like K-nearest neighbor and case-based learning to classify and predict data.
ECD002.CO5	Develop and implement advanced machine learning algorithms, including reinforcement learning, temporal difference learning, and first-order rule induction.

Unit I. Basics

Learning Problems Perspectives and Issues Concept Learning Version Spaces and Candidate eliminations – Inductive bias – Decision Tree learning – Representation – Algorithm – Heuristic Space Search.

Unit II. Neural Networks and Genetic Algorithms

Neural Network Representation Problems Perceptions Multilayer Networks and Back Propagation Algorithms – Advanced Topics – Genetic Algorithms Hypothesis Space Search– Genetic Programming – Models of Evolutions and Learning.

Unit III. Bayesian and Computational Learning

Bayes Theorem Concept Learning Maximum Likelihood Minimum Description Length Principle Bayes Optimal Classifier Gibbs Algorithm Naïve Bayes Classifier Bayesian Belief Network EM Algorithm Probability Learning Sample Complexity Finite and Infinite Hypothesis Spaces – Mistake Bound Model.

Unit IV. Instant Based Learning

K- Nearest Neighbor Learning Locally weighted Regression Radial Bases Functions – Case Based Learning.

Unit V. Advanced Learning

Learning Sets of Rules Sequential Covering Algorithm Learning Rule Set First Order Rules Sets of First Order Rules Induction on Inverted Deduction Inverting Resolution Analytical Learning Perfect Domain Theories Explanation Base Learning – FOCL Algorithm - Reinforcement Learning Task Learning Temporal Difference Learning.

Text Books and References

1. Tom M. Mitchell, "Machine Learning", McGraw-Hill, 2010.
2. Bishop, Christopher. Neural Networks for Pattern Recognition. New York, NY: Oxford University Press, 1995.
3. Ethem Alpaydin, (2004) "Introduction to Machine Learning (Adaptive Computation and Machine Learning)", The MIT Press.
4. Hastie, Tibshirani, J. H. Friedman, "The Elements of Statistical Learning", Springer(2nd ed.), 2009.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECD003 Data Mining

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECD003.CO1	Understand the concepts of data mining, knowledge discovery, and preprocessing techniques for improving data quality.
ECD003.CO2	Apply association rule mining techniques, including APRIORI and FP-growth algorithms, to generate frequent item sets and derive meaningful patterns.
ECD003.CO3	Analyze various classification techniques such as decision trees, Naive-Bayes, and K-nearest neighbor to evaluate and solve classification problems.
ECD003.CO4	Evaluate and implement clustering algorithms such as K-means and hierarchical clustering for data segmentation and outlier detection.
ECD003.CO5	Develop mining techniques for web, text, spatial, and multimedia data, along with the concept of data warehousing, to extract valuable insights from complex datasets.

UNIT-I Introduction to Data Mining:

Introduction to Data Mining, , KDD, Challenges, Data Mining Tasks, Data Preprocessing, Data Cleaning, Data Missing, Horizontal and Vertical Partitioning, Dimensionality Reduction, Feature Subset Selection, Discretization and Binaryzation, Data Transformation; Measures of Similarity and Dissimilarity- Basics.

UNIT – II Association Rules:

Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIORI Algorithm, The Partition Algorithms, FP-Growth Algorithms, Compact Representation of Frequent Item Set- Maximal Frequent Item Set, Closed Frequent Item Set.

UNIT – III Classification:

Problem Definition, General Approaches to solving a classification problem Evaluation of Classifiers, Classification techniques, Decision Trees-Decision tree Construction, Algorithm for Decision tree Induction; Naive-Bayes Classifier, Bayesian Belief Networks; K- Nearest neighbor classification-Algorithm and Characteristics.

UNIT – IV Clustering:

Problem Definition, Clustering Overview, Evaluation of Clustering Algorithms, Partitioning Clustering-K-Means Algorithm, K-Means Additional issues, PAM Algorithm; Hierarchical Clustering-Agglomerative Methods and divisive methods, Key Issues in Hierarchical Clustering, Strengths and Weakness; Outlier Detection: Outliers and Outlier Analysis, Outlier Detection Methods

UNIT – V Web and Text Mining:

Mining Object, Spatial, Multimedia, Text and Web Data: Multidimensional Analysis and Descriptive Mining of Complex Data Objects – Spatial Data Mining – Multimedia Data Mining – Text Mining – Mining the World Wide Web, Concept of Data Warehousing.

Textbook Books:

- 1) P.-N. Tan, M. Steinbach, and V. Kumar, Introduction to Data Mining, Addison Wesley, 2005.
- 2) Mohammed J. Zaki, Wagner Meira Jr., Data Mining and Analysis: Fundamental Concepts and Algorithms, Cambridge Press, 2014.
- 3) Data Mining --- Concepts and techniques, by Han and Kamber, Morgan Kaufmann, 2nd, 200 Other Supplemental Material:

Reference Books:

- 1) Principles of Data Mining, by Hand, Mannila, and Smyth, MIT Press, 2001. (ISBN: 0-262-08290-X).
- 2) The Elements of Statistical Learning --- Data Mining, Inference, and Prediction, by Hastie, Tibshirani, and Friedman, Springer, 2001. (ISBN: 0-387-95284-5).
- 3) Mining the Web --- Discovering Knowledge from Hypertext Data, by Chakrabarti, Morgan Kaufmann, 2003. (ISBN:1-55860-754-4).
- 4) I. H. Witten and E. Frank, Data Mining: Practical Machine Learning Tools and Techniques with Java Implementations, Morgan Kaufmann, 2nd ed. 2005.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECD004 Soft Computing

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECD004.CO1	Understand the fundamental concepts of soft computing, including the difference between soft and hard computing, and identify the components of soft computing systems.
ECD004.CO2	Analyze and compare different artificial neural network architectures and their limitations across various domains.
ECD004.CO3	Apply supervised learning algorithms, including the perceptron and backpropagation networks, to solve real-world problems in computer vision and pattern recognition.
ECD004.CO4	Evaluate unsupervised learning techniques such as Hebbian learning, Kohonen networks, and fuzzy logic for developing intelligent systems.
ECD004.CO5	Design and implement genetic algorithms and genetic programming techniques to solve optimization problems and develop adaptive systems.

UNIT- I Soft Computing:

Overview of Soft Computing, Difference between Soft and Hard computing, Brief descriptions of different components of soft computing including Artificial intelligence systems Neural networks, fuzzy logic, genetic algorithms, An overview of hybrid soft computing approaches.

UNIT-II Artificial Neural Networks:

Artificial neural networks Vs Biological neural networks, ANN architecture, Basic building block of an artificial neuron, Activation functions, Introduction to Early ANN architectures (basics only)- McCulloch & Pitts model, Perceptron, ADALINE, and MADALINE, limitations of ANNs in various domains, potential societal impacts.

UNIT - III Supervised Learning:

Introduction and how brain works, Neuron as a simple computing element, The perceptron, Backpropagation networks: architecture, multilayer perceptron, backpropagation learning-input layer, accelerated learning in multilayer perceptron, The Hopfield network, Bidirectional associative memories (BAM), RBF Neural Network, computer vision tasks.

UNIT- IV Unsupervised Learning:

Hebbian Learning, Generalized Hebbian learning algorithm, Competitive learning, Self-Organizing Computational Maps: Kohonen Network. Fuzzy Logic Crisp & fuzzy sets fuzzy relations fuzzy conditional statements fuzzy rules fuzzy algorithm. Fuzzy logic controller, Generative Adversarial Networks (GANs)

UNIT-V Genetic Algorithms:

Basic concepts, encoding, fitness function, reproduction-Roulette wheel, Boltzmann, tournament, rank, and steady state selections, Convergence of GA, Applications of GA case studies. Introduction to genetic programming- basic concepts, basics of program synthesis

Textbook Books:

- 1) R. Rajasekaran and G. A and Vijayalakshmi Pa, Neural Networks, Fuzzy Logic, and Genetic Algorithms: Synthesis and Applications, Prentice Hall of India.
- 2) D. E. Goldberg, Genetic Algorithms in Search, Optimisation, and Machine Learning, Addison-Wesley.

Reference Books:

- 1) L. Fausett, Fundamentals of Neural Networks, Prentice Hall.
- 2) T. Ross, Fuzzy Logic with Engineering Applications, Tata McGraw Hill.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECD006 Data Analytics

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECD006.CO1	Explain the phases, types, and importance of data quality and exploratory data analysis in business intelligence contexts.
ECD006.CO2	Apply Big Data concepts, Hadoop ecosystem components, and MapReduce techniques for effective data processing and management.
ECD006.CO3	Develop interactive dashboards using data visualization tools and frameworks like dc.js to present data insights.
ECD006.CO4	Analyze machine learning models, including supervised and unsupervised learning algorithms, for predictive and exploratory analytics.
ECD006.CO5	Evaluate ethical considerations in data science, such as privacy, informed consent, and inclusion, to ensure responsible analytics practices.

Unit I. Introduction - Data Analytics

Types – Phases - Quality and Quantity of data – Measurement - Exploratory data analysis - Business Intelligence.

Unit II. BIG DATA Big Data and Cloud technologies - Introduction to HADOOP: Big Data, Apache Hadoop, MapReduce - Data Serialization - Data Extraction - Stacking Data - Dealing with data.

Unit III. Data Visualization - Introduction to data visualization – Data visualization options – Filters – Dashboard development tools – Creating an interactive dashboard with dc.js - summary

Unit IV. Analytics And Machine Learning - Machine learning – Modeling Process – Training model – Validating model – Predicting new observations –Supervised learning algorithms – Unsupervised learning algorithms.

Unit V. Ethics And Recent Trends - Data Science Ethics – Doing good data science – Owners of the data - Valuing different aspects of privacy - Getting informed consent - The Five Cs – Diversity – Inclusion – Future Trends.

Text Books and References

1. Davy Cielen, Arno D. B. Meysman, Mohamed Ali, Introducing Data Science, Manning Publications Co., 1st edition, 2016.
2. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, An Introduction to Statistical Learning: with Applications in R, Springer, 1st edition, 2013.
3. Bart Baesens, Analytics in a Big Data World: The Essential Guide to Data Science and its Applications, Wiley.

4. D J Patil, Hilary Mason, Mike Loukides, Ethics and Data Science, O' Reilly, 1st edition, 2018.
5. Dr Anil Maheshwari, Data Analytics Made Accessible, Publisher: Amazon.com Services LLC.
6. Joel Grus, Data Science from Scratch: First Principles with Python, O'Reilly, 1st edition, 2015.
7. Cathy O'Neil, Rachel Schutt, Doing Data Science, Straight Talk from the Frontline, O' Reilly, 1st edition, 2013.
8. Jure Leskovec, AnandRajaraman, Jeffrey David Ullman, Mining of Massive Datasets, Cambridge University Press, 2nd edition, 2014.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECA008 Neural Networks and Deep Learning

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECD008.CO1	Understand the foundational concepts of artificial neural networks, including neuron models, architectures, and learning methods.
ECD008.CO2	Apply feed-forward neural network algorithms, such as backpropagation and delta rule, to solve classification and regression problems.
ECD008.CO3	Analyze the characteristics and functionalities of advanced ANN architectures like Kohonen Self-Organizing Maps, ART networks, and Associative Memories.
ECD008.CO4	Evaluate deep learning models, including strategies for training and regularizing deep neural networks, addressing vanishing/exploding gradients and optimization challenges.
ECD008.CO5	Design convolutional neural networks (CNNs) and implement them for real-world applications like object detection and content-based image retrieval.

Unit I . Introduction to Artificial Neural Networks

Fundamentals of Neural Networks – Model of Artificial Neuron – Neural Network Architectures – Learning Methods – Taxonomy of Neural Network Architectures – Applications

Unit II . Feed Forward Neural Networks

Perceptron Models, Discrete, Continuous and Multi-Category –Training Algorithms: Discrete and Continuous Perceptron Networks – Limitations of the Perceptron – Model. Credit Assignment Problem – Generalized Delta Rule, Derivation of Back propagation (BP) Training, and Summary of Back propagation Algorithm –Kolmogorov Theorem.

Unit III . Other ANN Architectures

Associative Memory – Exponential BAM – Associative Memory For Real Coded Pattern Pairs – Applications Adaptive Resonance Theory – Introduction – ART 1 – ART2 – Applications – Neural Networks Based On Competition – Kohonen Self Organizing Maps – Learning Vector Quantization – Counter Propagation Networks – Industrial Applications

Unit IV . Deep Learning

Deep Feed Forward network, regularizations, training deep models, dropouts, Training Deep Neural Networks using Back Propagation-Setup and initialization issues, vanishing and exploding Gradient problems, Gradient- Descent Strategies.

Unit V. Convolutional Neural Network

Convolutional Neural Network, Basic structure of Convolutional Network, Case studies: Alex net, VGGNet, GoogLeNet, Applications of CNN– Object Detection, Content based image Retrieval.

Text Books and References

- 1.CharuC.Aggarwal “Neural Networks and Deep learning” Springer International Publishing, 2018.
- 2.Satish Kumar, “Neural Networks, A Classroom Approach”, Tata McGraw -Hill, 2007.
- 3.Simon Haykin, “Neural Networks, A Comprehensive Foundation”, 2nd Edition, Addison Wesley Longman, 2001.
- 4.Bishop, Christopher M. Pattern Recognition and Machine Learning. Springer, 2006.
- 5.Duda, Richard, Peter Hart, and David Stork. Pattern Classification. 2nd ed. New York, NY: Wiley- Interscience, 2000.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECA001 Image Processing

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECA001.CO1	Explain the fundamental steps and techniques in digital image processing, including image enhancement in spatial and frequency domains.
ECA001.CO2	Apply segmentation techniques such as edge detection, thresholding, and region growing for analyzing image features.
ECA001.CO3	Analyze various feature extraction methods like edge detection, shape descriptors, and texture features to extract relevant information from images.
ECA001.CO4	Evaluate image registration techniques and fusion methods, including point matching, affine transformations, and wavelet-based fusion.
ECA001.CO5	Design and implement 3D image visualization methods, including ray tracing, volumetric display, and surface measurements, for enhanced visual analysis.

Unit I. Digital Image Processing

Steps in digital image processing-Elements of visual perception- brightness adaptation, Mach band effect. Image enhancement in spatial and frequency domain, Histogram equalization.

Unit II. Segmentation

Edge detection, Thresholding, Region growing, Fuzzy clustering, Watershed algorithm, Active contour models, and Texture feature based segmentation, Graph based segmentation, Wavelet based Segmentation - Applications of image segmentation.

Unit III. Feature Extraction

First and second order edge detection operators, Phase congruency, Localized feature extraction - detecting image curvature, shape features, Hough transform, shape skeletonization, Boundary descriptors, Moments, Texture descriptors- Autocorrelation, Co-occurrence features, Run length features, Fractal model based features, Gabor filter, wavelet features.

Unit IV. Registration And Image Fusion

Registration - Preprocessing, Feature selection - points, lines, regions and templates Feature correspondence - Point pattern matching, Line matching, Region matching, Template matching. Transformation functions - Similarity transformation and Affine Transformation. Resampling – Nearest Neighbour and Cubic Splines. Image Fusion - Overview of image fusion, pixel fusion, and wavelet based fusion -region based fusion.

Unit V. 3D Image Visualization

Sources of 3D Data sets, Slicing the Data set, Arbitrary section planes, The use of color, Volumetric display, Stereo Viewing, Ray tracing, Reflection, Surfaces, Multiple connected surfaces, Image processing in 3D, Measurements on 3D images.

Text Books and References

1. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing', Pearson, Education, Inc., Second Edition, 2004.
2. Mark Nixon, Alberto Aguado, "Feature Extraction and Image Processing", Academic Press, 2008.
3. Ardeshtir Goshtasby, "2D and 3D Image registration for Medical, Remote Sensing and Industrial Applications", John Wiley and Sons, 2005.
4. John C. Russ, "The Image Processing Handbook", CRC Press, 2007.
5. Anil K. Jain, Fundamentals of Digital Image Processing', Pearson Education, Inc., 2002.
6. Rick S. Blum, Zheng Liu, "Multisensor image fusion and its Applications", Taylor Francis, 2006.

Faculty: Faculty of Computing and Information Technology**Program:** B.Tech (CSE)**Semester:** V/VI/VII/VIII**Session:** ODD/EVEN**Subject Code & Name:** ECA002 Digital Signal Processing**Credits:** 3**Course Outcome:** At the end of the course, the students will be able to:

ECA002.CO1	Understand the fundamental concepts of discrete-time signals, systems, and their frequency domain representations.
ECA002.CO2	Analyze and apply Fourier series and Fourier transforms to compute the discrete Fourier transform (DFT) and analyze its properties.
ECA002.CO3	Design and implement realization structures for FIR and IIR digital filters using direct, cascade, and parallel forms.
ECA002.CO4	Evaluate the frequency response of FIR digital filters and apply various design techniques such as Fourier method and window techniques.
ECA002.CO5	Apply multirate signal processing techniques, including decimation, interpolation, and sampling rate conversion, to solve real-world signal processing problems.

Unit I. Introduction

Introduction to Digital Signal Processing: Discrete time signals & systems, linear shift invariant systems, stability and causality, discrete time systems described by difference equations, Frequency domain representation of discrete time signals and systems.

Unit II. Fourier series And Fourier Transforms

Fourier series And Fourier Transforms: Discrete Fourier series representation of periodic sequences, Properties of discrete Fourier series, Discrete Fourier transforms: frequency domain sampling, linear convolution of sequences using DFT, Computation of DFT, Relationship of DFT to other transforms, Properties of DFT,

Unit III. Realization of Digital Filters

Realization of Digital Filters: Structures for FIR systems: Direct form structure, Cascade form structures, Structures for IIR systems: Direct form structures, Signal flow graphs and transposed structures, cascade form structures, Parallel form structures.

Unit IV. FIR Digital Filters

FIR Digital Filters: Characteristics of FIR Digital Filters, Frequency Response. Design of FIR Filters: Fourier Method. Digital Filters using Window Techniques, Frequency Sampling Technique, Comparison of IIR & FIR filters.

Unit V. Multirate Digital Signal Processing

Multirate Digital Signal Processing: Introduction, down sampling, Decimation, up sampling, Interpolation, Sampling Rate Conversion, Applications of Multi-Rate Signal Processing.

Text Books and References

1. John G. Proakis, Dimitris G. Manolakis (2007), Digital Signal Processing, Principles, Algorithms, and Applications, Pearson Education / PHI, India.
2. A.V. Oppenheim, R. W. Schaffer (2009), Discrete Time Signal Processing, Prentice Hall of India, New Delhi.
3. Andreas Antoniou (2006), Digital Signal Processing, Tata McGraw Hill, New Delhi.
4. M. H. Hayes (2007), Schaums Outlines of Digital Signal Processing, Tata McGraw Hill, India.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECA003 Cloud Computing

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECA003.CO1	Understand the core concepts, service models, and deployment models of cloud computing, including its architecture and properties.
ECA003.CO2	Apply virtualization techniques such as hypervisors, VM provisioning, and migration to effectively manage cloud resources.
ECA003.CO3	Analyze service management aspects, including SLAs, billing, and large-scale data processing, to ensure efficient cloud operations.
ECA003.CO4	Evaluate task scheduling algorithms for cloud environments, such as MapReduce and deadline-based scheduling, to optimize resource utilization.
ECA003.CO5	Design secure cloud systems by addressing virtualization risks, data privacy concerns, and implementing access control mechanisms.

Unit I. Introduction

Overview of Computing Paradigms: Grid Computing, Cluster Computing, Distributed Computing, Utility Computing. Cloud Computing Cloud Computing (NIST Model) - Properties and Characteristics of Cloud. Cloud Computing Architecture - Cloud computing stack Service Models (XaaS): Infrastructure as aService (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS) Deployment Models: Public cloud, Private cloud, Hybrid cloud. Data Center Architecture.

Unit II. Cloud Resource Virtualization

Introduction to virtualization Different approaches to virtualization Hypervisors Machine Image Virtual Machine (VM) Process VM vs System VM Resource Virtualization: Server, Storage, Network Full Virtualization vs Para Virtualization Operating System Support for Virtualization Virtual Machine (resource) Provisioning and Manageability VM Placement, VM Migration.

Unit III. Service Management in Cloud Computing

Service Level Agreements (SLAs) Billing and Accounting Economics of scaling Managing

Data: Database and Data Stores in Cloud, Large Scale Data Processing.

Unit IV. Task Scheduling in Cloud

Scheduling Algorithms for Computing Clouds Fair Queuing Start Time Fair Queuing
Borrowed Virtual Time Cloud Scheduling Subject to Deadlines Scheduling Map Reduce
Applications Subject to Deadlines.

Unit V. Cloud Security

Cloud Security Risks, Trust, Operating System Security, VM Security, Security of
Virtualization, Security Risks Posted by Shared Images, Security Risks Posted by Management
OS, Data privacy and security Issues, Identity & Access Management, Access Control,
Authentication in cloud computing, Case Study - Microsoft Azure, Amazon EC2.

Text Books and References

- 1.Dan C Marinescu, Cloud Computing, Theory and Practice, MK Elsevier.
2. Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Cloud Computing: Principles and Paradigms, Wiley.
- 3.Barrie Sosinsky, Cloud Computing Bible, Wiley.
- 4.Jim Smith, Ravi Nair, Virtual Machines: Versatile Platforms for Systems and Processes, MK Elsevier.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECA004 Human Computer Interaction

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECA004.CO1	Understand the fundamentals of human-computer interaction, including the human I/O channels, memory, reasoning, computer devices, and interaction paradigms
ECA004.CO2	Apply interactive design principles, prototyping, and evaluation techniques to the software development process, emphasizing usability engineering.
ECA004.CO3	Analyze cognitive models and socio-organizational issues in HCI, along with communication and collaboration models in digital environments.
ECA004.CO4	Evaluate mobile HCI concepts, including mobile ecosystems, platforms, application frameworks, and mobile design elements.
ECA004.CO5	Design effective web interfaces by utilizing principles such as drag-and-drop, direct selection, and process flow in web applications.

Unit I. Foundations of HCI

The Human: I/O channels – Memory – Reasoning and problem solving; The computer: Devices – Memory – processing and networks; Interaction: Models – frameworks – Ergonomics – styles – elements – interactivity- Paradigms.

Unit II. Design & Software Process

Interactive Design basics – process – scenarios – navigation – screen design – Iteration and prototyping. HCI in software process – software life cycle – usability engineering – Prototyping in practice – design rationale. Design rules – principles, standards, guidelines, rules. Evaluation Techniques – Universal Design.

Unit III. Models and Theories

Cognitive models –Socio-Organizational issues and stake holder requirements –Communication and collaboration models-Hypertext, Multimedia and WWW.

Unit IV. Mobile HCI: Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools.

Unit V. Web Interface Design

Designing Web Interfaces – Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow. Case Studies.

Text Books and References

2. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, “Human Computer Interaction”, 3rd Edition, Pearson Education, 2004.

3. 2.Brian Fling, “Mobile Design and Development”, First Edition, O’Reilly Media Inc., 2009
4. Bill Scott and Theresa Neil, “Designing Web Interfaces”, First Edition, O’Reilly, 2009.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECA006 Computer Graphics

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECA006.CO1	Understand the basics of computer graphics, different graphics systems and applications of computer graphics.
ECA006.CO2	Apply matrix representations, transformations, and viewing pipelines to create and manipulate graphical objects in 2D and 3D spaces.
ECA006.CO3	Analyze the efficiency and applicability of various 2D and 3D clipping algorithms for graphical rendering tasks.
ECA006.CO4	Evaluate the use of different geometric primitives and transformations to achieve realistic 3D object representations and projections.
ECA006.CO5	Design complex graphical systems using advanced techniques like polygon clipping, composite transformations, and 3D object viewing.

Unit I. Introduction

Introduction and Line Generation: Types of computer graphics, Graphic Displays- Random scan displays, Raster scan displays, Frame buffer and video controller, Points and lines, Line drawing algorithms, Midpoint circle generating algorithm, and parallel version of these algorithms.

Unit II. Transformations

Basic transformation, Matrix representations and homogenous coordinates, Composite transformations, Reflections and shearing. Windowing and Clipping: Viewing pipeline, viewing transformations.

Unit III. Clipping

2-D Clipping algorithms-Line clipping algorithms such as Cohen Sutherland line clipping algorithm, Liang Barsky algorithm, Line clipping against non-rectangular clip windows; Polygon clipping – Sutherland Hodgeman polygon clipping, Weiler and Atherton polygon clipping, Curve clipping, Text clipping, 3-D geometric primitives, 3-D Object representation, 3-D Transformation, 3-D viewing, projections, 3-D Clipping.

Unit IV. Curves and Surfaces

Quadric surfaces, Spheres, Ellipsoid, Blobby objects, introductory concepts of Spline, Bspline and Bezier curves and surfaces.

Unit V.Methods and Models

Hidden Lines and Surfaces, Back Face Detection algorithm, Depth buffer method, A- buffer method, Scan line method, basic illumination models – Ambient light, Diffuse reflection, Specular reflection and Phong model.

Text Books and References

1. Amrendra N Sinha and Arun D Udai,” Computer Graphics”, TMH.
2. Steven Harrington, “Computer Graphics: A Programming Approach”, TMH.
3. Rogers, “Procedural Elements of Computer Graphics”, McGraw Hill.

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECA008 Optimization Technique

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECA008.CO1	Explain the phases, types, and importance of data quality and exploratory data analysis in business intelligence contexts.
ECA008.CO2	Apply Big Data concepts, Hadoop ecosystem components, and MapReduce techniques for effective data processing and management.
ECA008.CO3	Develop interactive dashboards using data visualization tools and frameworks like dc.js to present data insights.
ECA008.CO4	Analyze machine learning models, including supervised and unsupervised learning algorithms, for predictive and exploratory analytics.
ECA008.CO5	Evaluate ethical considerations in data science, such as privacy, informed consent, and inclusion, to ensure responsible analytics practices.

Unit I. Introduction to Optimization

Historical Development, Engineering applications of Optimization, Design vector and constraints, Constraint surface, Objective function, Classification of Optimization Problems.

Unit II. Classical Optimization Techniques

Single variable optimization, Constrained and unconstrained multi-variable optimization, Direct substitution method, Lagrange's method of multipliers, Karush-Kuhn-Tucker conditions.

Linear Programming

Statement of an LP problem, Graphical Solution of an LP problem, Simplex method, Dual simplex method

Unit III. Non-linear Programming

One-dimensional minimization method Unimodal function, unrestricted search, Exhaustive search, Dichotomous search, Interval halving method, Fibonacci method, Golden section method, direct root methods.

Unit IV. Non-linear Programming

Unconstrained Optimization Techniques Direct Search Methods: Random search methods, Grid search method, Univariate method, Hookes and Jeeves' method, Powell's method Indirect Search Methods: Steepest descent method, Fletcher-Reeves method, Newton's method.

Unit. V

Non-linear Programming

Constrained Optimization Techniques Direct Methods: Random search method, Sequential linear programming Indirect methods: Transformation techniques, Exterior penalty function method, Interior penalty function method

Evolutionary Algorithms

An overview of evolutionary algorithms, Simulated annealing algorithm, Genetic algorithm, Particle swarm optimization.

Text Books and References

- 1.Mokhtar S. Bazaaraa, Hanif D. Shirali and M.C.Shetty, “Nonlinear Programming, Theory and Algorithms”, John Wiley & Sons, New York (2004).
- 2.S. S. Rao, “Engineering Optimization: Theory and Practice”, 4th Edition, John Wiley & Sons (2009).
- 3.Kwang Y. Lee, Mohamed A. El-Sharkawi, “Modern heuristic optimization techniques: theory and applications”, Kluwer (2008).
- 4.Hamdy A. Taha, “Operations Research: An Introduction”, 8th Edition, Pearson Education (2008).
- 5.G. V. Reklaitis, A. Ravindran, K. M. Ragsdell, “Engineering Optimization: Methods and Applications”, Wiley (2006).
- 6.Michael C. Bartholomew-Biggs, “Nonlinear optimization with engineering applications”, Springer (2008).

Faculty: Faculty of Computing and Information Technology

Program: B.Tech (CSE)

Semester: V/VI/VII/VIII

Session: ODD/EVEN

Subject Code & Name: ECA010 Cryptography and Network Security

Credits: 3

Course Outcome: At the end of the course, the students will be able to:

ECA010.CO1	Explain the fundamental concepts of security, including types of attacks, cryptographic techniques, and the principles of cryptography.
ECA010.CO2	Apply symmetric key cryptographic algorithms such as DES, AES, and Blowfish, along with hash functions like MD5 and SHA, to secure data.
ECA010.CO3	Analyze asymmetric key cryptographic methods, including RSA and digital signatures, for secure communication.
ECA010.CO4	Evaluate authentication mechanisms, key management techniques, and Public Key Infrastructure (PKI) to ensure system security.
ECA010.CO5	Implement security protocols like SSL, SET, and IPSec to protect data in networks and prevent intrusion.

Unit I- Introduction to the Concept of Security:

Introduction, The Need of Security, Security Approaches, Principal of Security, Types of Attacks

Cryptographic Techniques:

Introduction, Plain Text and Cipher Text, Substitution Techniques, Transposition Techniques, Encryption and Decryption, Symmetric and Asymmetric Key Cryptography, Steganography, Key Range and Key Size, Possible Types of Attacks

Unit II- Symmetric Key Cryptography:

Introduction, Algorithm Types and Models, An Overview of Symmetric Key Cryptography, Data Encryption Standard(DES), International Data Encryption Algorithm(IDEA), Blowfish, Advanced Encryption Standard(AES), Hash Functions- MD5 & SHA, Message Authentication Codes

Unit III-Asymmetric Key Cryptography:

Introduction, An Overview of Asymmetric Key Cryptography, The RSA Algorithm, Symmetric and Asymmetric Key Cryptography Together, Digital Signature

Unit IV-Authentication and Public Key Infrastructure (PKI):

User Authentication: Password, Challenge Response, Introduction to PKI, Digital Certificates, Key Management techniques,

Unit V-Security Protocols:

Basic Concepts, Kerberos, Security Socket Layer(SSL), Secure Electronic Transaction(SET), Email Security- PGP, S/MIME, IP Security (IPSec).

Network Security:

Brief Introduction to TCP/IP, Firewalls, IP Security, Virtual Private Networks (VPN), Intrusion Detection

Text Book:

1. Forouzan B. A., Mukhopadhyay D., “Cryptography and Network Security”, 3rd Edition, McGraw Higher Education, 2016.

Reference Books:

1. Stallings W., “Cryptography and Network Security: Principles and Practice”, 7th Edition, Pearson, 2017.
2. Kahate A., “Cryptography and Network Security”, 3rd Edition, McGraw Hill Education, New Delhi, 2013.
3. C. P. Pfleeger, S. L. Pfleeger; Security in Computing, Prentice Hall of India, 2006